Model Analysis

Table of contents

1 Data analysis /Dataframe import and filters

1.1 DataFrame import and filters

The data has been imported with the following(s) function(s):

```python
def build_df():
    if flag_IR:
        df, features_engineered = import_data_IR()
        if flag_IR_maskP:
            df = mask_period_IR(df)
        if flag_IR_balR:
            df = mask_balanceHC_random_IR(df, n_samples=None)
    if flag_ER:
        df, features_engineered = import_data_ER()

    cols_to_check = target_col + features_X
    mask = np.isfinite(df[cols_to_check]).all(axis=1)
    df = df[mask]
```

```python
    cols = list(dict.fromkeys(target_col + features_X  +
                              feats_to_balance + feats_grouping +
                              feats_reference+ feats_postproc))
    df = df[cols]
    #df['campaign'] = "ALL"

    return df
```

```python
def import_data_IR():
    if target_col[0] == "i_NH3":
        file_path = "csv_decomposed/innova/df_signal_decomposition_NH3_junesept.csv"
        #file_path = "csv/innova/switched/df_signal_decomposition_10_50_200.csv"
    elif target_col[0] == "i_CO2":
        file_path = "csv_decomposed/innova/df_signal_decomposition_CO2_current.csv"
    else:
        raise ValueError("Target is not supported")

    df = pd.read_csv(file_path, sep=";")
    df["datetime"] = pd.to_datetime(df["datetime"])
    df.sort_values("datetime").reset_index(drop=True)

    if target_col[0] == "i_NH3":
        df["campaign"] = df["datetime"].apply(assign_campaign)
        df = df[df["campaign"] != "other"].copy()
        df["innova_point"] = df.apply(assign_innova_point, axis=1)
        df = df[df["innova_point"] != "no"].copy()
        with open("csv_decomposed/innova/features_decomposition_NH3_list_junesept.json", "r"
        #with open("csv/innova/switched/features_decomposition_list_10_50_200.json", "r")as
            dict_features = json.load(f)
    elif target_col[0] == "i_CO2":
        df["campaign"] = df["datetime"].apply(assign_campaign_iCO2)
        df = df[df["campaign"] != "other"].copy()
        df["innova_point"] = df.apply(assign_innova_point_iCO2, axis=1)
        df = df[df["innova_point"] != "no"].copy()
        with open("csv/innova/features_decomposition_CO2_list.json", "r") as f:
            dict_features = json.load(f)
    else:
        raise ValueError("The target col do not match any existing function")


    df["hot"] = df.apply(assign_hot_season,axis=1)
    df["umidity_range"] = df.apply(assign_humidity, axis = 1)
```

```
    #Removing of rabbit 4
    df["column"] = df.apply(assign_colonna, axis = 1)
    df['height'] = df.apply(assign_altezza, axis = 1)



    return df,dict_features
```

```
def assign_innova_point(row):
    camp = row["campaign"]
    rid = row["id_rabbit"]

    # if camp == "2025_Jun":
    #     mapping = {2: "sl11", 4: "sl5", 6: "sl4", 1: "sl3", 5: "no"}
    # elif camp == "2025_Sep":
    #     mapping = {2: "sl11", 4: "sl5", 3: "sl4", 6: "sl3", 5: "no"}
    if camp == "2025_Apr":
        #mapping = {1: "no", 2: "no", 4: "no", 6: "sl5"}
        #mapping = {1: "no", 2: "sl5", 4: "no", 6: "sl5"}
        mapping = {1: "no", 2: "no", 4: "no", 6: "no"}
    elif camp == "2025_Jun":
        mapping = {2: "sl11", 4: "sl5", 1: "sl3", 6: "sl4", 5: "no"}
        #mapping = {2: "sl11", 4: "no", 1: "sl4", 6: "sl3", 5: "no"}
    elif camp == "2025_Sep":
        mapping = {2: "sl11", 4: "sl5", 6: "sl3", 3: "sl4", 5: "no"}
        #mapping = {2: "sl11", 4: "no", 6: "sl4", 3: "sl3", 5: "no"}
    elif camp == "2026_Gen":
        #mapping = {1: "no", 4: "no", 5: "sl11", 6: "no"}
        #mapping = {1: "sl11", 4: "no", 5: "sl11", 6: "sl11"}
        mapping = {1: "no", 4: "no", 5: "no", 6: "no"}
    else:
        mapping = {}

    return mapping.get(rid, "no")
```

Function print :

1.2 Metadata

Nombre de lignes df	7280

Nombre de lignes df clean	7280
Nombre de features	77
% de NaN	0.0
Taille dataset entraînement	5096
Taille dataset test	2184
Métrique utilisée	r2

Column target : i_NH3

Features_X : r_NH3, r_temperature, r_umidity, r_CO2, r_THI, r_NH3_st_8h_mean, r_NH3_st_8h_std, r_NH3_st_8h_rms, r_NH3_st_8h_median, r_NH3_st_8h_min, r_NH3_st_8h_max, r_NH3_st_8h_diff_last, r_NH3_st_8h_slope, r_NH3_st_12h_mean, r_NH3_st_12h_std, r_NH3_st_12h_rms, r_NH3_st_12h_median, r_NH3_st_12h_min, r_NH3_st_12h_max, r_NH3_st_12h_diff_last, r_NH3_st_12h_slope, r_NH3_st_24h_mean, r_NH3_st_24h_std, r_NH3_st_24h_rms, r_NH3_st_24h_median, r_NH3_st_24h_min, r_NH3_st_24h_max, r_NH3_st_24h_diff_last, r_NH3_st_24h_slope, r_temperature_st_8h_mean, r_temperature_st_8h_std, r_temperature_st_8h_rms, r_temperature_st_8h_median, r_temperature_st_8h_min, r_temperature_st_8h_max, r_temperature_st_8h_diff_last, r_temperature_st_8h_slope, r_temperature_st_12h_mean, r_temperature_st_12h_std, r_temperature_st_12h_rms, r_temperature_st_12h_median, r_temperature_st_12h_min, r_temperature_st_12h_max, r_temperature_st_12h_diff_last, r_temperature_st_12h_slope, r_temperature_st_24h_mean, r_temperature_st_24h_std, r_temperature_st_24h_rms, r_temperature_st_24h_median, r_temperature_st_24h_min, r_temperature_st_24h_max, r_temperature_st_24h_diff_last, r_temperature_st_24h_slope, r_umidity_st_8h_mean, r_umidity_st_8h_std, r_umidity_st_8h_rms, r_umidity_st_8h_median, r_umidity_st_8h_min, r_umidity_st_8h_max, r_umidity_st_8h_diff_last, r_umidity_st_8h_slope, r_umidity_st_12h_mean, r_umidity_st_12h_std, r_umidity_st_12h_rms, r_umidity_st_12h_median, r_umidity_st_12h_min, r_umidity_st_12h_max, r_umidity_st_12h_diff_last, r_umidity_st_12h_slope, r_umidity_st_24h_mean, r_umidity_st_24h_std, r_umidity_st_24h_rms, r_umidity_st_24h_median, r_umidity_st_24h_min, r_umidity_st_24h_max, r_umidity_st_24h_diff_last, r_umidity_st_24h_slope

feats_to_balance :

feats_grouping :

feats_postproc : datetime, id_channel, id_rabbit, campaign

feats_ref :

id_cols : campaign

2 Data correlation and clustering

2.1 Clustermap and cluster list

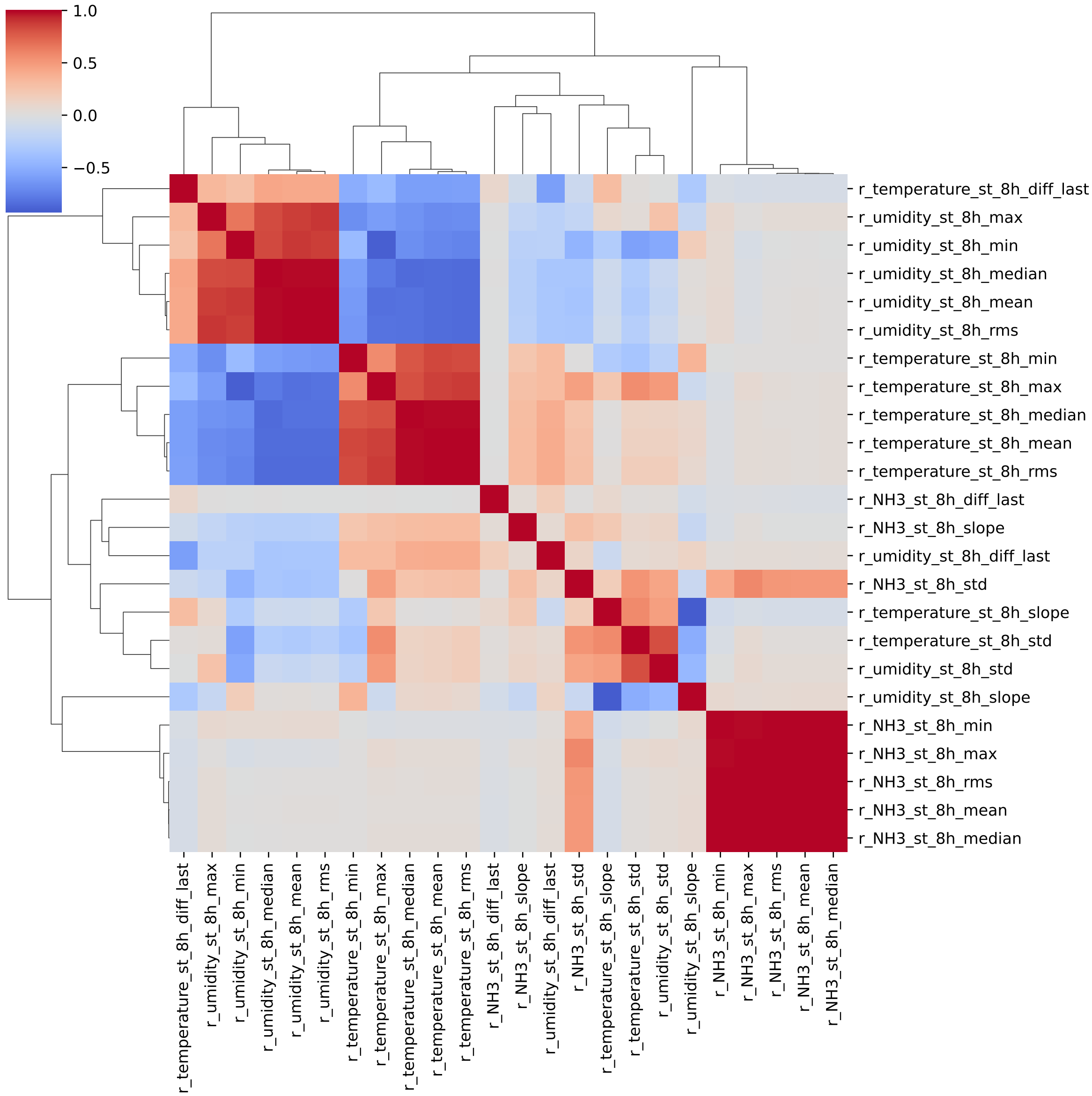
The following section has been done with a threshold choosen of : 0.8

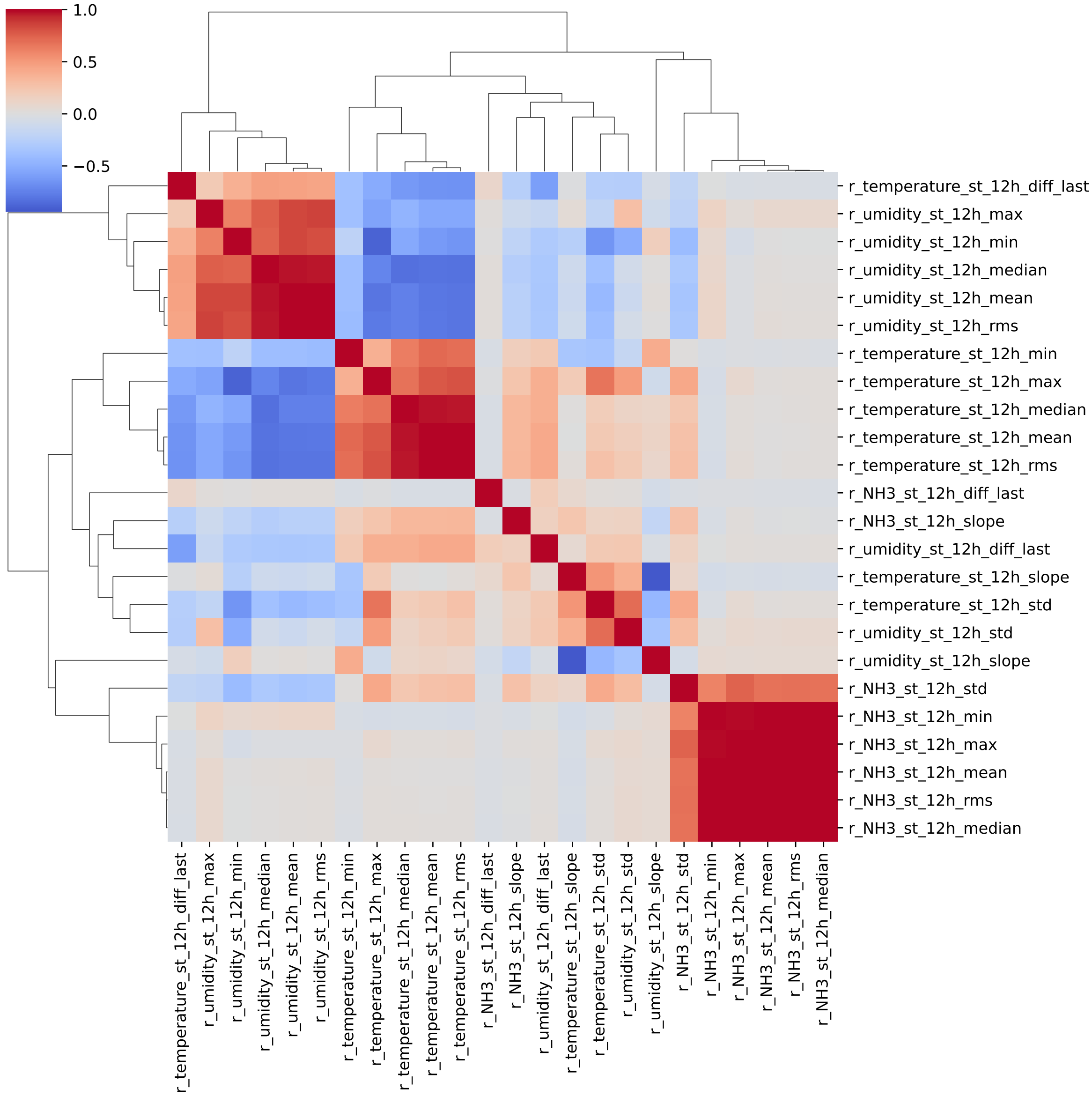
cluster 1: r, *, N, H, 3 ;* ***cluster 2:*** *r,* , t, e, m, p, e, r, a, t, u, r, e ; **cluster 3:** r, *, u, m, i, d, i, t, y ;* ***cluster 4:*** *r,* , C, O, 2 ; **cluster 5:** r, _, T, H, I ; **cluster 6:** r_NH3_st_8h_rms, r_NH3_st_8h_mean, r_NH3_st_8h_max, r_NH3_st_8h_median, r_NH3_st_8h_min ; **cluster 7:** r_NH3_st_8h_std ; **cluster 8:** r_NH3_st_8h_diff_last ; **cluster 9:** r_NH3_st_8h_slope ; **cluster 10:** r_umidity_st_8h_mean, r_umidity_st_8h_max, r_umidity_st_8h_min, r_temperature_st_8h_mean, r_temperature_st_8h_min, r_temperature_st_8h_max, r_temperature_st_8h_median, r_umidity_st_8h_rms, r_temperature_st_8h_rms, r_umidity_st_8h_median ; **cluster 11:** r_temperature_st_8h_std, r_umidity_st_8h_std ; **cluster 12:** r_temperature_st_8h_diff_last ; **cluster 13:** r_umidity_st_8h_slope, r_temperature_st_8h_slope ; **cluster 14:** r_umidity_st_8h_diff_last ; **cluster 15:** r_NH3_st_12h_max, r_NH3_st_12h_median, r_NH3_st_12h_rms, r_NH3_st_12h_min, r_NH3_st_12h_mean ; **cluster 16:** r_NH3_st_12h_std ; **cluster 17:** r_NH3_st_12h_diff_last ; **cluster 18:** r_NH3_st_12h_slope ; **cluster 19:** r_temperature_st_12h_median, r_temperature_st_12h_rms, r_umidity_st_12h_mean, r_umidity_st_12h_rms, r_temperature_st_12h_mean, r_umidity_st_12h_median, r_umidity_st_12h_max, r_temperature_st_12h_max, r_umidity_st_12h_min ; **cluster 20:** r_temperature_st_12h_std ; **cluster 21:** r_temperature_st_12h_min ; **cluster 22:** r_temperature_st_12h_diff_last ; **cluster 23:** r_temperature_st_12h_slope, r_umidity_st_12h_slope ; **cluster 24:** r_umidity_st_12h_std ; **cluster 25:** r_umidity_st_12h_diff_last ; **cluster 26:** r_NH3_st_24h_median, r_NH3_st_24h_max, r_NH3_st_24h_rms, r_NH3_st_24h_min, r_NH3_st_24h_std, r_NH3_st_24h_mean ; **cluster 27:** r_NH3_st_24h_diff_last ; **cluster 28:** r_NH3_st_24h_slope ; **cluster 29:** r_umidity_st_24h_rms, r_umidity_st_24h_median, r_temperature_st_24h_rms, r_umidity_st_24h_max, r_temperature_st_24h_mean, r_temperature_st_24h_max, r_temperature_st_24h_median, r_umidity_st_24h_min, r_temperature_st_24h_std, r_umidity_st_24h_mean ; **cluster 30:** r_temperature_st_24h_min ; **cluster 31:** r_temperature_st_24h_diff_last ; **cluster 32:** r_umidity_st_24h_slope, r_temperature_st_24h_slope ; **cluster 33:** r_umidity_st_24h_std ; **cluster 34:** r_umidity_st_24h_diff_last ;

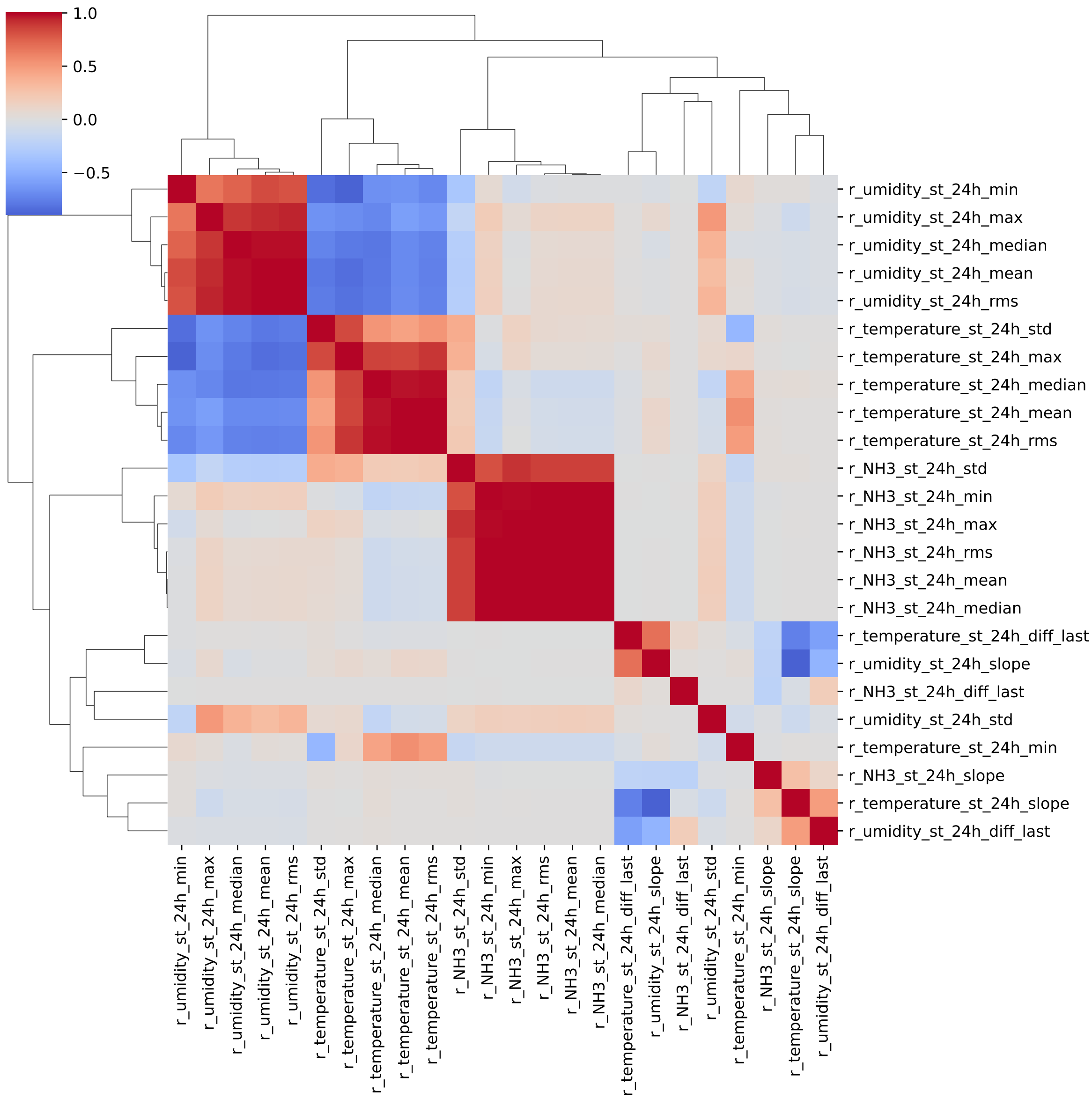
2.2 Features selection

Only the first items of the clusters is choosen for the rest of the study

Choosen features_X: r_NH3, r_temperature, r_umidity, r_CO2, r_THI, r_NH3_st_8h_rms, r_NH3_st_8h_std, r_NH3_st_8h_diff_last, r_NH3_st_8h_slope, r_umidity_st_8h_mean, r__temperature__st__8h__std, r__temperature__st__8h__diff__last, r__umidity__st__8h__slope, r__umidity__st__8h__diff__last, r__NH3__st__12h__max, r__NH3__st__12h__std, r__NH3__st__12h__diff__last, r__NH3__st__12h__slope, r__temperature__st__12h__median, r__temperature__st__12h__std, r__temperature__st__12h__min, r__temperature__st__12h__diff__last, r__temperature__st__12h__slope, r__umidity__st__12h__std, r__umidity__st__12h__diff__last, r__NH3__st__24h__median, r__NH3__st__24h__diff__last, r__NH3__st__24h__slope, r__umidity__st__24h__rms, r__temperature__st__24h__min, r__temperature__st__24h__diff__last, r__umidity__st__24h__slope, r__umidity__st__24h__std, r__umidity__st__24h__diff__last




1.0
0.5
0.0
−0.5
r_temperature_st_12h_diff_last
r_umidity_st_12h_max
r_umidity_st_12h_min
r_umidity_st_12h_median
r_umidity_st_12h_mean
r_umidity_st_12h_rms
r_temperature_st_12h_min
r_temperature_st_12h_max
r_temperature_st_12h_median
r_temperature_st_12h_mean
r_temperature_st_12h_rms
r_NH3_st_12h_diff_last
r_NH3_st_12h_slope
r_umidity_st_12h_diff_last
r_temperature_st_12h_slope
r_temperature_st_12h_std
r_umidity_st_12h_std
r_umidity_st_12h_slope
r_NH3_st_12h_std
r_NH3_st_12h_min
r_NH3_st_12h_max
r_NH3_st_12h_mean
r_NH3_st_12h_rms
r_NH3_st_12h_median



3 Model(s) analysis

3.1 Metadata

Features_X: r_NH3, r_temperature, r_umidity, r_CO2, r_THI, r_NH3_st_8h_rms, r_NH3_st_8h_std, r_NH3_st_8h_diff_last, r_NH3_st_8h_slope, r_umidity_st_8h_mean, r_temperature_st_8h_std, r_temperature_st_8h_diff_last, r_umidity_st_8h_slope, r_umidity_st_8h_diff_last, r_NH3_st_12h_max, r_NH3_st_12h_std, r_NH3_st_12h_diff_last, r_NH3_st_12h_slope, r_temperature_st_12h_median, r_temperature_st_12h_std, r_temperature_st_12h_min, r_temperature_st_12h_diff_last, r_temperature_st_12h_slope, r_umidity_st_12h_std, r_umidity_st_12h_diff_last, r_NH3_st_24h_median, r_NH3_st_24h_diff_last, r_NH3_st_24h_slope, r_umidity_st_24h_rms, r_temperature_st_24h_min, r_temperature_st_24h_diff_last, r_umidity_st_24h_slope, r_umidity_st_24h_std, r_umidity_st_24h_diff_last

Len(Features_X): 34

Config Gridsearch:{'show': True, 'search_type': 'optuna', 'model_name': None, 'model': None, 'param_grid': None, 'best_param': None, 'cv_func': ShuffleSplit(n_splits=5, random_state=42, test_size=0.3, train_size=None), 'save_models': True, 'sumup-table': True}

3.2 Explanation legend tables

Columns

- rank: ranking regarding cross validation test
- train_cv: mean score train of the CV
- test_cv: mean score test of the CV
- test: score on the testing dataset
- diff: test-test_cv
- r_test: test/test_cv
- r_train: train_cv/test_cv

Colors:

[Orange] r_test < 0.95 (under performance of the test)

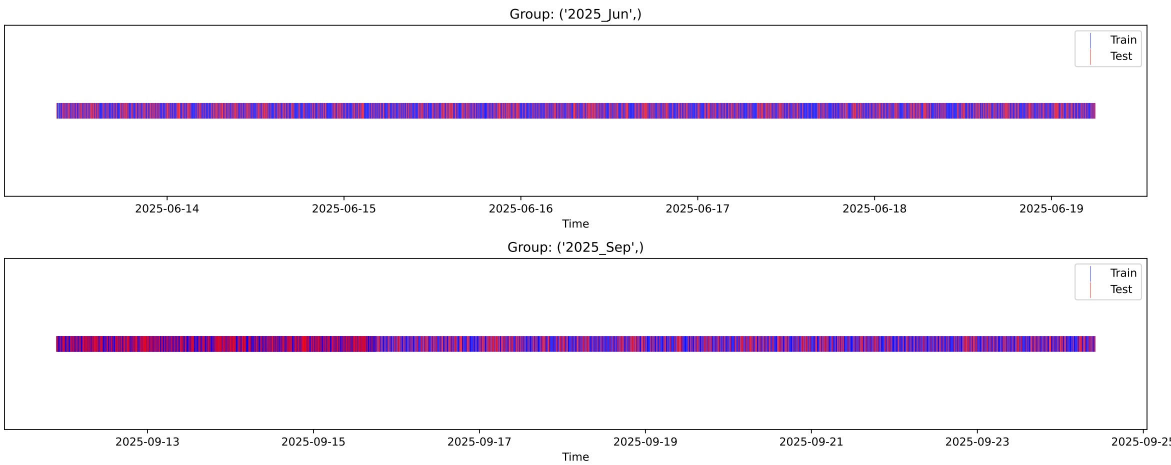
[Green] r_test > 1.05 (over performance of the test)

[Red] r_train not in [0.95, 1.05] (gap between train/test)

3.3 Repartition of train and test

3.3.1 External Split: X_train vs X_test

External Split: Train: 5096 samples Test: 2184 samples



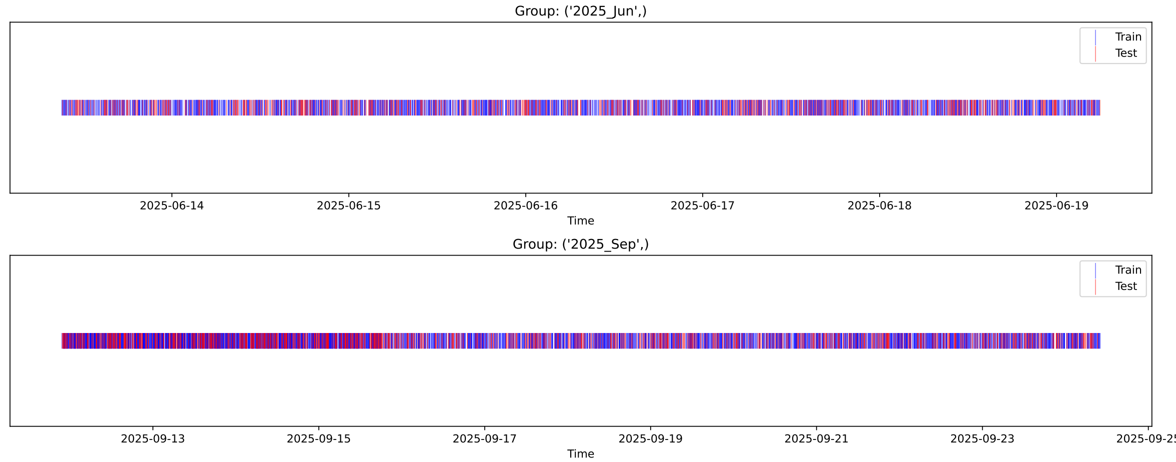
3.3.2 Internal CV Folds

Fold 1/5

Train: 3567 samples Test: 1529 samples Excluded: 0 samples Window exclusion: 0h

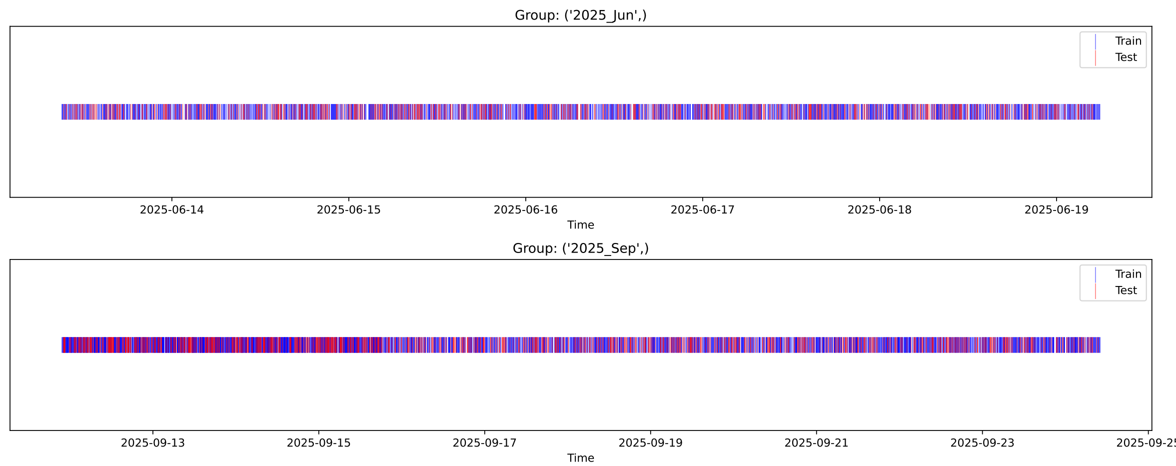
Fold 2/5

Train: 3567 samples Test: 1529 samples Excluded: 0 samples Window exclusion: 0h



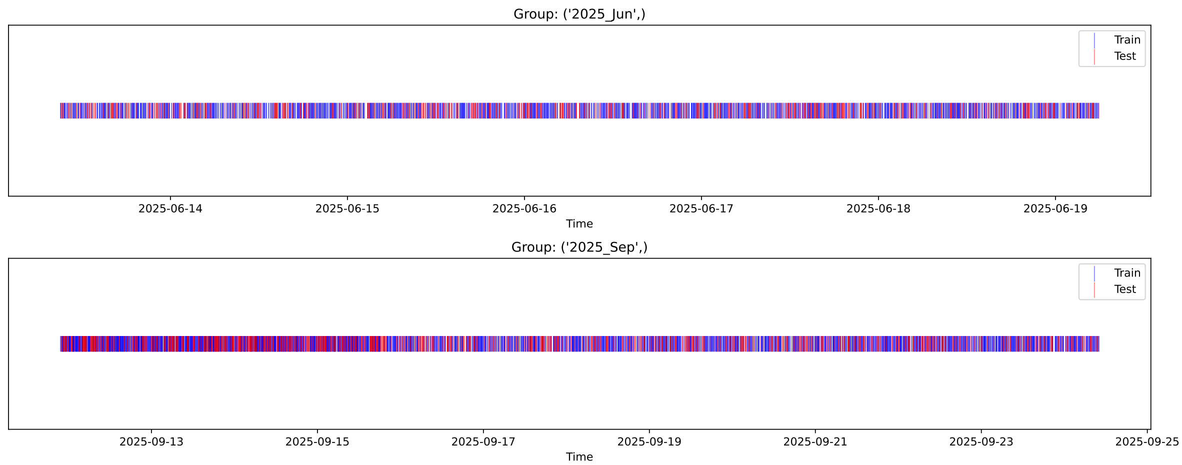
Fold 3/5

Train: 3567 samples Test: 1529 samples Excluded: 0 samples Window exclusion: 0h



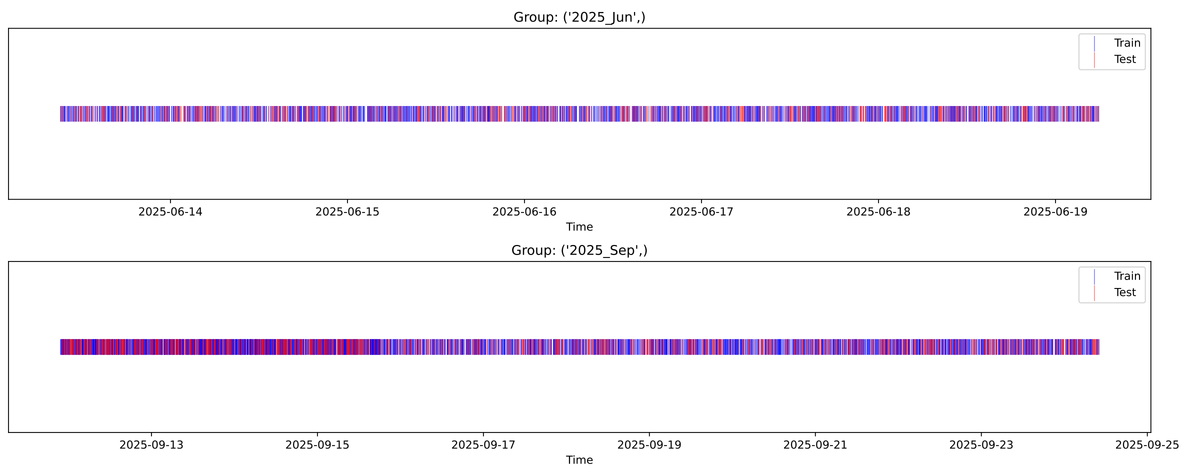
Fold 4/5

Train: 3567 samples Test: 1529 samples Excluded: 0 samples Window exclusion: 0h



Fold 5/5

Train: 3567 samples Test: 1529 samples Excluded: 0 samples Window exclusion: 0h



3.4 Model : Linear

3.4.1 Gridsearch

Distribution y_train vs y_test: y_train: mean=2.362, std=0.816, min=0.959, max=13.845 y_test: mean=2.359, std=0.781, min=1.062, max=7.822

Fold 0 — y_test: mean=2.36, std=0.87, y_train: mean=2.36, std=0.79

Fold 1 — y_test: mean=2.38, std=0.84, y_train: mean=2.36, std=0.81

Fold 2 — y_test: mean=2.36, std=0.83, y_train: mean=2.36, std=0.81

Fold 3 — y_test: mean=2.36, std=0.81, y_train: mean=2.36, std=0.82

Fold 4 — y_test: mean=2.39, std=0.85, y_train: mean=2.35, std=0.80

train scores CV: [0.6263669 0.61843533 0.62047629 0.5833112 0.61719278]

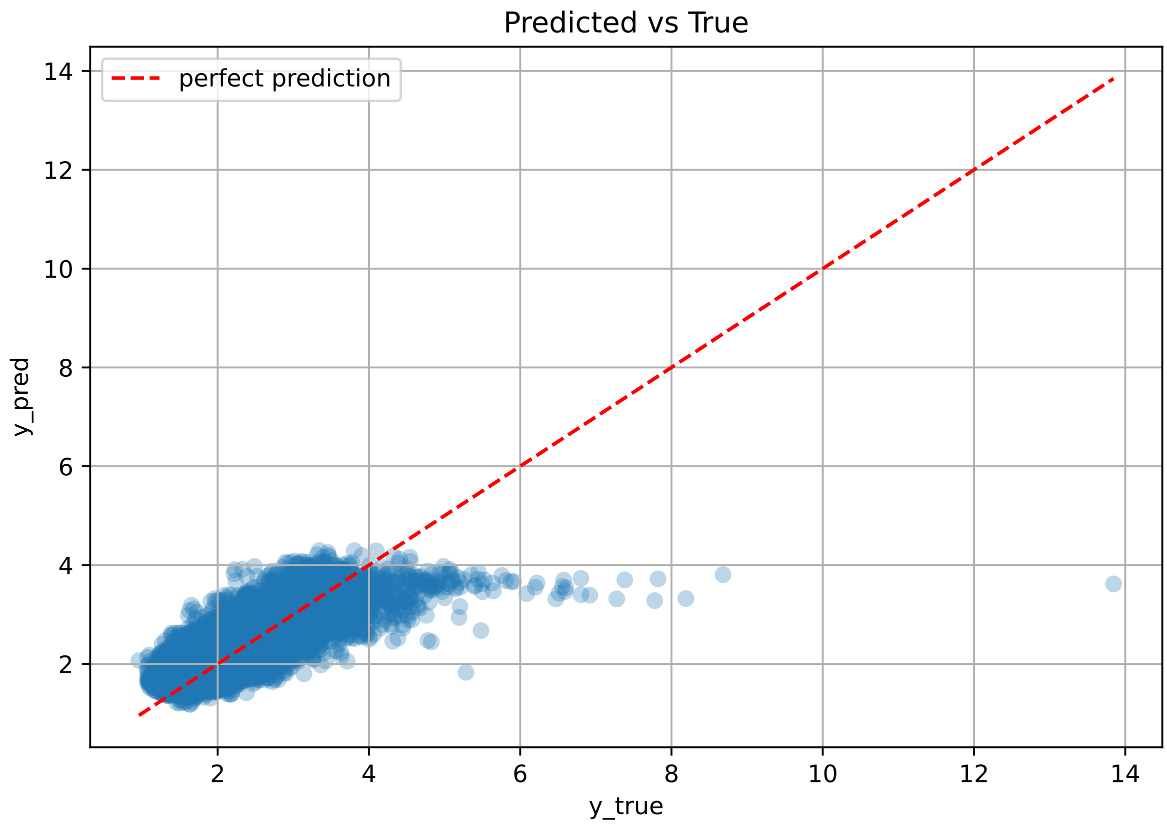
test scores CV: [0.52935201 0.5415593 0.53387921 0.61990031 0.54589262]

rank	mean_train_cv	mean_test_cv	std_test_cv	test_score_cv	test	diff	r_test	r_train
1.0	0.6132	0.5541	0.0334	0.6152	0.6164	0.0623	1.1123	1.1065

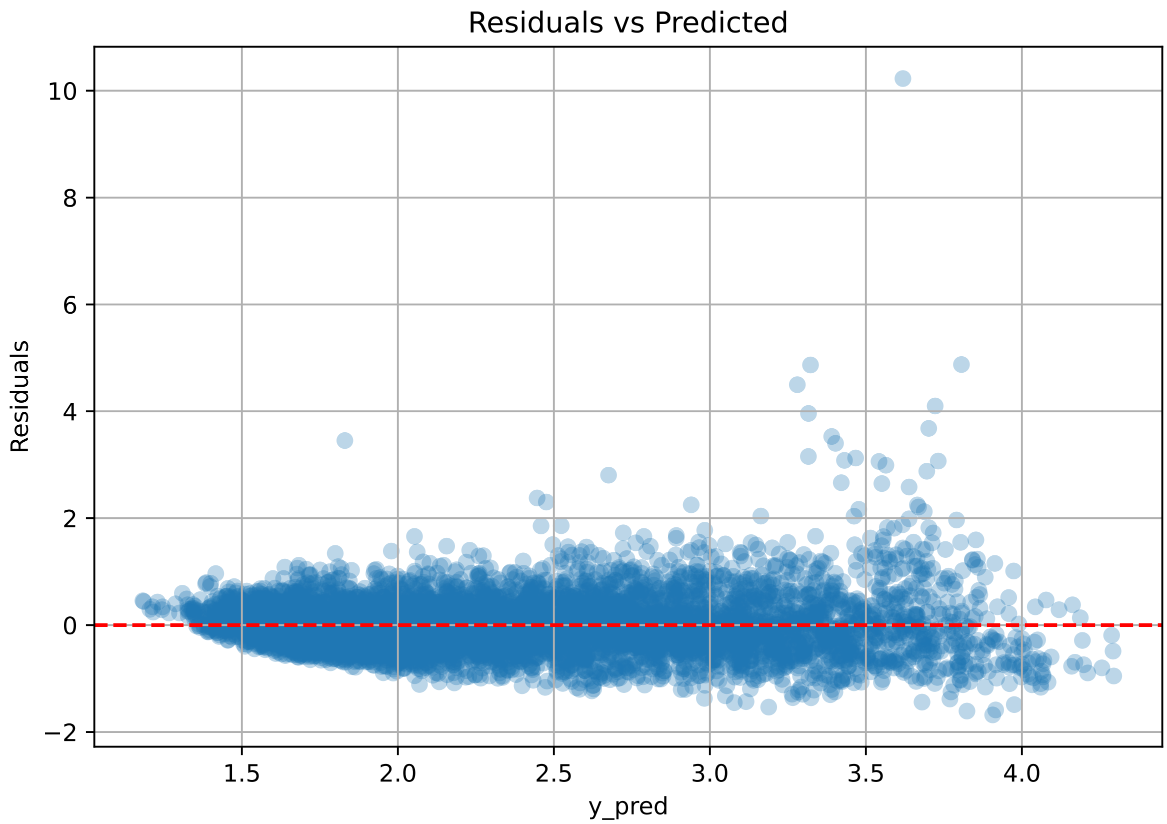
Coefficients:

coef_r__CO2 = -0.0329, coef_r__NH3 = 0.0126, coef_r__NH3__st__12h__diff__last = 0.0065, coef_r__NH3__st__12h__max = 0.8176, coef_r__NH3__st__12h__slope = 0.0121, coef_r__NH3__st__12h__std = -0.0488, coef_r__NH3__st__24h__diff__last = 0.0065, coef_r__NH3__st__24h__median = -0.6555, coef_r__NH3__st__24h__slope = -0.0165, coef_r__NH3__st__8h__diff__last = 0.0065, coef_r__NH3__st__8h__rms = -0.1808, coef_r__NH3__st__8h__slope = 0.0352, coef_r__NH3__st__8h__std = -0.0226, coef_r__THI = 1.0141, coef_r__temperature = -0.4719, coef_r__temperature__st__12h__diff__last = 0.0022, coef_r__temperature__st__12h__median = -0.1058, coef_r__temperature__st__12h__min = -0.1120, coef_r__temperature__st__12h__slope = -0.3408, coef_r__temperature__st__12h__std = -0.0555, coef_r__temperature__st__24h__diff__last = 0.0022, coef_r__temperature__st__24h__min = -0.0695, coef_r__temperature__st__8h__diff__last = 0.0022, coef_r__temperature__st__8h__std = -0.0476, coef_r__umidity = -0.7675, coef_r__umidity__st__12h__diff__last = -0.0127, coef_r__umidity__st__12h__std = -0.1687, coef_r__umidity__st__24h__diff__last = -0.0127, coef_r__umidity__st__24h__rms = -0.2013, coef_r__umidity__st__24h__slope = -0.1928, coef_r__umidity__st__24h__std = 0.0667, coef_r__umidity__st__8h__diff__last = -0.0127, coef_r__umidity__st__8h__mean = 0.5096, coef_r__umidity__st__8h__slope = 0.1847, intercept = 2.3622,

3.4.2 Scatter


Predicted vs True
perfect prediction
y_pred
y_true
14
12
10
8
6
4
2
2
4
6
8
10
12
14

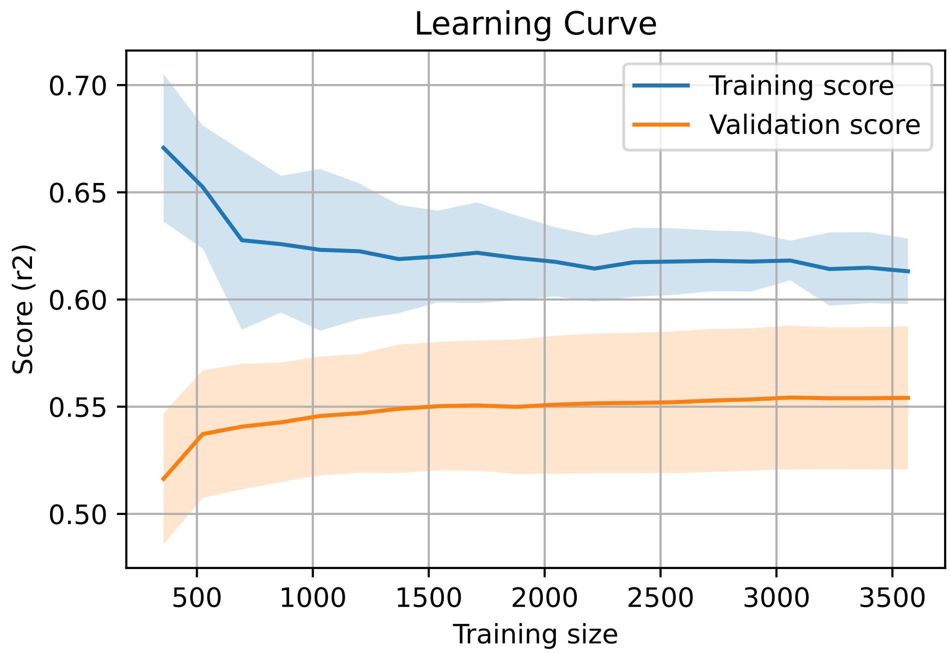
3.4.3 Scatter residuals



3.4.4 Learning curve

Len(group) : 7280, **Len(training)** : 5096,

Len(training_real) : 3568, **Len(test_of_training)** : 1528, **Len(test)** : 2184

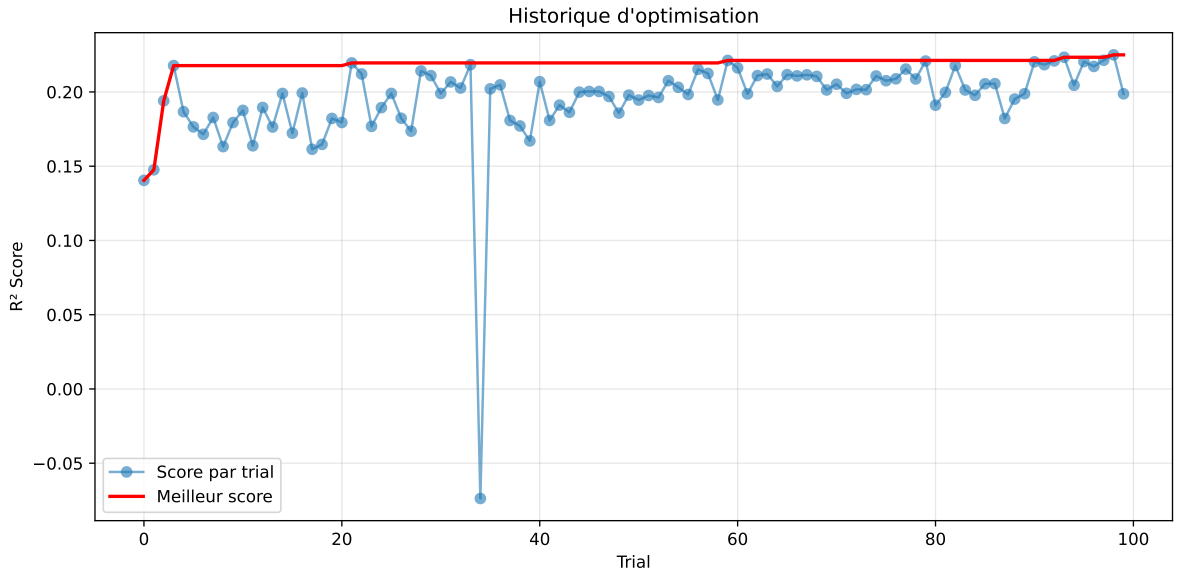


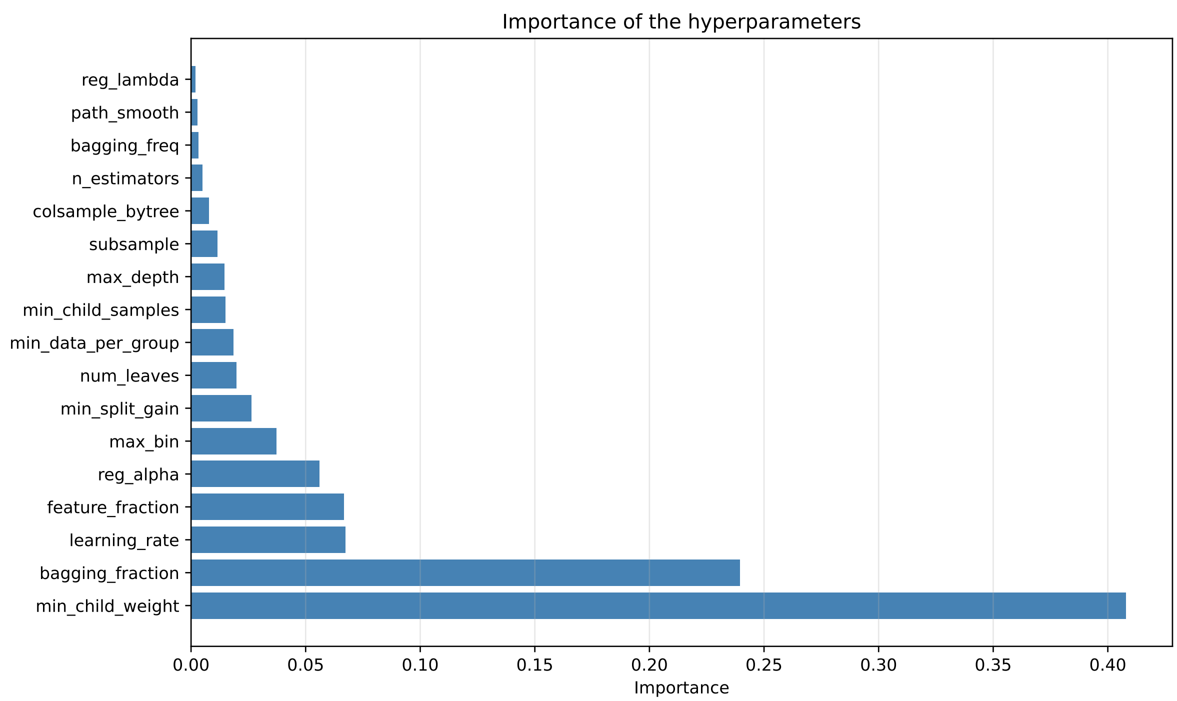
3.5 Model : LGBM

3.5.1 Gridsearch

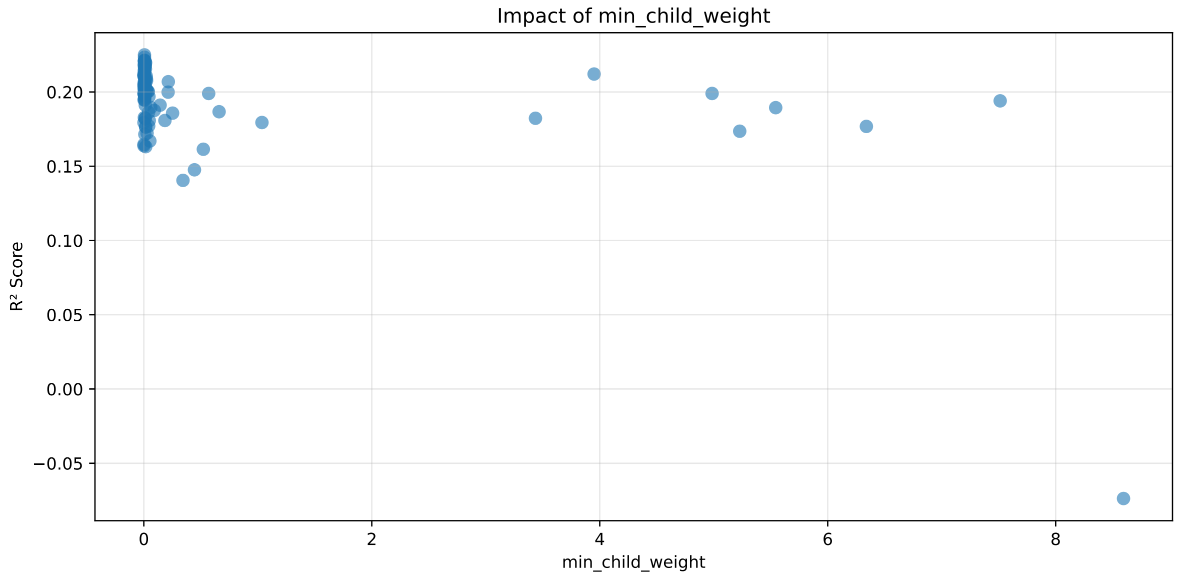
Distribution y_train vs y_test: y_train: mean=2.362, std=0.816, min=0.959, max=13.845 y_test: mean=2.359, std=0.781, min=1.062, max=7.822

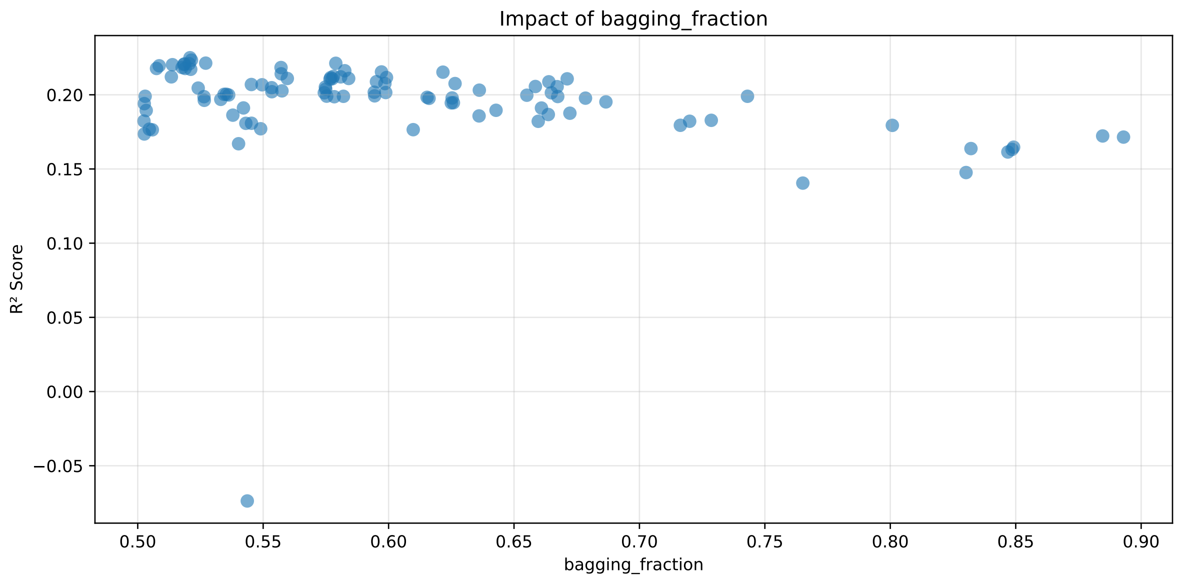
==================== TOP Importance FEATURES ==================== feature importance r_umidity 214 r_temperature_st_24h_min 176 r_NH3_st_12h_max 164 r_NH3_st_24h_median 145 r_temperature 140 r_CO2 139 r_temperature_st_12h_median 127 r_temperature_st_8h_diff_last 112 r_umidity_st_12h_std 92 r_THI 63 r_temperature_st_12h_slope 63 r_umidity_st_8h_diff_last 62 r_NH3_st_8h_slope 60 r_temperature_st_12h_diff_last 57 r_umidity_st_24h_rms 55 r_NH3 50 r_temperature_st_12h_std 47 r_umidity_st_8h_slope 40 r_NH3_st_8h_rms 40 r_umidity_st_24h_slope 37 r_temperature_st_24h_diff_last 31 r_NH3_st_8h_std 31 r_NH3_st_12h_slope 28 r_umidity_st_12h_diff_last 25 r_umidity_st_8h_mean 23 r_NH3_st_8h_diff_last 23 r_umidity_st_24h_std 17 r_NH3_st_12h_diff_last 14 r_umidity_st_24h_diff_last 14 r_NH3_st_24h_diff_last 13 r_temperature_st_8h_std 13 r_NH3_st_12h_std 8 r_temperature_st_12h_min 7 r_NH3_st_24h_slope 2

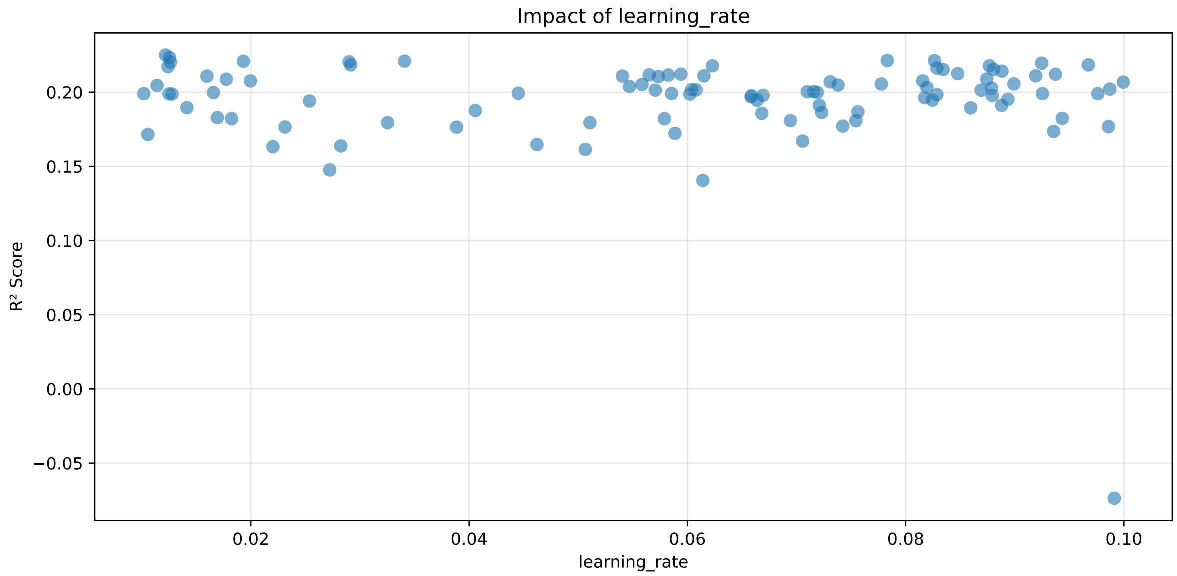


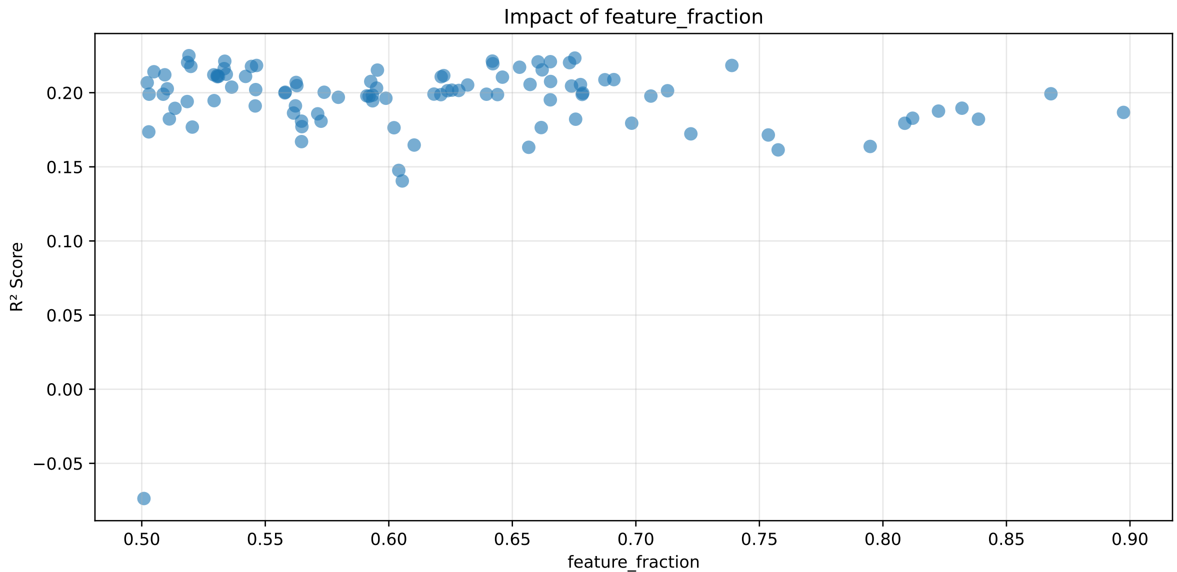


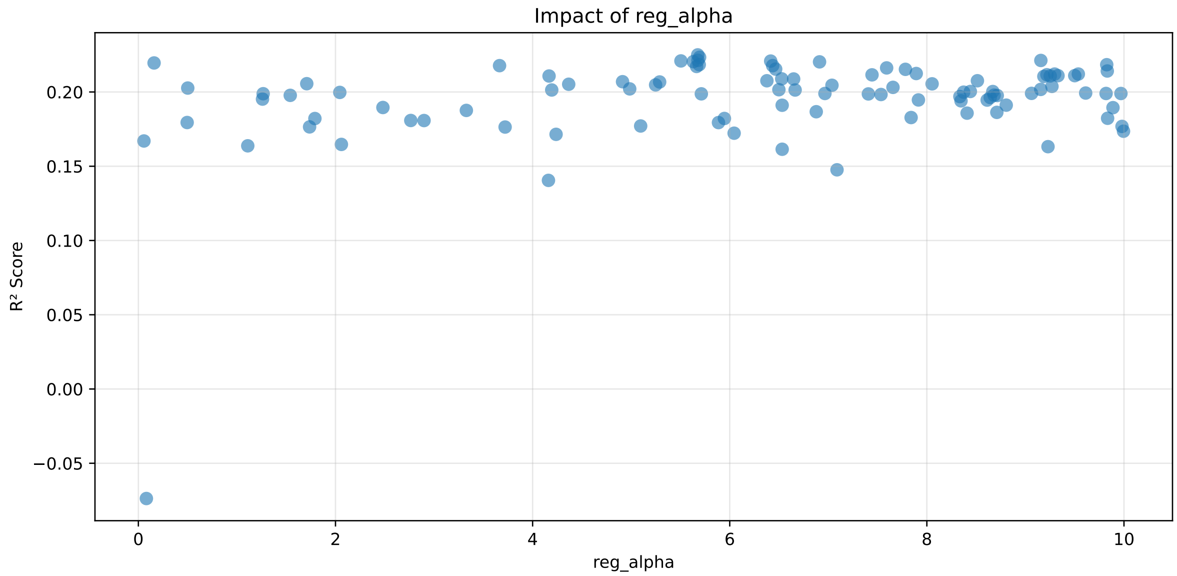
Paramètres avec >5% d'importance : ['min_child_weight', 'bagging_fraction', 'learning_rate', 'feature_fraction', 'reg_alpha']


Impact of min_child_weight
R² Score
0.20
0.15
0.10
0.05
0.00
−0.05
0
2
4
6
8
min_child_weight


Impact of bagging_fraction
R² Score
0.20
0.15
0.10
0.05
0.00
−0.05
0.50
0.55
0.60
0.65
0.70
0.75
0.80
0.85
0.90
bagging_fraction


Impact of learning_rate
R² Score
0.20
0.15
0.10
0.05
0.00
−0.05
0.02
0.04
0.06
0.08
0.10
learning_rate


Impact of feature_fraction
R² Score
0.20
0.15
0.10
0.05
0.00
−0.05
0.50
0.55
0.60
0.65
0.70
0.75
0.80
0.85
0.90
feature_fraction



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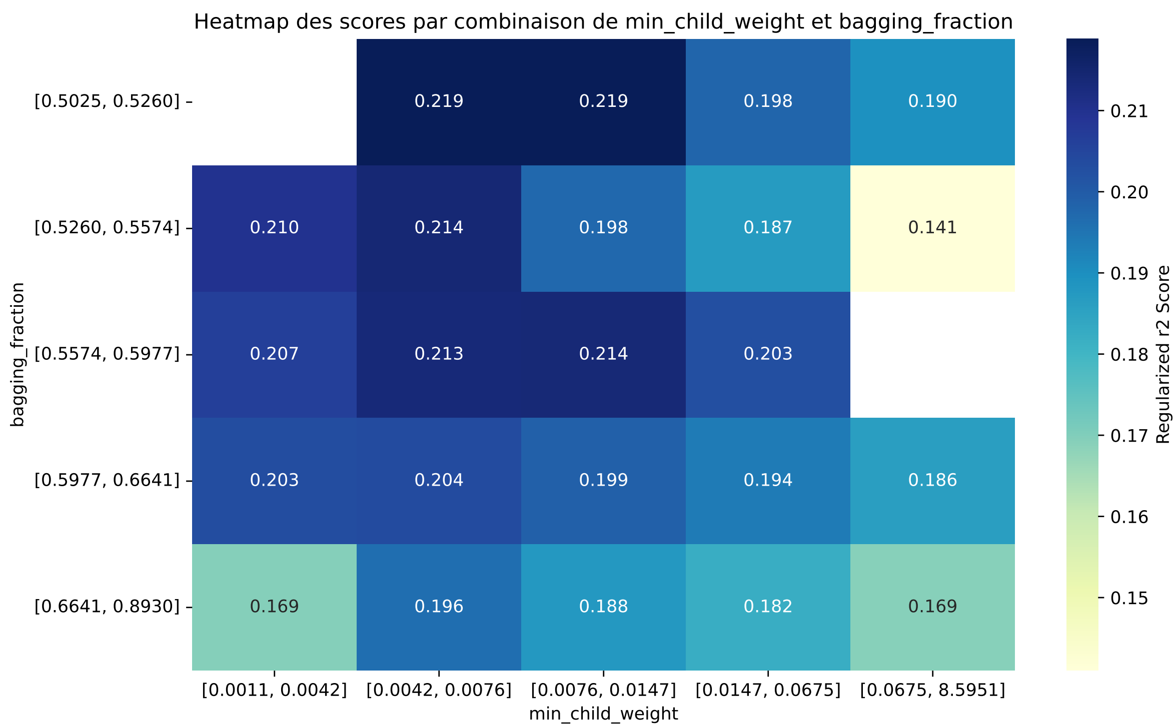
The default value of observed=False is deprecated and will change to observed=True in a futu



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.6648	0.5915	0.6662	0.0292	0.225	0.692	0.1005	1.1699	1.1239
2	0.6707	0.5962	0.671	0.0295	0.2234	0.6961	0.0999	1.1676	1.1251
3	0.6909	0.6126	0.6862	0.0294	0.2213	0.7116	0.099	1.1616	1.1277
4	0.6702	0.5954	0.6704	0.0247	0.2212	0.6963	0.1009	1.1695	1.1257
5	0.6443	0.5738	0.6483	0.0286	0.2209	0.678	0.1042	1.1816	1.123
6	0.6595	0.5864	0.661	0.0255	0.2208	0.695	0.1087	1.1853	1.1247
7	0.6783	0.602	0.6775	0.0289	0.2204	0.7007	0.0987	1.1639	1.1268
8	0.6759	0.6	0.6745	0.0299	0.2203	0.7012	0.1013	1.1688	1.1266
9	0.7246	0.6404	0.7139	0.0285	0.2195	0.7334	0.093	1.1452	1.1314
10	0.6476	0.5761	0.6513	0.0314	0.2184	0.6799	0.1038	1.1802	1.1242
---	---	---	---	---	---	---	---	---	---
100	0.9085	0.7448	0.8216	0.0327	-0.0737	0.8333	0.0885	1.1188	1.2198

Hyperparameters:

Rank 1: {’reg___n_estimators’: 800, ’reg___max_depth’: 3, ’reg___learning_rate’: 0.01218127913000503, ’reg___num_leaves’: 37, ’reg___subsample’: 0.6525568679502188, ’reg___colsample_bytree’: 0.8788492486821922, ’reg___min_child_samples’: 39, ’reg___reg_alpha’: 5.67531654975634, ’reg___reg_lambda’: 49.85139340136498, ’reg___min_split_gain’: 3.348333797245976, ’reg___path_smooth’: 0.2713548155978176, ’reg___min_child_weight’: 0.005747188405893943, ’reg___feature_fraction’: 0.5190936336037869, ’reg___bagging_fraction’: 0.520840050700289, ’reg___bagging_freq’: 1, ’reg___max_bin’: 77, ’reg___min_data_per_group’: 55}

Rank 2: {’reg___n_estimators’: 800, ’reg___max_depth’: 3, ’reg___learning_rate’: 0.012551007964225118, ’reg___num_leaves’: 48, ’reg___subsample’: 0.6440354661456525, ’reg___colsample_bytree’: 0.8821263376084472, ’reg___min_child_samples’: 39, ’reg___reg_alpha’: 5.694354561129914, ’reg___reg_lambda’: 41.73483698298009, ’reg___min_split_gain’: 3.3199244044807794, ’reg___path_smooth’: 1.037084768991396, ’reg___min_child_weight’: 0.005949432066706034, ’reg___feature_fraction’: 0.6753017199368421, ’reg___bagging_fraction’: 0.5214072721639136, ’reg___bagging_freq’: 1, ’reg___max_bin’: 74, ’reg___min_data_per_group’: 195}

Rank 3: {’reg___n_estimators’: 800, ’reg___max_depth’: 7, ’reg___learning_rate’: 0.07832236794717945, ’reg___num_leaves’: 17, ’reg___subsample’: 0.6505592316210604, ’reg___colsample_bytree’: 0.8233948104016543, ’reg___min_child_samples’: 40, ’reg___reg_alpha’: 5.677120371920951, ’reg___reg_lambda’: 49.629393554926715, ’reg___min_split_gain’: 3.1325189250110155, ’reg___path_smooth’: 1.6863586830538926, ’reg___min_child_weight’: 0.005757876674520515, ’reg___feature_fraction’: 0.6419198105594301, ’reg___bagging_fraction’: 0.5270964057588011, ’reg___bagging_freq’: 1, ’reg___max_bin’: 77, ’reg___min_data_per_group’: 182}

Rank 4: {’reg___n_estimators’: 800, ’reg___max_depth’: 3, ’reg___learning_rate’: 0.08263673685156896, ’reg___num_leaves’: 15, ’reg___subsample’: 0.7474765161566356, ’reg___colsample_bytree’: 0.681081083954438, ’reg___min_child_samples’: 47, ’reg___reg_alpha’: 9.156633277975077, ’reg___reg_lambda’: 35.95844675272035, ’reg___min_split_gain’: 2.4662723489366565, ’reg___path_smooth’: 0.3719286796695438, ’reg___min_child_weight’: 0.007347514388741595, ’reg___feature_fraction’: 0.5336035501393481, ’reg___bagging_fraction’: 0.5789576944192628, ’reg___bagging_freq’: 2, ’reg___max_bin’: 82, ’reg___min_data_per_group’: 191}

Rank 5: {’reg___n_estimators’: 700, ’reg___max_depth’: 3, ’reg___learning_rate’: 0.03408228517733389, ’reg___num_leaves’: 17, ’reg___subsample’: 0.6603236463784025, ’reg___colsample_bytree’: 0.8662804685829802, ’reg___min_child_samples’: 43, ’reg___reg_alpha’: 5.504902336117944, ’reg___reg_lambda’: 49.685081465421796, ’reg___min_split_gain’: 3.2459774942899617, ’reg___path_smooth’: 0.6129846160209685, ’reg___min_child_weight’: 0.005415660786474711, ’reg___feature_fraction’: 0.665401849607051, ’reg___bagging_fraction’: 0.5205314604247163, ’reg___bagging_freq’: 6, ’reg___max_bin’: 74, ’reg___min_data_per_group’: 195}

Rank 6: {’reg___n_estimators’: 700, ’reg___max_depth’: 3, ’reg___learning_rate’: 0.019324360542728358, ’reg___num_leaves’: 20, ’reg___subsample’: 0.6694119906009437, ’reg___colsample_bytree’: 0.7288513835054868, ’reg___min_child_samples’: 44, ’reg___reg_alpha’: 6.415481057271482, ’reg___reg_lambda’: 42.14094241357398, ’reg___min_split_gain’: 2.722181979442417, ’reg___path_smooth’: 0.6245293674740554, ’reg___min_child_weight’: 0.0057433806328736, ’reg___feature_fraction’: 0.6604201244089304, ’reg___bagging_fraction’: 0.5186720299132749, ’reg___bagging_freq’: 2, ’reg___max_bin’: 70, ’reg___min_data_per_group’: 192}

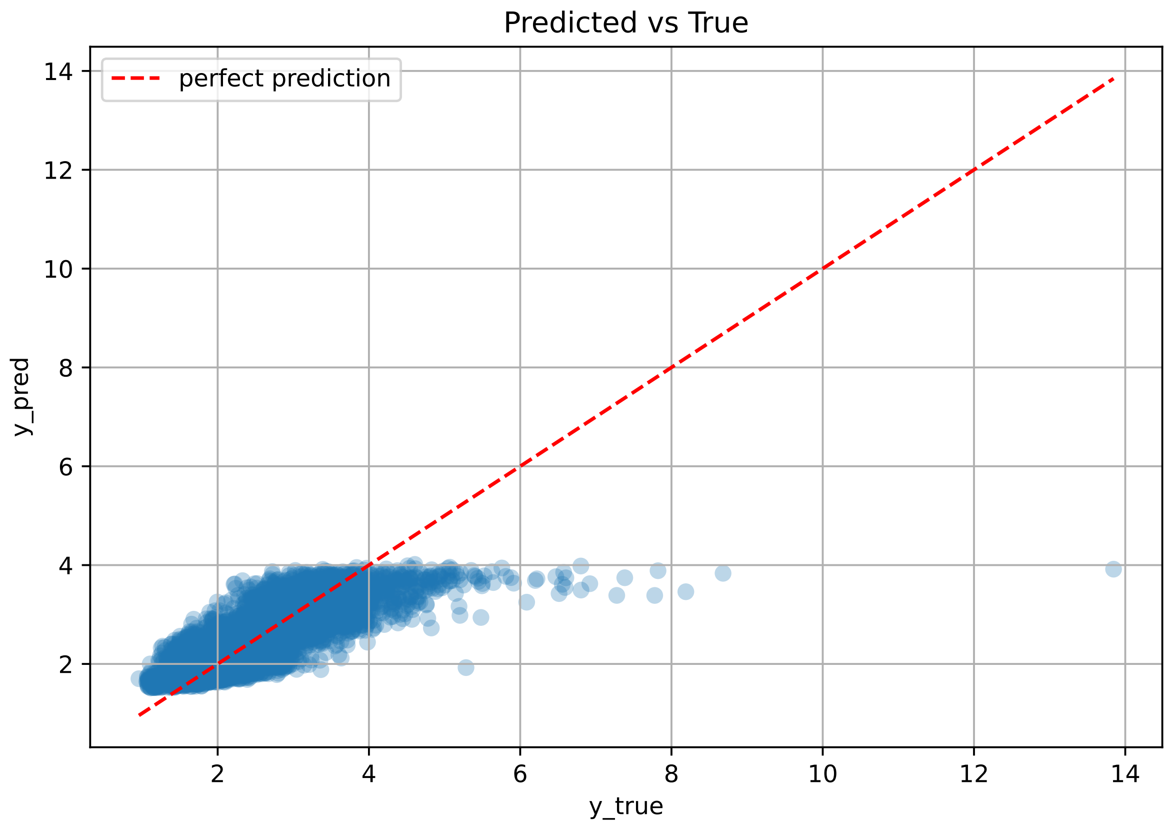
Rank 7: {’reg___n_estimators’: 800, ’reg___max_depth’: 3, ’reg___learning_rate’: 0.029005695669437004, ’reg___num_leaves’: 17, ’reg___subsample’: 0.6500686918294422, ’reg___colsample_bytree’: 0.8765205849426196, ’reg___min_child_samples’: 43, ’reg___reg_alpha’: 5.630233917306312, ’reg___reg_lambda’: 49.859116910586934, ’reg___min_split_gain’: 3.1259913970745528, ’reg___path_smooth’: 1.6701548451059298, ’reg___min_child_weight’: 0.005267344948904633, ’reg___feature_fraction’: 0.5185664819930924, ’reg___bagging_fraction’: 0.5183486460531485, ’reg___bagging_freq’: 1, ’reg___max_bin’: 75, ’reg___min_data_per_group’: 182}

Rank 8: {’reg___n_estimators’: 700, ’reg___max_depth’: 3, ’reg___learning_rate’: 0.012640739138160084, ’reg___num_leaves’: 49, ’reg___subsample’: 0.6521824166930028, ’reg___colsample_bytree’: 0.8438395369681695, ’reg___min_child_samples’: 43, ’reg___reg_alpha’: 6.911185153209188, ’reg___reg_lambda’: 49.60103990512455, ’reg___min_split_gain’: 1.9384210818187166, ’reg___path_smooth’: 0.5826754688148646, ’reg___min_child_weight’: 0.013818815361100136, ’reg___feature_fraction’: 0.6731512480896573, ’reg___bagging_fraction’: 0.5138437024995817, ’reg___bagging_freq’: 6, ’reg___max_bin’: 73, ’reg___min_data_per_group’: 195}

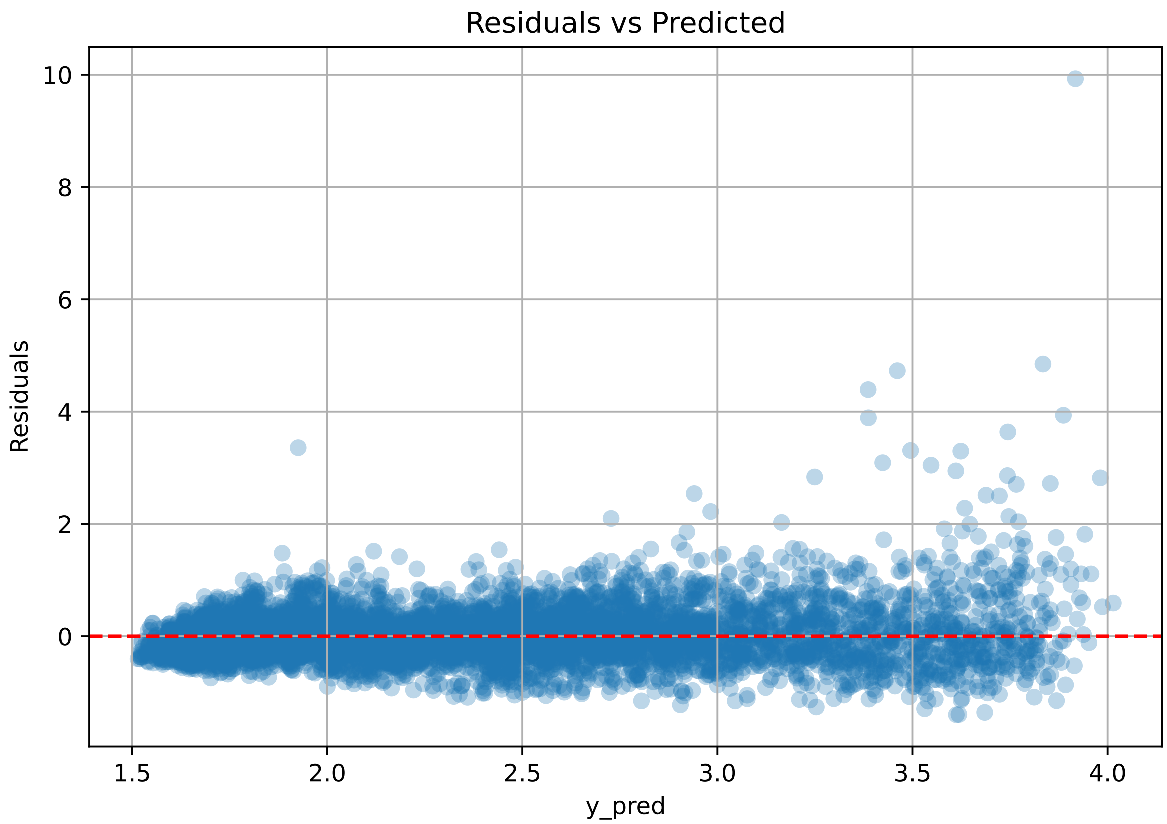
Rank 9: {'reg__n_estimators': 600, 'reg__max_depth': 3, 'reg__learning_rate': 0.09246626803369758, 'reg__num_leaves': 16, 'reg__subsample': 0.7748619128407985, 'reg__colsample_bytree': 0.7294992598596568, 'reg__min_child_samples': 50, 'reg__reg_alpha': 0.15939755315039816, 'reg__reg_lambda': 30.273784180011088, 'reg__min_split_gain': 1.5041511811876878, 'reg__path_smooth': 2.767237403044698, 'reg__min_child_weight': 0.015342609915388425, 'reg__feature_fraction': 0.642113302174314, 'reg__bagging_fraction': 0.5085577383208572, 'reg__bagging_freq': 4, 'reg__max_bin': 76, 'reg__min_data_per_group': 198}

Rank 10: {'reg__n_estimators': 700, 'reg__max_depth': 3, 'reg__learning_rate': 0.029149412633666855, 'reg__num_leaves': 48, 'reg__subsample': 0.6481322037568842, 'reg__colsample_bytree': 0.8722066966779304, 'reg__min_child_samples': 43, 'reg__reg_alpha': 5.690773529376283, 'reg__reg_lambda': 48.661781193049606, 'reg__min_split_gain': 3.3676594486866955, 'reg__path_smooth': 1.6843626549065642, 'reg__min_child_weight': 0.005533677032682071, 'reg__feature_fraction': 0.738859333237362, 'reg__bagging_fraction': 0.5176896826794558, 'reg__bagging_freq': 2, 'reg__max_bin': 74, 'reg__min_data_per_group': 195}

3.5.2 Scatter


Predicted vs True
perfect prediction
y_pred
y_true
14
12
10
8
6
4
2
2
4
6
8
10
12
14

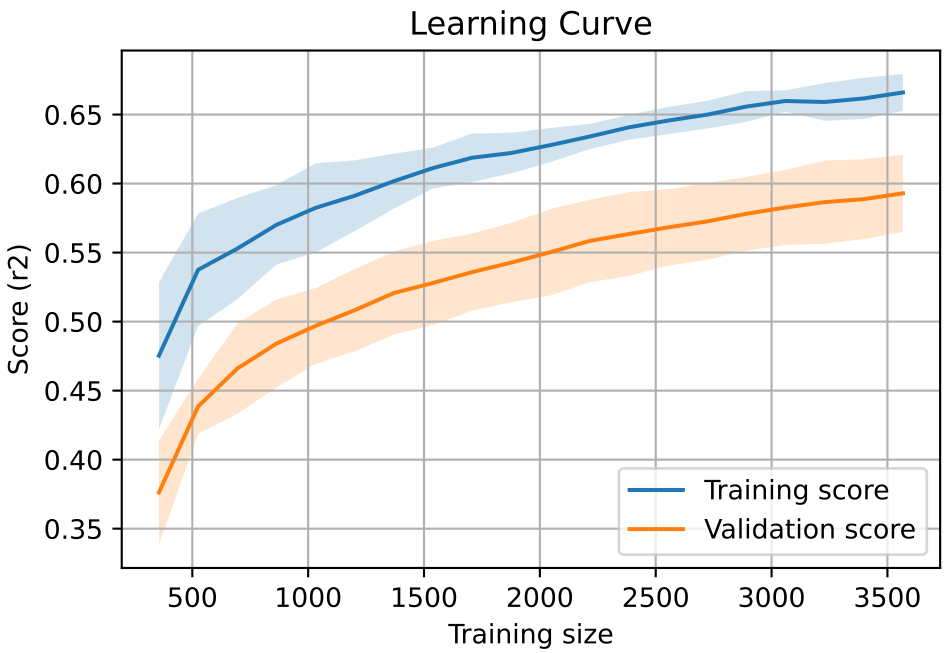
3.5.3 Scatter residuals



3.5.4 Learning curve

Len(group) : 7280, **Len(training)** : 5096,

Len(training_real) : 3568, **Len(test_of_training)** : 1528, **Len(test)** : 2184

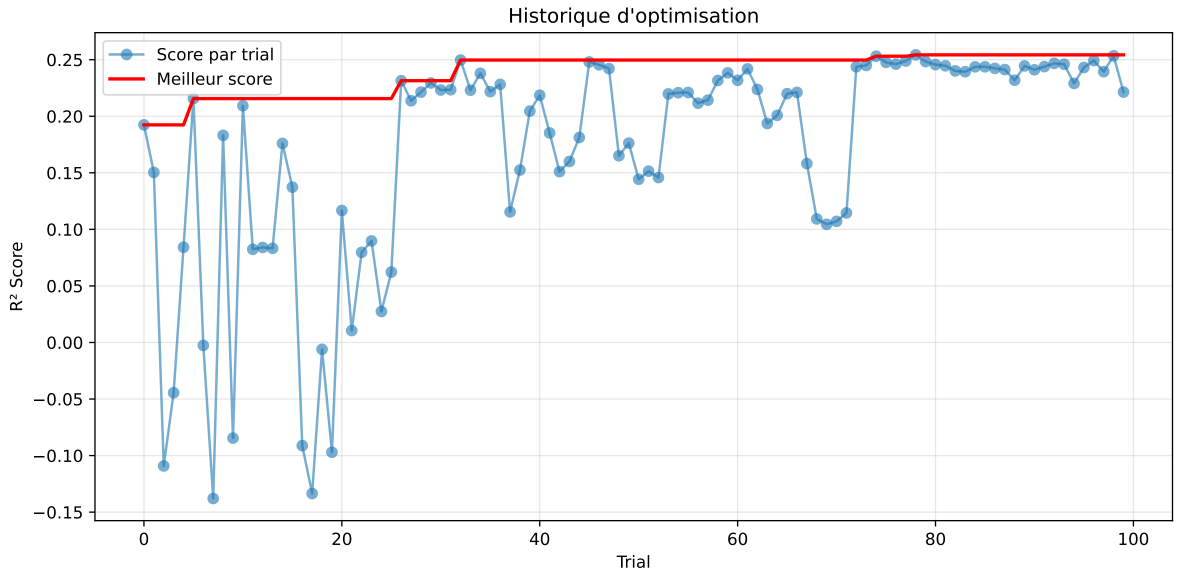


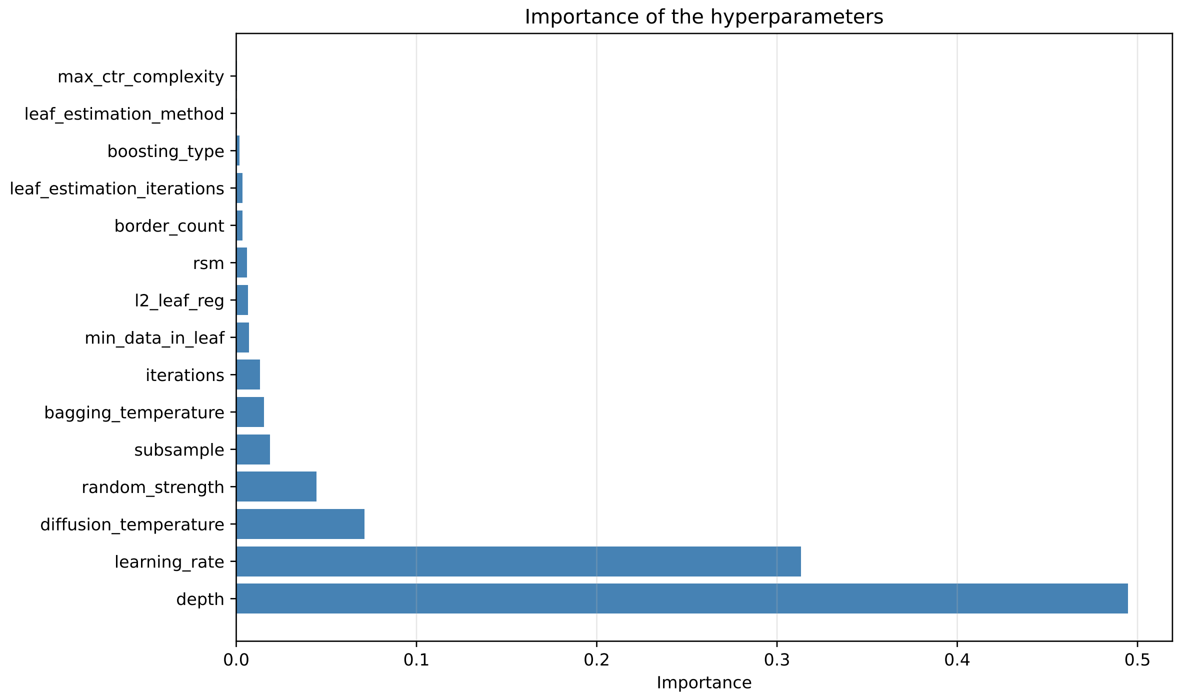
3.6 Model : CatBoost

3.6.1 Gridsearch

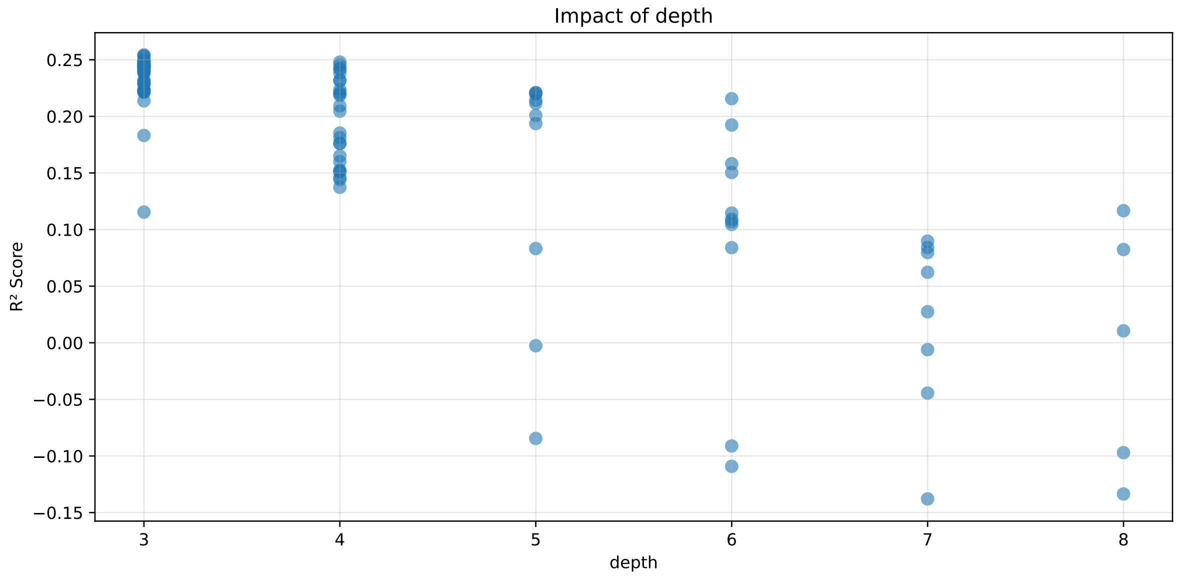
Distribution y_train vs y_test: y_train: mean=2.362, std=0.816, min=0.959, max=13.845 y_test: mean=2.359, std=0.781, min=1.062, max=7.822

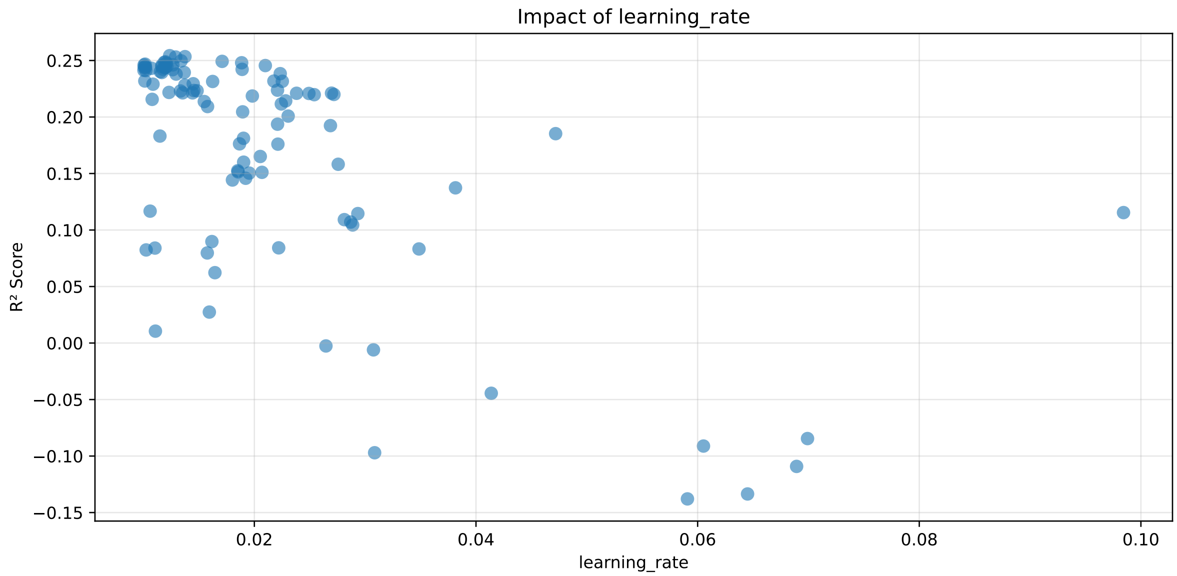
==================== TOP Importance FEATURES ==================== feature importance r_umidity 26.988717 r_temperature 13.044325 r_temperature_st_8h_diff_last 10.588744 r_temperature_st_24h_diff_last 6.485236 r_temperature_st_12h_slope 5.569428 r_temperature_st_12h_diff_last 5.444985 r_THI 4.727774 r_umidity_st_12h_std 3.265461 r_temperature_st_12h_median 3.169858 r_umidity_st_8h_slope 3.061925 r_umidity_st_24h_slope 2.809878 r_umidity_st_24h_rms 2.477361 r_umidity_st_8h_diff_last 2.078767 r_umidity_st_8h_mean 1.622699 r_umidity_st_24h_diff_last 1.498409 r_umidity_st_12h_diff_last 1.309082 r_NH3_st_8h_rms 0.724096 r_temperature_st_24h_min 0.651256 r_temperature_st_12h_std 0.650824 r_NH3_st_24h_median 0.580063 r_NH3_st_8h_diff_last 0.439121 r_NH3_st_8h_std 0.376545 r_NH3 0.346385 r_temperature_st_12h_min 0.336577 r_CO2 0.331132 r_NH3_st_12h_max 0.255649 r_NH3_st_24h_diff_last 0.245435 r_NH3_st_12h_diff_last 0.217407 r_temperature_st_8h_std 0.182740 r_NH3_st_12h_slope 0.164166 r_NH3_st_8h_slope 0.113409 r_NH3_st_12h_std 0.100306 r_umidity_st_24h_std 0.092139 r_NH3_st_24h_slope 0.050102

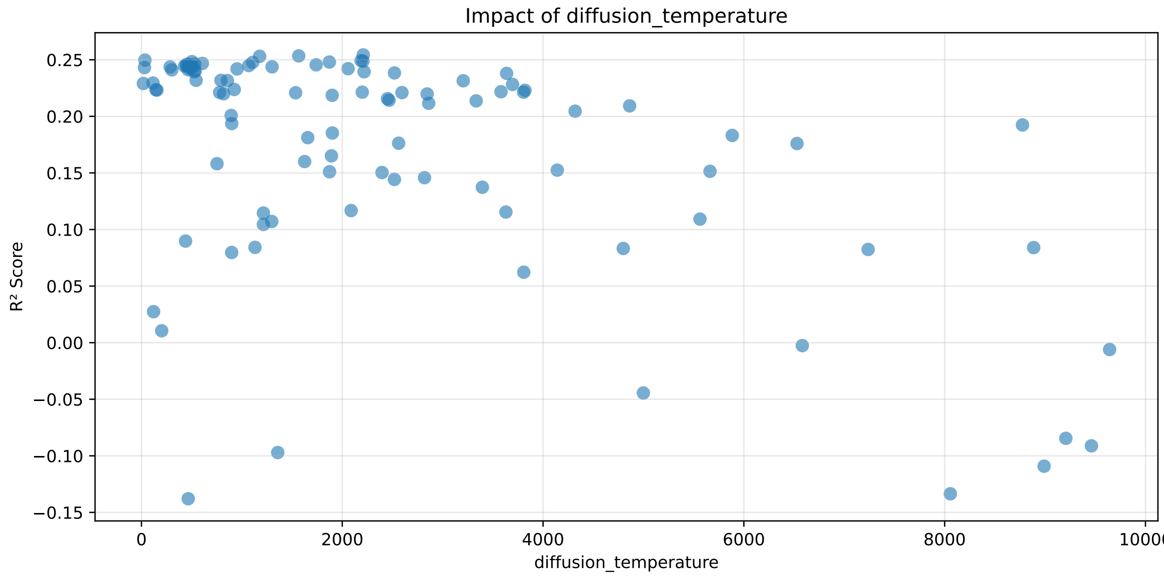




Paramètres avec >5% d'importance : ['depth', 'learning_rate', 'diffusion_temperature']


Impact of depth
R² Score
0.25
0.20
0.15
0.10
0.05
0.00
−0.05
−0.10
−0.15
3
4
5
6
7
8
depth


Impact of learning_rate
R² Score
0.25
0.20
0.15
0.10
0.05
0.00
−0.05
−0.10
−0.15
0.02
0.04
0.06
0.08
0.10
learning_rate




```
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The default value of observed=False is deprecated and will change to observed=True in a futu
```
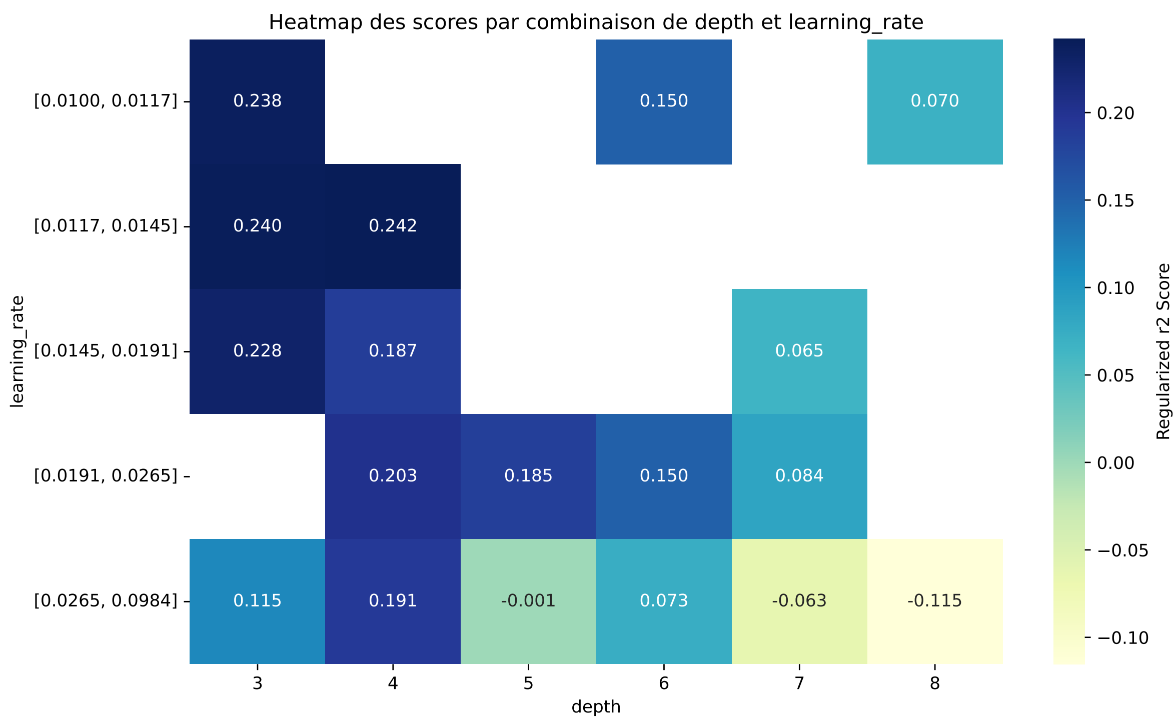



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.5595	0.5086	0.5768	0.0262	0.2543	0.583	0.0743	1.1461	1.1
2	0.5699	0.5172	0.5866	0.0276	0.2534	0.5912	0.0741	1.1432	1.102
3	0.6007	0.5428	0.6138	0.0293	0.2531	0.6186	0.0758	1.1397	1.1067
4	0.6619	0.5932	0.66	0.029	0.2497	0.6663	0.0731	1.1232	1.1158
5	0.6528	0.5855	0.6571	0.028	0.2492	0.6607	0.0752	1.1284	1.1149
6	0.558	0.5065	0.5758	0.0282	0.2488	0.579	0.0725	1.1432	1.1017
7	0.549	0.4989	0.566	0.0274	0.2484	0.5711	0.0722	1.1448	1.1004
8	0.6335	0.5692	0.6399	0.0305	0.248	0.6487	0.0795	1.1397	1.1129
9	0.5978	0.5394	0.6101	0.0282	0.2477	0.6137	0.0743	1.1376	1.1082
10	0.5214	0.4756	0.5403	0.0271	0.2468	0.5481	0.0725	1.1524	1.0962
---	---	---	---	---	---	---	---	---	---
100	0.9499	0.7686	0.8402	0.0345	-0.1379	0.8458	0.0772	1.1004	1.2359

Hyperparameters:

Rank 1: {'reg___iterations': 200, 'reg___depth': 3, 'reg___learning_rate': 0.012358473244334525, 'reg___l2_leaf_reg': 29.585177187474386, 'reg___bagging_temperature': 1.7230466740402473, 'reg___random_strength': 3.097169825378701, 'reg___subsample': 0.7418801857442515, 'reg___min_data_in_leaf': 46, 'reg___leaf_estimation_iterations': 6, 'reg___rsm': 0.5266388122581638, 'reg___border_count': 40, 'reg___leaf_estimation_method': 'Gradient', 'reg___boosting_type': 'Ordered', 'reg___max_ctr_complexity': 1, 'reg___diffusion_temperature': 2209.086217588144}

Rank 2: {'reg___iterations': 200, 'reg___depth': 3, 'reg___learning_rate': 0.013739836305708748, 'reg___l2_leaf_reg': 30.73592166178596, 'reg___bagging_temperature': 2.7639090733224028, 'reg___random_strength': 3.3727283395196475, 'reg___subsample': 0.8618972927942741, 'reg___min_data_in_leaf': 44, 'reg___leaf_estimation_iterations': 4, 'reg___rsm': 0.7135644643652702, 'reg___border_count': 72, 'reg___leaf_estimation_method': 'Newton', 'reg___boosting_type': 'Ordered', 'reg___max_ctr_complexity': 1, 'reg___diffusion_temperature': 1566.5173715149992}

Rank 3: {'reg___iterations': 300, 'reg___depth': 3, 'reg___learning_rate': 0.012899221110716879, 'reg___l2_leaf_reg': 23.585814904291468, 'reg___bagging_temperature': 1.5821615001086182, 'reg___random_strength': 3.1004274572336534, 'reg___subsample': 0.7785899877158444, 'reg___min_data_in_leaf': 45, 'reg___leaf_estimation_iterations': 4, 'reg___rsm': 0.5273836478302821, 'reg___border_count': 38, 'reg___leaf_estimation_method': 'Gradient', 'reg___boosting_type': 'Ordered', 'reg___max_ctr_complexity': 2, 'reg___diffusion_temperature': 1177.0473259013777}

Rank 4: {'reg___iterations': 700, 'reg___depth': 3, 'reg___learning_rate': 0.013393753402446596, 'reg___l2_leaf_reg': 12.427036589299012, 'reg___bagging_temperature': 0.9101886938742494, 'reg___random_strength': 1.6741030014714364, 'reg___subsample': 0.8410021670822877, 'reg___min_data_in_leaf': 44, 'reg___leaf_estimation_iterations': 2, 'reg___rsm':

0.613311005187111, ’reg__border_count’: 59, ’reg__leaf_estimation_method’: ‘Gradient’, ’reg__boosting_type’: ‘Ordered’, ’reg__max_ctr_complexity’: 2, ’reg__diffusion_temperature’: 34.270948259120814}

Rank 5: {’reg__iterations’: 400, ’reg__depth’: 3, ’reg__learning_rate’: 0.01709211351085708, ’reg__l2_leaf_reg’: 31.410045388235318, ’reg__bagging_temperature’: 1.639518829526811, ’reg__random_strength’: 3.4988957986544515, ’reg__subsample’: 0.8597663820201108, ’reg__min_data_in_leaf’: 22, ’reg__leaf_estimation_iterations’: 4, ’reg__rsm’: 0.7105636255718161, ’reg__border_count’: 47, ’reg__leaf_estimation_method’: ‘Newton’, ’reg__boosting_type’: ‘Plain’, ’reg__max_ctr_complexity’: 1, ’reg__diffusion_temperature’: 2187.3192742741435}

Rank 6: {’reg__iterations’: 200, ’reg__depth’: 3, ’reg__learning_rate’: 0.011976735613164037, ’reg__l2_leaf_reg’: 20.229359933126826, ’reg__bagging_temperature’: 1.6596804797482094, ’reg__random_strength’: 3.0608021561542156, ’reg__subsample’: 0.7364475322551363, ’reg__min_data_in_leaf’: 45, ’reg__leaf_estimation_iterations’: 4, ’reg__rsm’: 0.7368870338827248, ’reg__border_count’: 39, ’reg__leaf_estimation_method’: ‘Gradient’, ’reg__boosting_type’: ‘Ordered’, ’reg__max_ctr_complexity’: 1, ’reg__diffusion_temperature’: 2206.8089548013345}

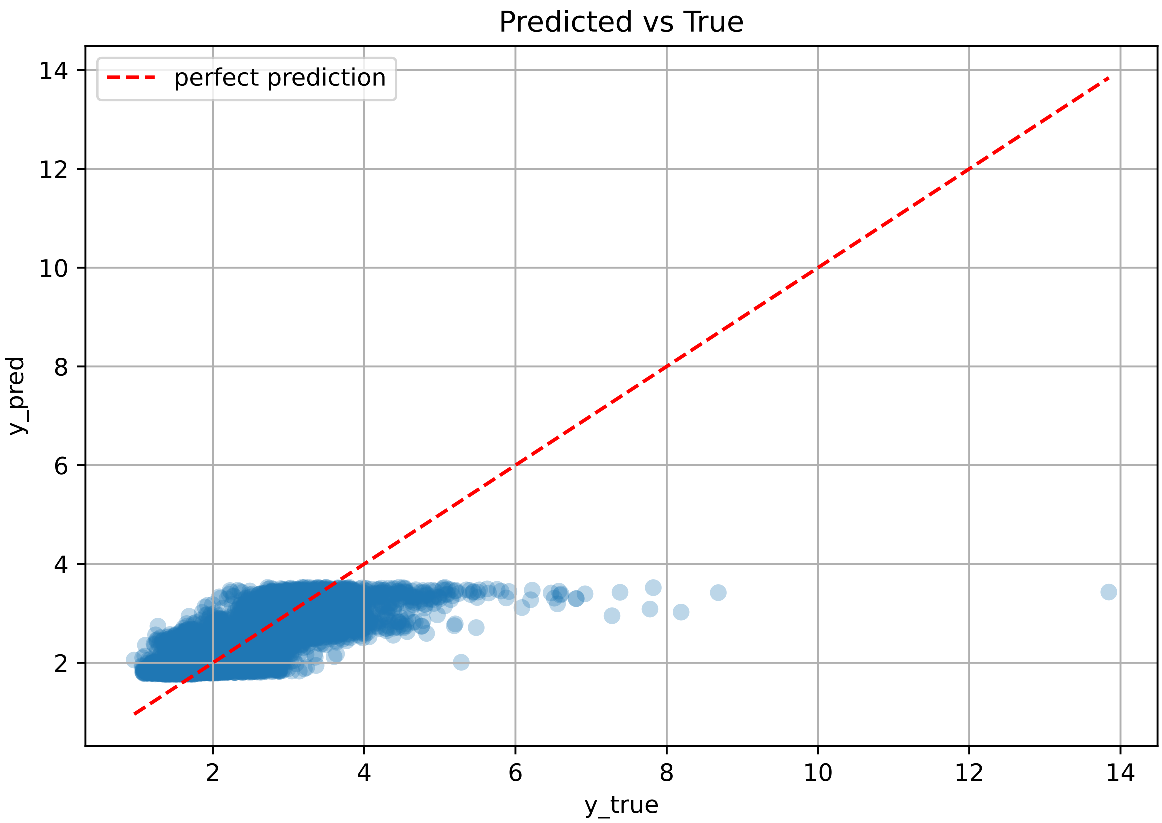
Rank 7: {’reg__iterations’: 200, ’reg__depth’: 3, ’reg__learning_rate’: 0.011871451993238743, ’reg__l2_leaf_reg’: 27.40261702544104, ’reg__bagging_temperature’: 1.6328086222841036, ’reg__random_strength’: 3.527312519678312, ’reg__subsample’: 0.7379820036380682, ’reg__min_data_in_leaf’: 27, ’reg__leaf_estimation_iterations’: 6, ’reg__rsm’: 0.7448703384486108, ’reg__border_count’: 39, ’reg__leaf_estimation_method’: ‘Newton’, ’reg__boosting_type’: ‘Plain’, ’reg__max_ctr_complexity’: 1, ’reg__diffusion_temperature’: 502.91142547219755}

Rank 8: {’reg__iterations’: 200, ’reg__depth’: 4, ’reg__learning_rate’: 0.018853719663285546, ’reg__l2_leaf_reg’: 13.321723776780209, ’reg__bagging_temperature’: 1.2863296962324449, ’reg__random_strength’: 2.741289048019854, ’reg__subsample’: 0.746925558603246, ’reg__min_data_in_leaf’: 36, ’reg__leaf_estimation_iterations’: 3, ’reg__rsm’: 0.5056093358240784, ’reg__border_count’: 48, ’reg__leaf_estimation_method’: ‘Gradient’, ’reg__boosting_type’: ‘Ordered’, ’reg__max_ctr_complexity’: 3, ’reg__diffusion_temperature’: 1871.506297092522}

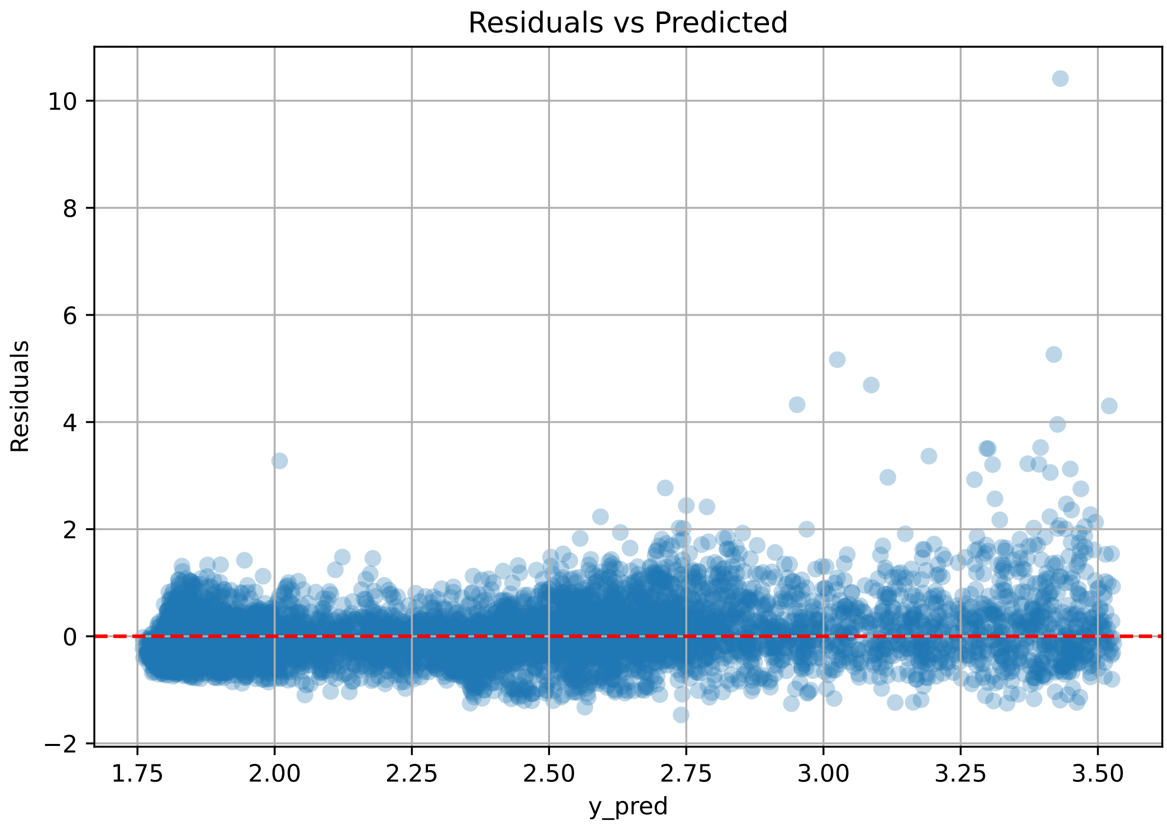
Rank 9: {’reg__iterations’: 300, ’reg__depth’: 3, ’reg__learning_rate’: 0.012055135809592483, ’reg__l2_leaf_reg’: 25.251064221417153, ’reg__bagging_temperature’: 0.5114380055344129, ’reg__random_strength’: 3.0646893147065466, ’reg__subsample’: 0.7847589229269054, ’reg__min_data_in_leaf’: 45, ’reg__leaf_estimation_iterations’: 6, ’reg__rsm’: 0.7478051524495396, ’reg__border_count’: 38, ’reg__leaf_estimation_method’: ‘Newton’, ’reg__boosting_type’: ‘Ordered’, ’reg__max_ctr_complexity’: 1, ’reg__diffusion_temperature’: 1107.3543486107542}

Rank 10: {’reg__iterations’: 200, ’reg__depth’: 3, ’reg__learning_rate’: 0.01014028073345461, ’reg__l2_leaf_reg’: 31.208469971267842, ’reg__bagging_temperature’: 0.5951419962160727, 'reg___random_strength': 3.8518285070751306, 'reg___subsample': 0.6932696655976172, 'reg___min_data_in_leaf': 22, 'reg___leaf_estimation_iterations': 8, 'reg___rsm': 0.7605408294211489, 'reg___border_count': 53, 'reg___leaf_estimation_method': 'Newton', 'reg___boosting_type': 'Plain', 'reg___max_ctr_complexity': 1, 'reg___diffusion_temperature': 606.4449367808409}

3.6.2 Scatter



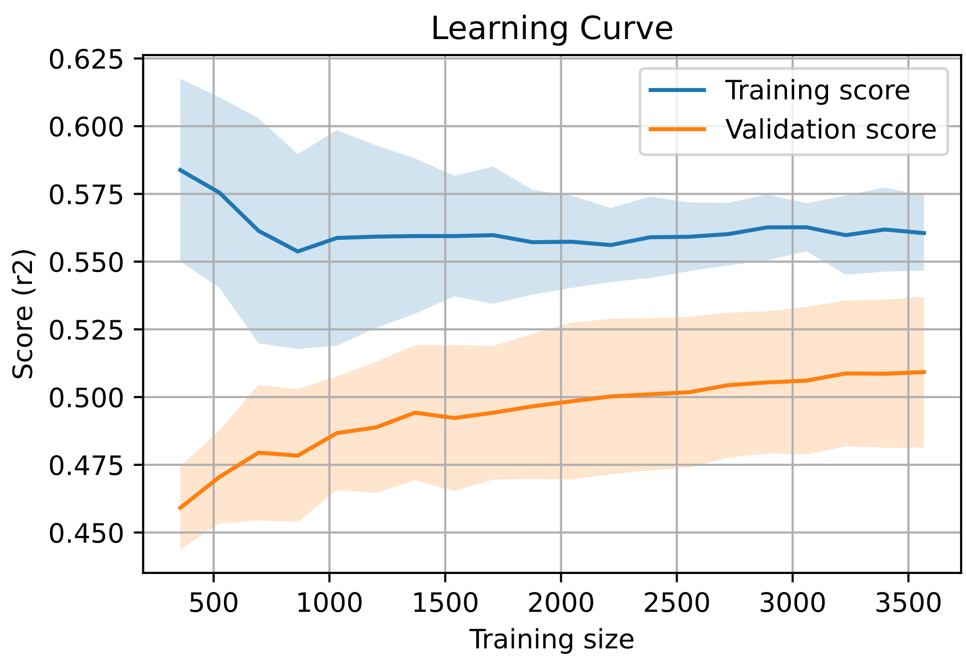
3.6.3 Scatter residuals



3.6.4 Learning curve

Len(group) : 7280, **Len(training)** : 5096,

Len(training_real) : 3568, **Len(test_of_training)** : 1528, **Len(test)** : 2184



3.7 Sumup-table

model	mean_train_cv	mean_test_cv	std_test_cv	test_cv	test	diff	r_test	r_train	pen_score
Linear	0.613	0.554	0.033	0.615	0.616	0.062	1.112	1.106	nan
LGBM	0.665	0.592	0.029	0.666	0.692	0.1	1.17	1.124	0.225
CatBoost	0.56	0.509	0.026	0.577	0.583	0.074	1.146	1.1	0.2543

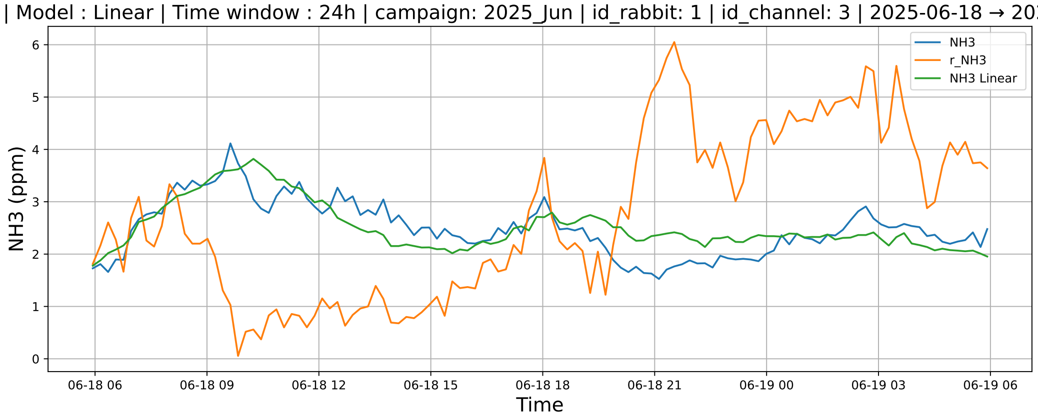
4 Plot

4.1 Standalone plots

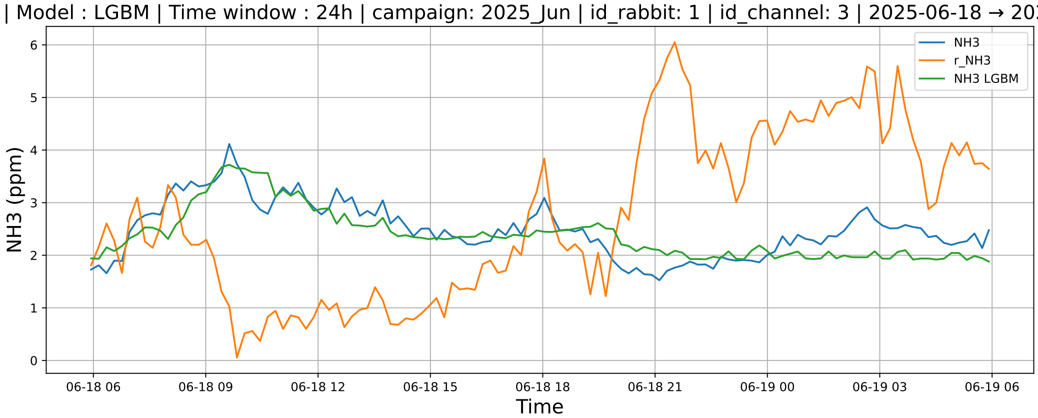
4.1.1 Group: campaign: 2025_Jun,id_rabbit: 1,id_channel: 3

4.1.1.1 Time window : 24

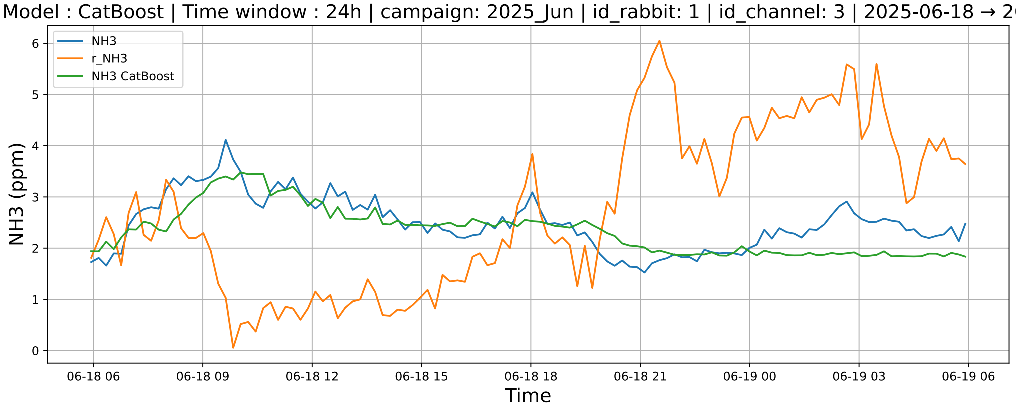
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19



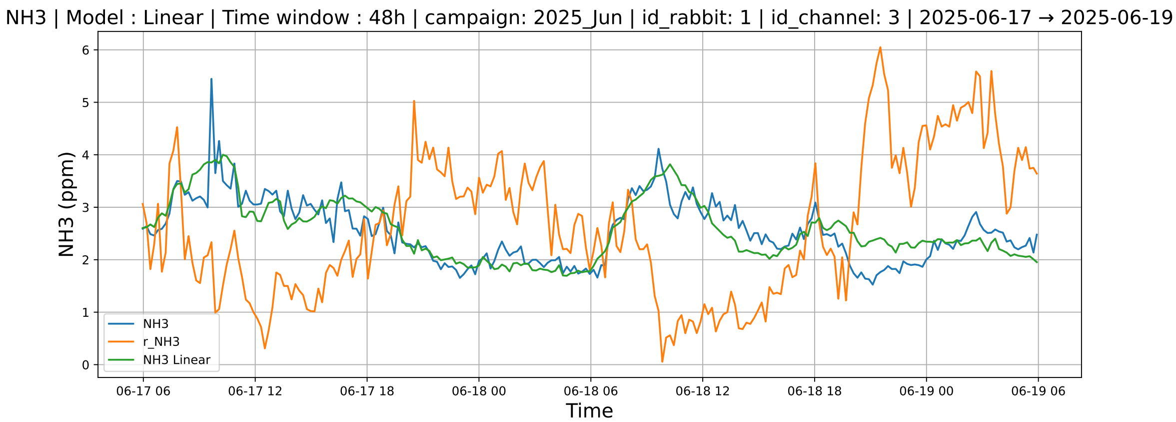
NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19

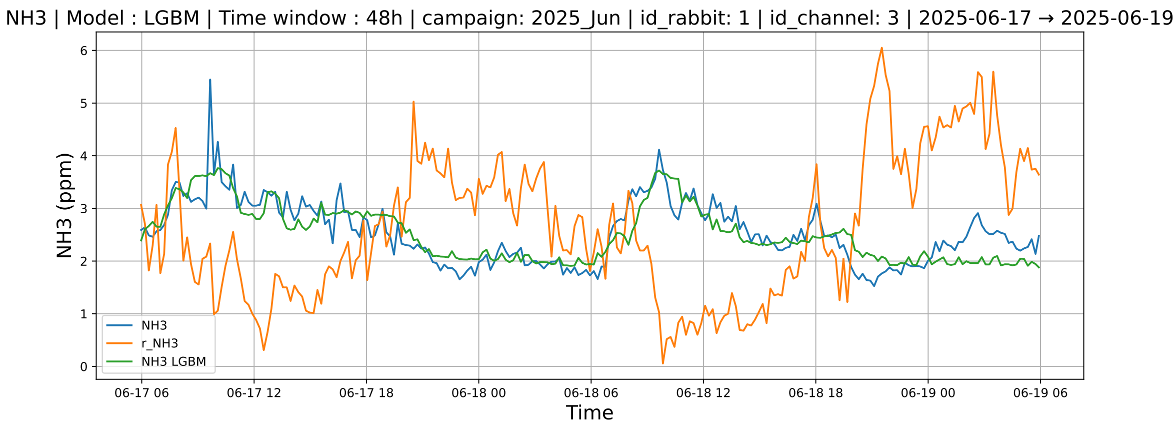


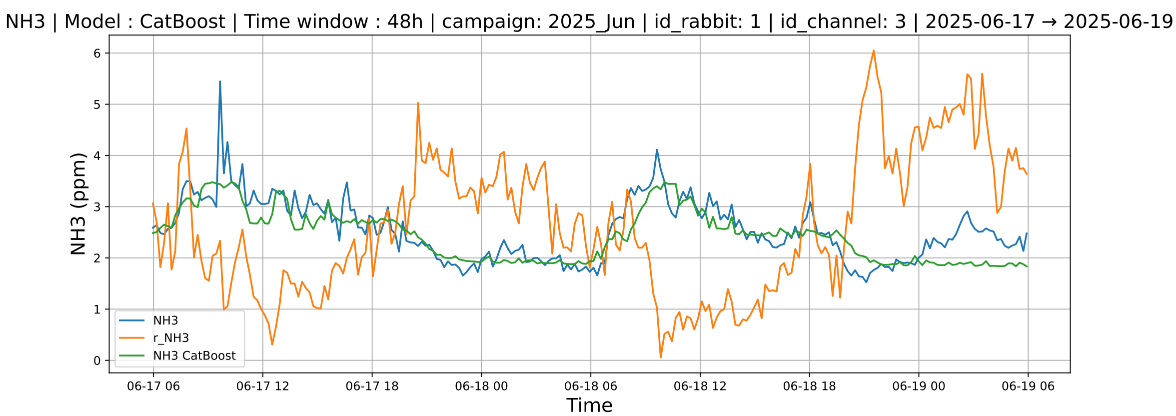
NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19



4.1.1.2 Time window : 48

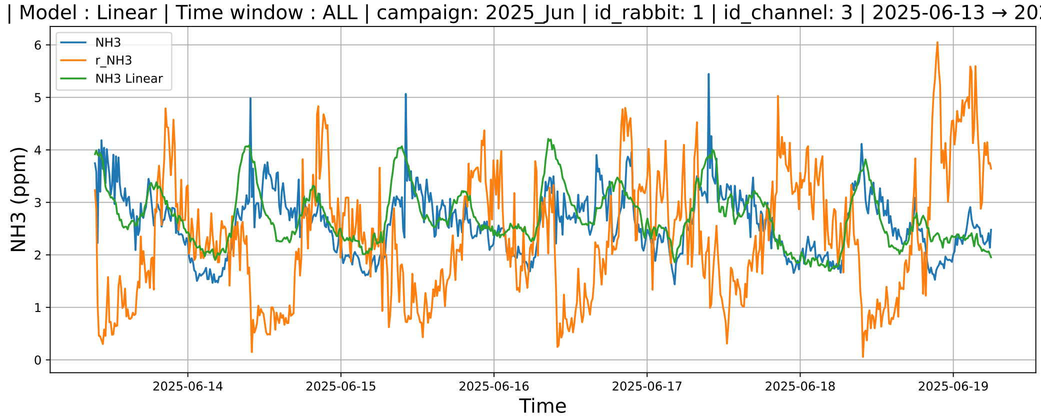




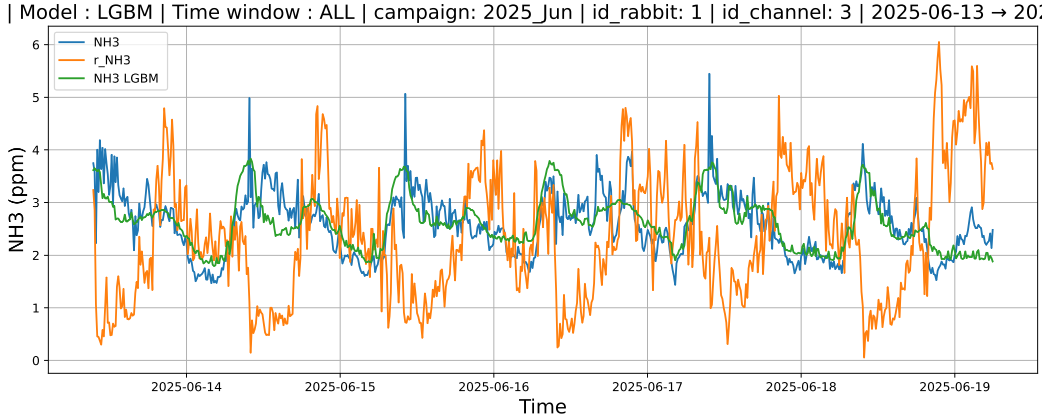


4.1.1.3 Time window : ALL

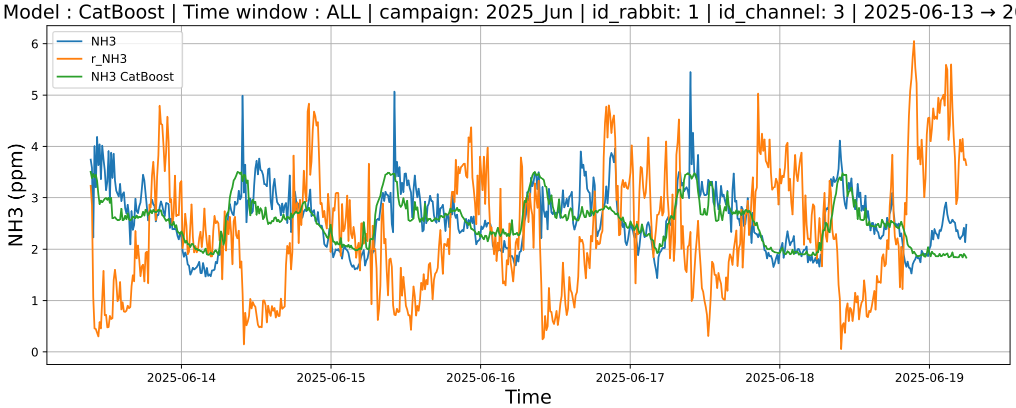
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-13 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-13 → 2025-06-19

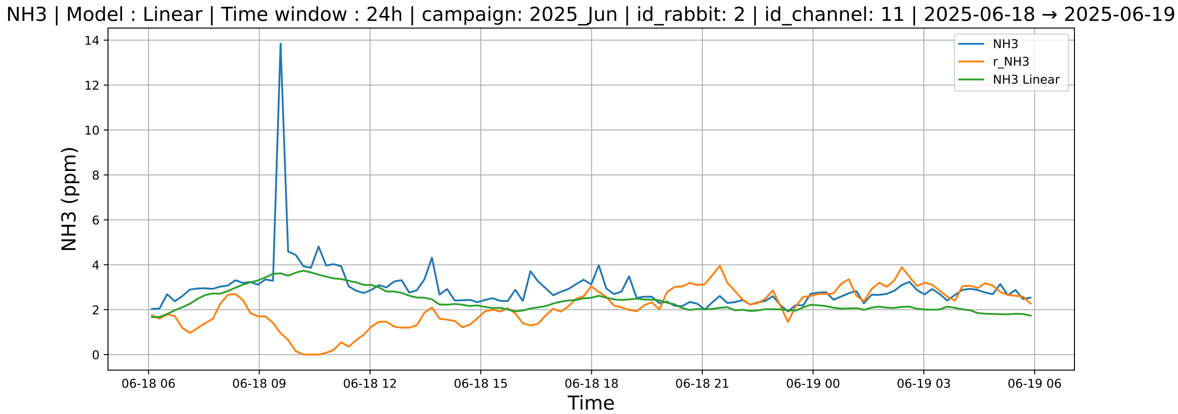


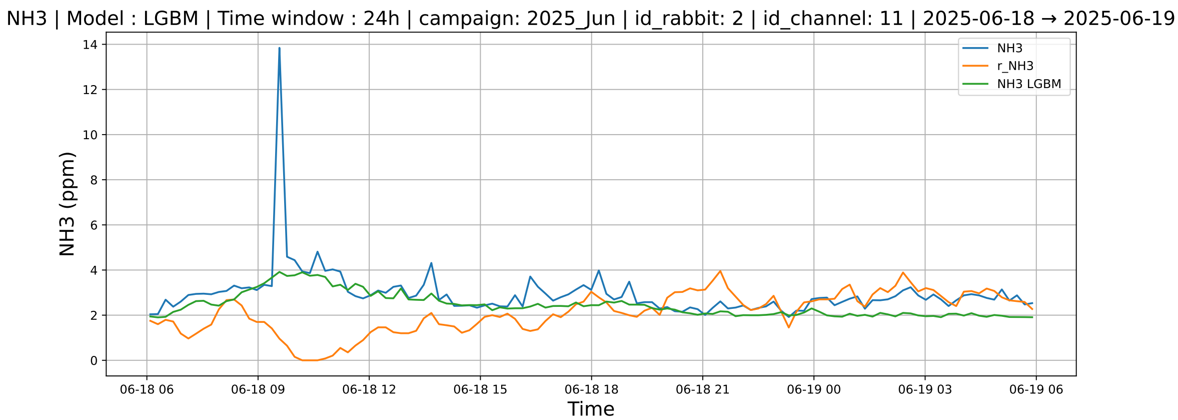
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-13 → 2025-06-19

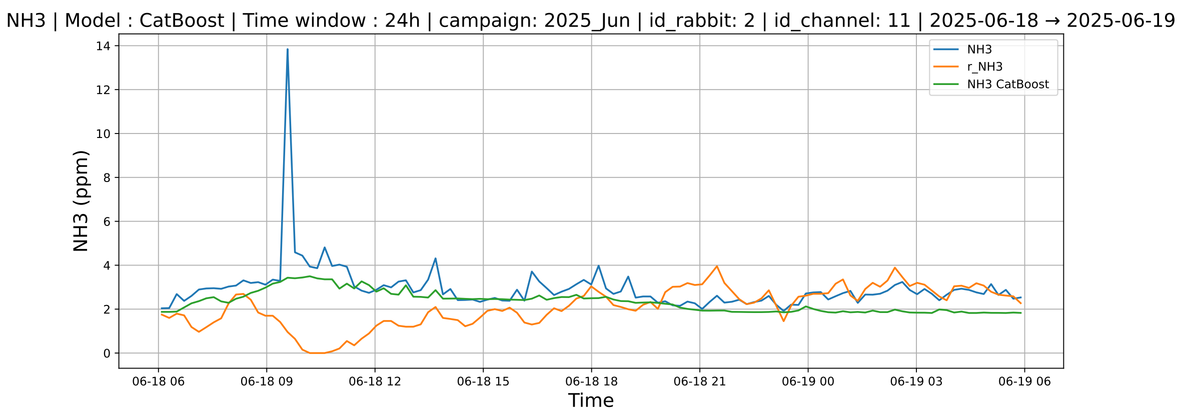


4.1.2 Group: campaign: 2025_Jun,id_rabbit: 2,id_channel: 11

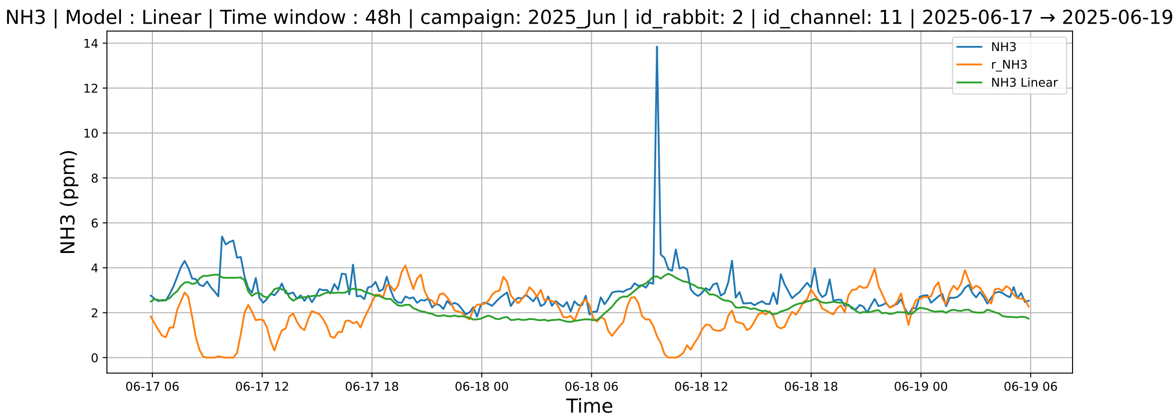
4.1.2.1 Time window : 24

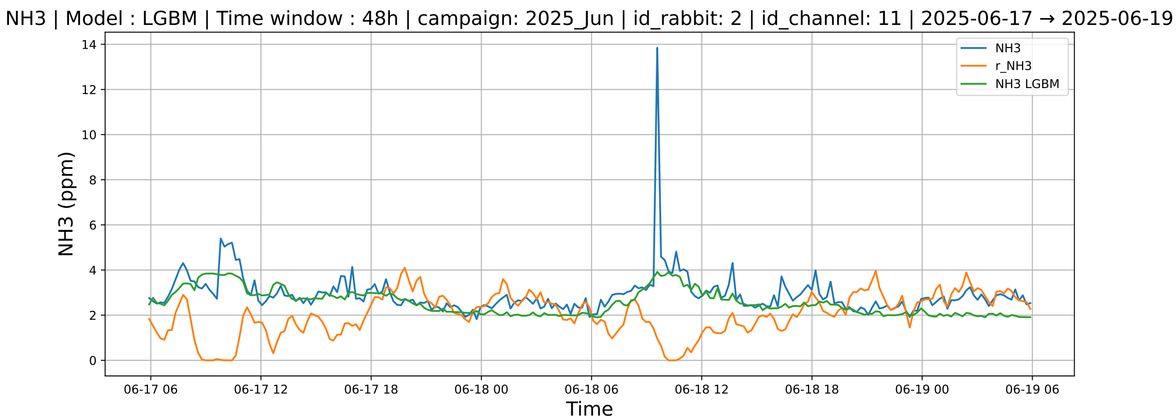


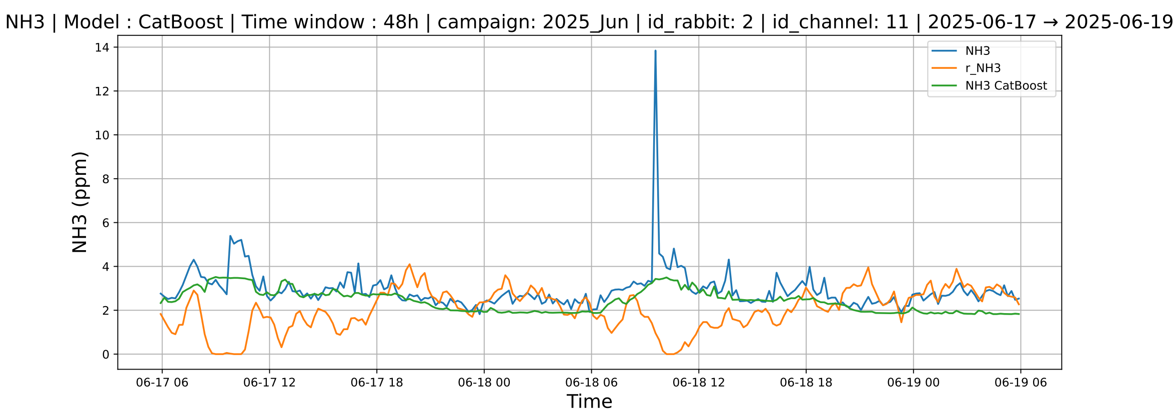




4.1.2.2 Time window : 48

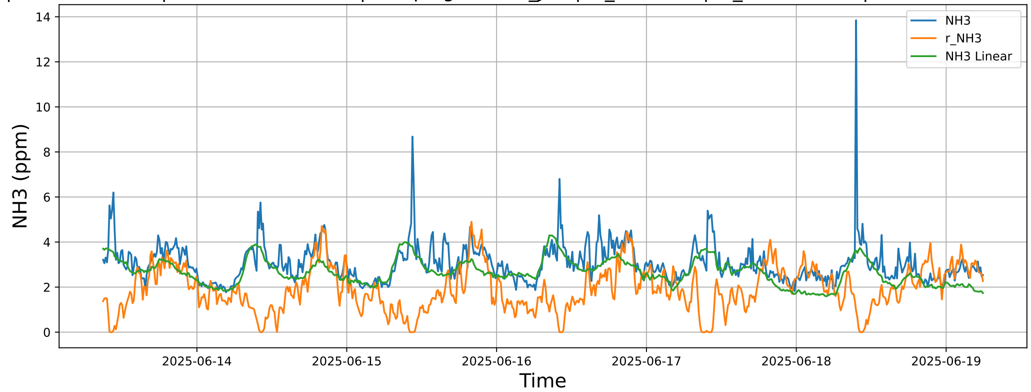




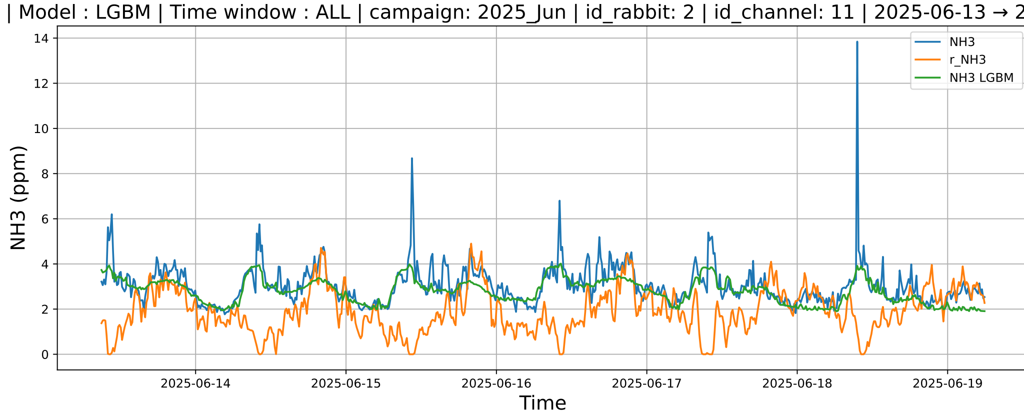


4.1.2.3 Time window : ALL

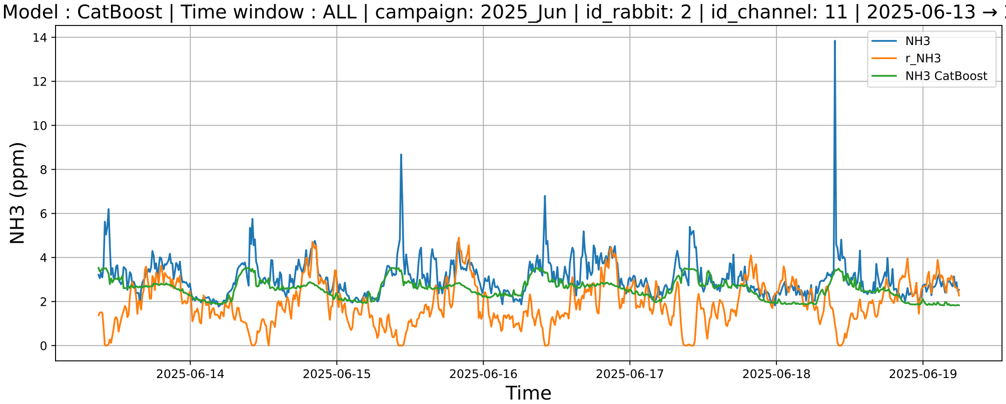
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-13 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-13 → 2025-06-19

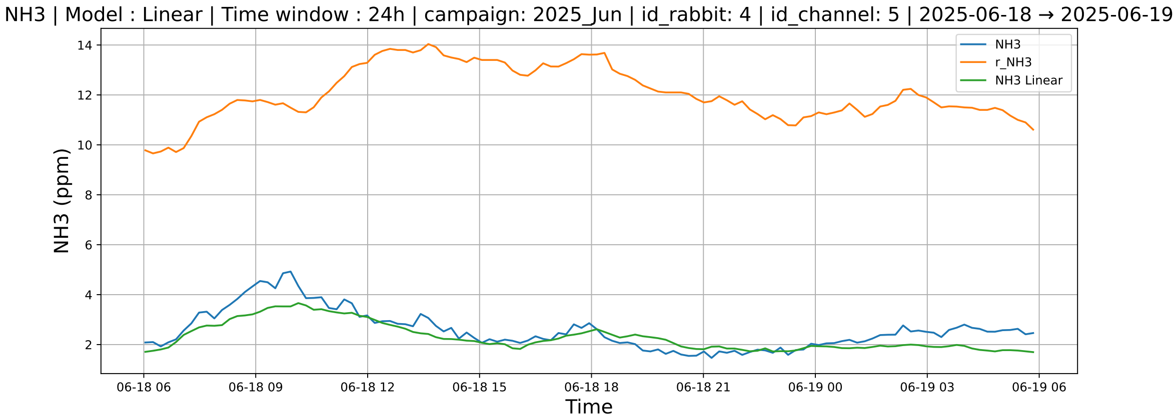


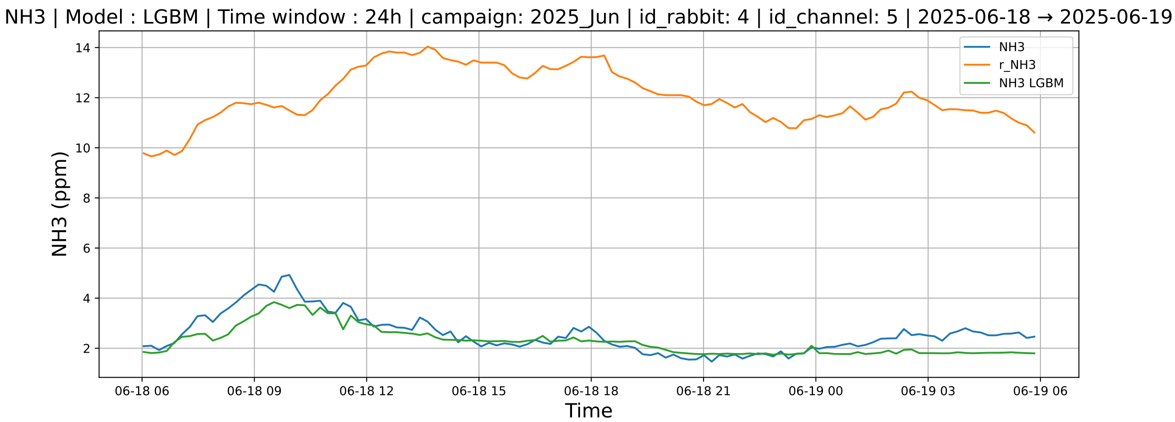
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-13 → 2025-06-19

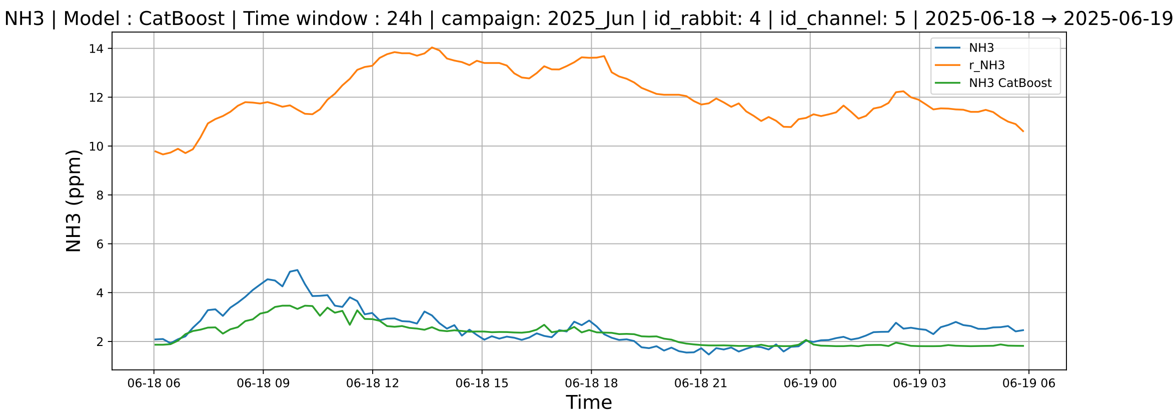


4.1.3 Group: campaign: 2025_Jun,id_rabbit: 4,id_channel: 5

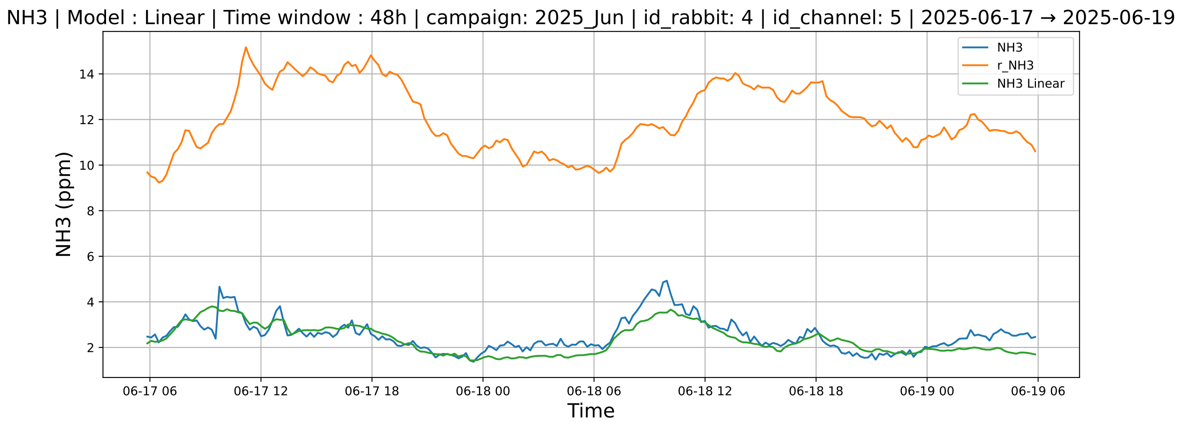
4.1.3.1 Time window : 24

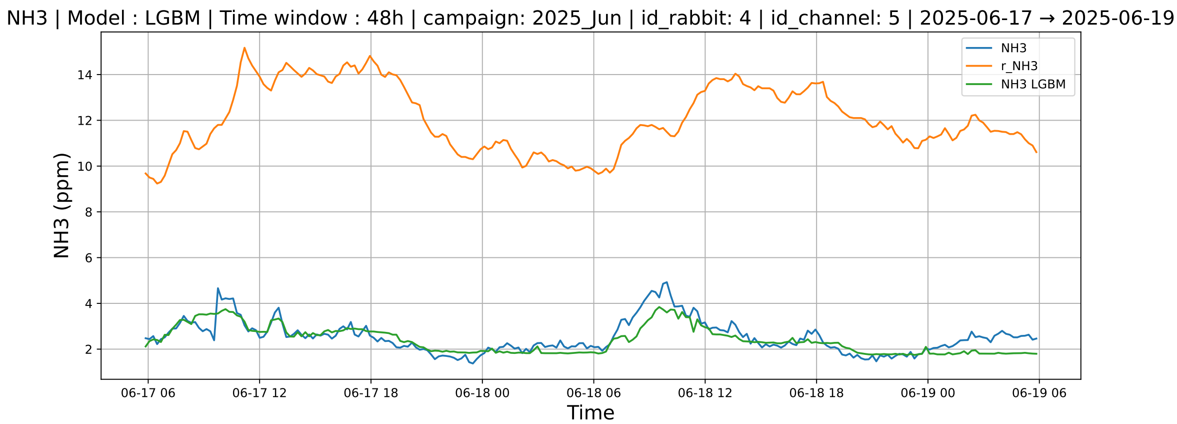


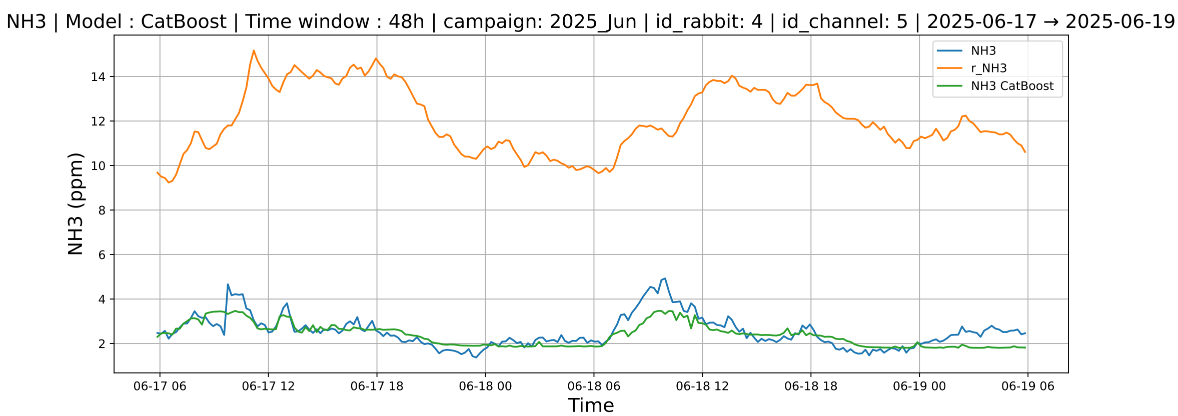




4.1.3.2 Time window : 48

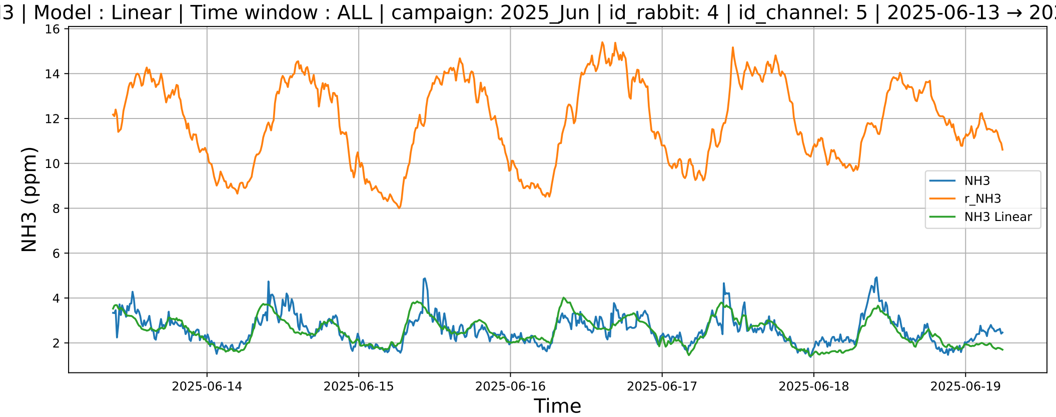




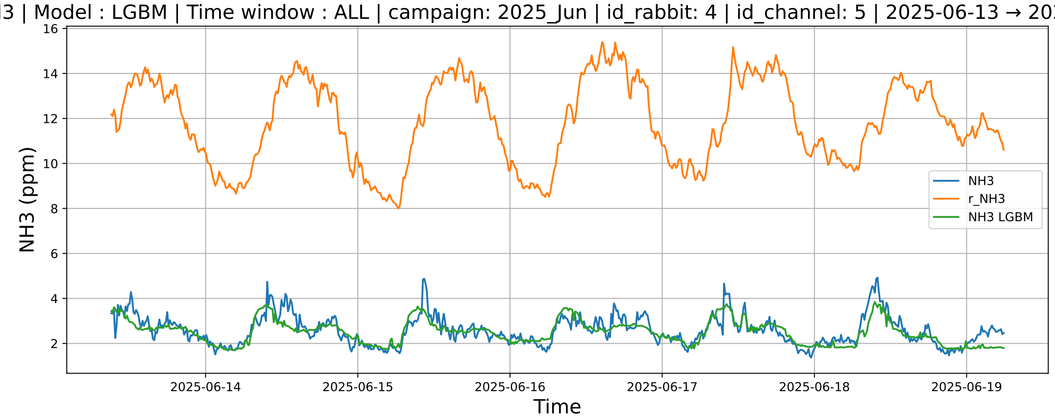


4.1.3.3 Time window : ALL

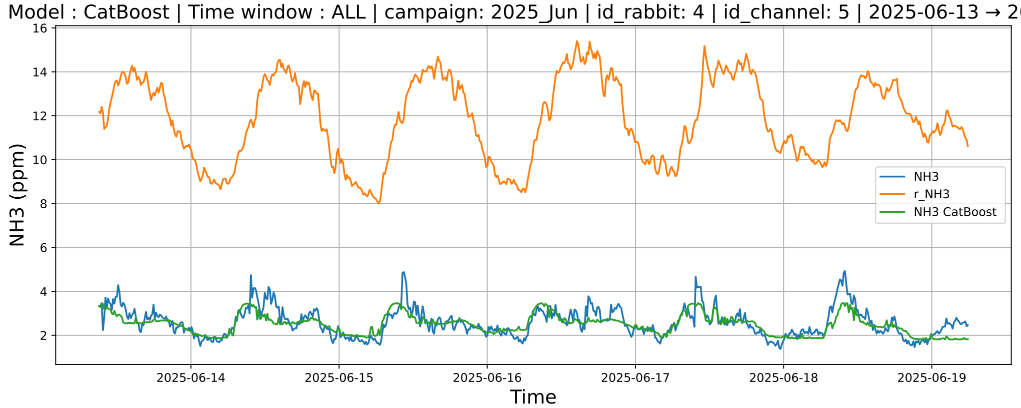
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-13 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-13 → 2025-06-19

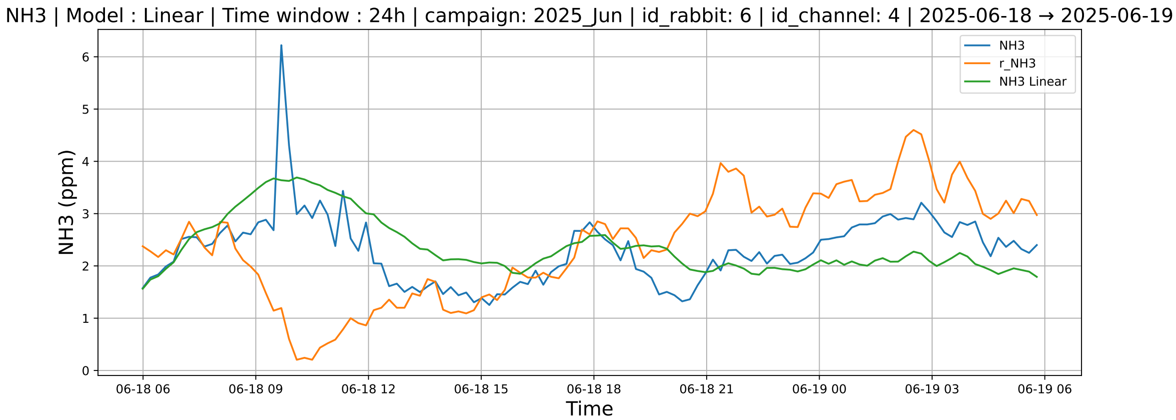


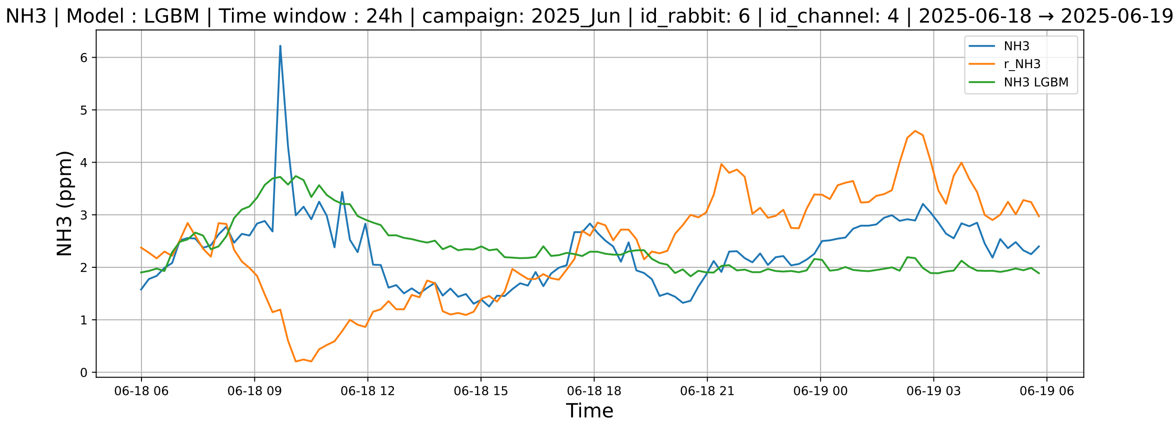
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-13 → 2025-06-19

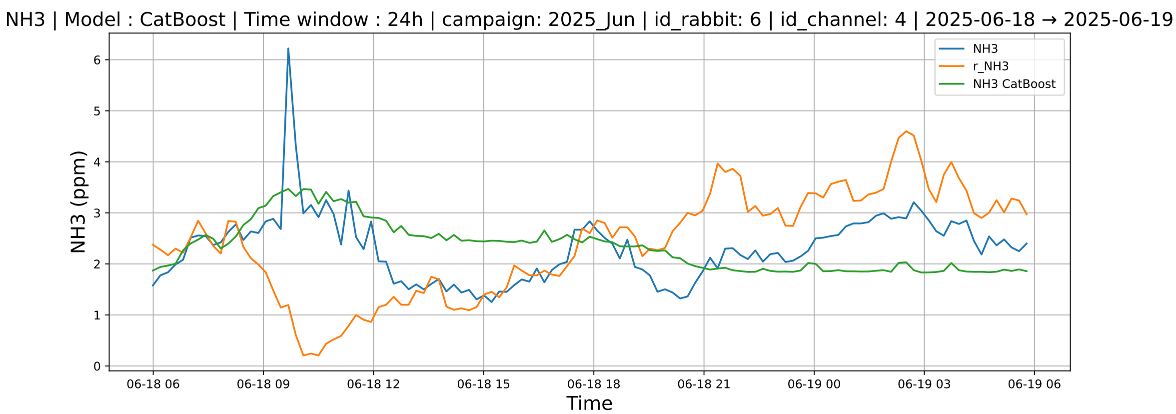


4.1.4 Group: campaign: 2025_Jun,id_rabbit: 6,id_channel: 4

4.1.4.1 Time window : 24

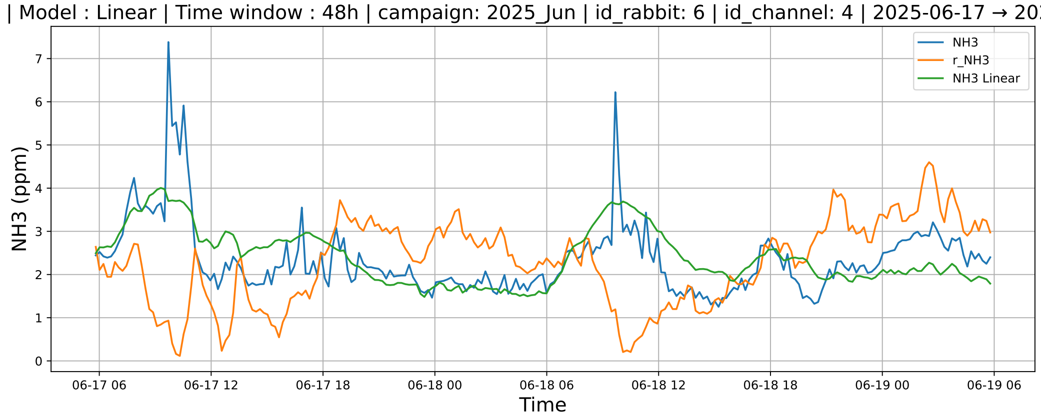




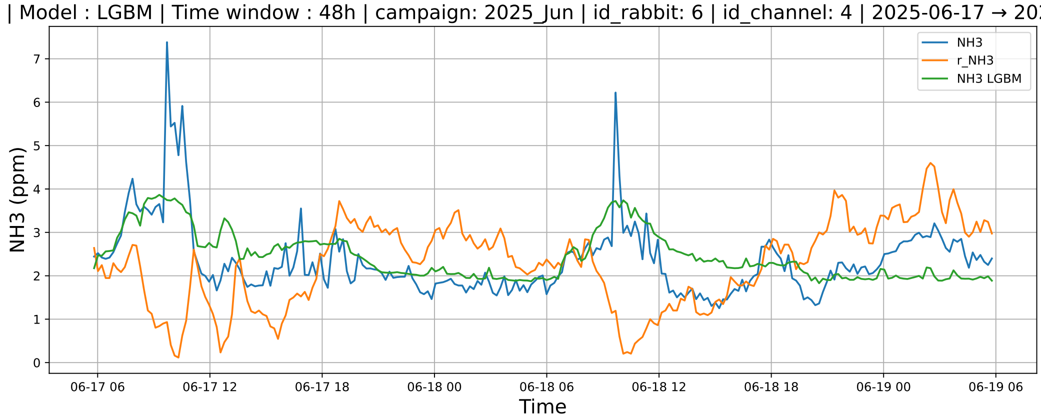


4.1.4.2 Time window : 48

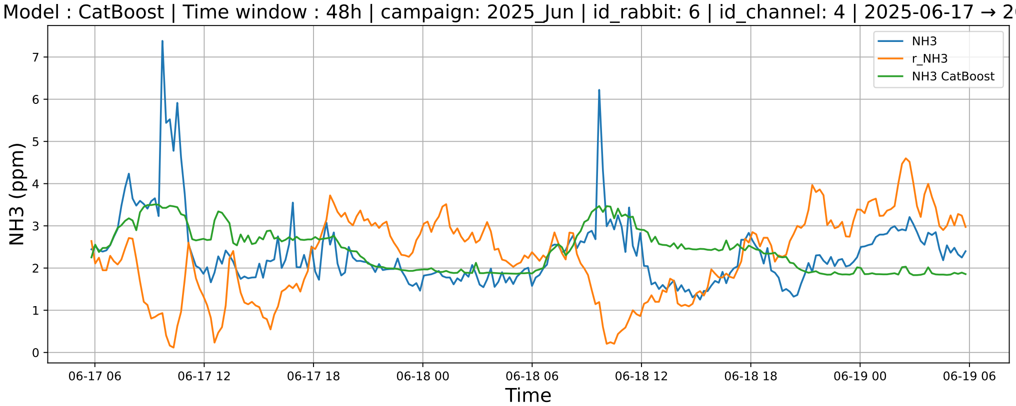
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19



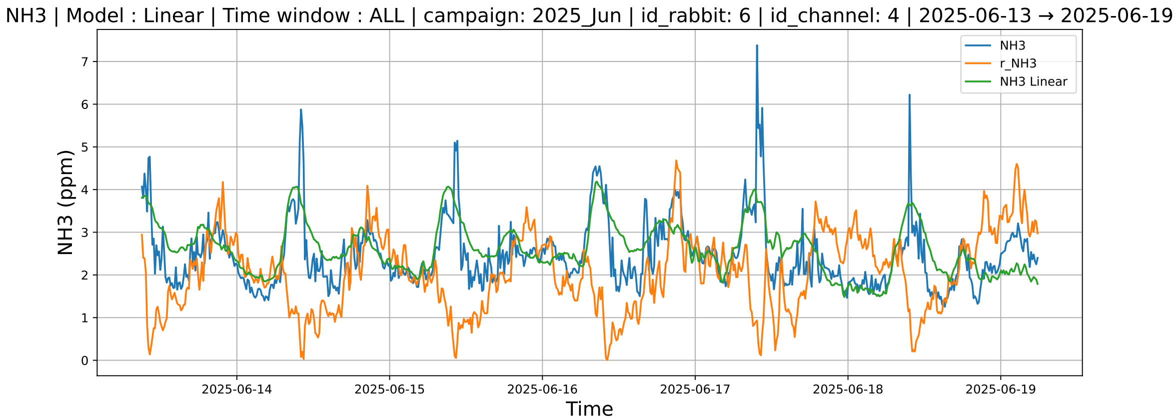
NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19

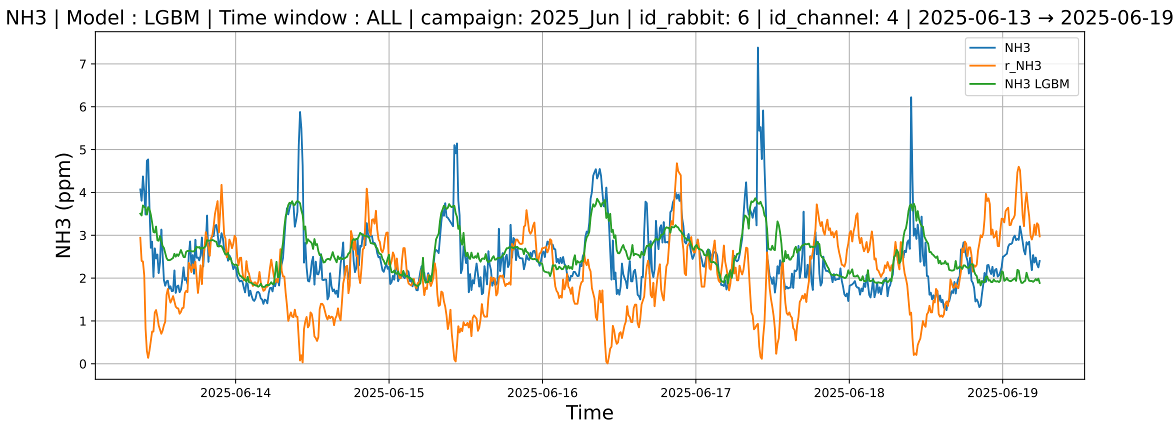


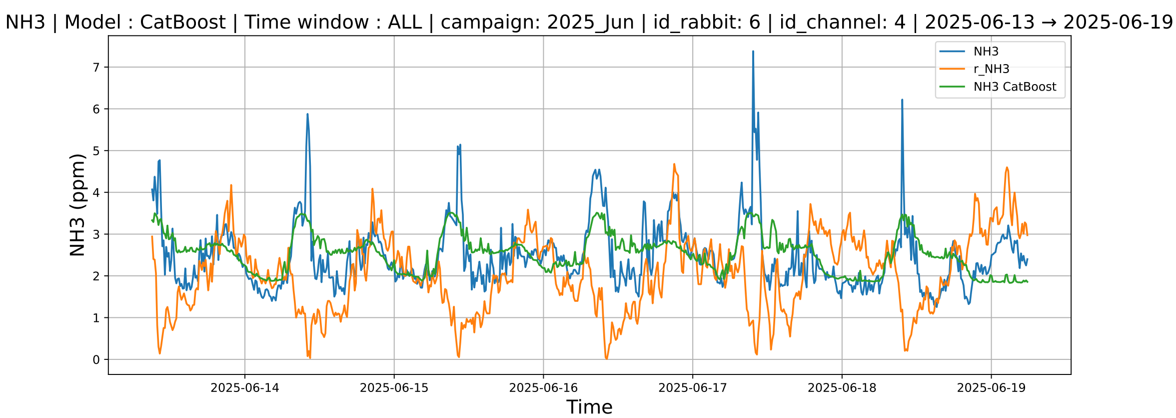
NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19



4.1.4.3 Time window : ALL



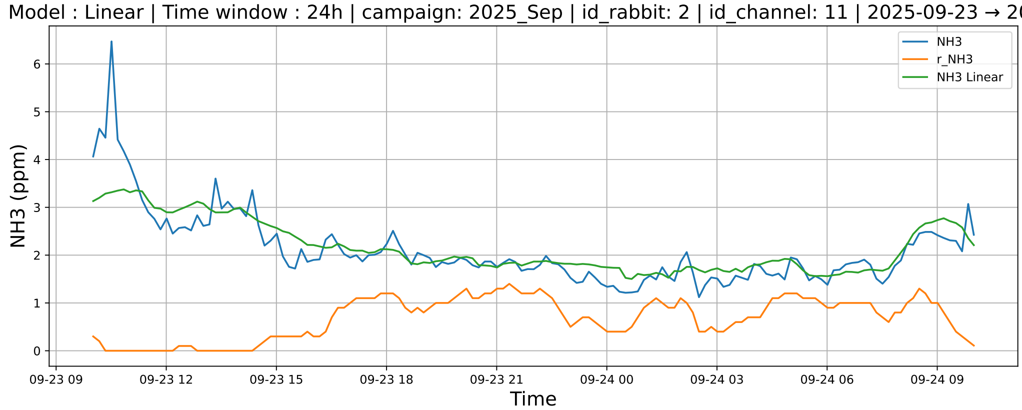




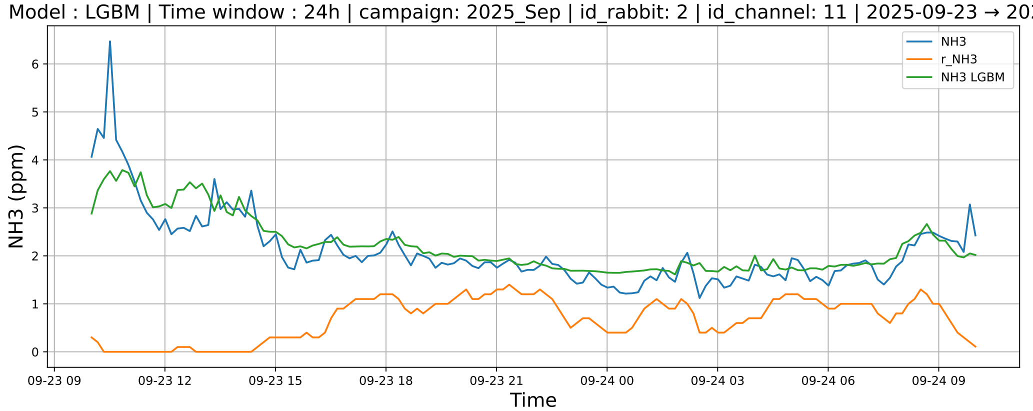
4.1.5 Group: campaign: 2025_Sep,id_rabbit: 2,id_channel: 11

4.1.5.1 Time window : 24

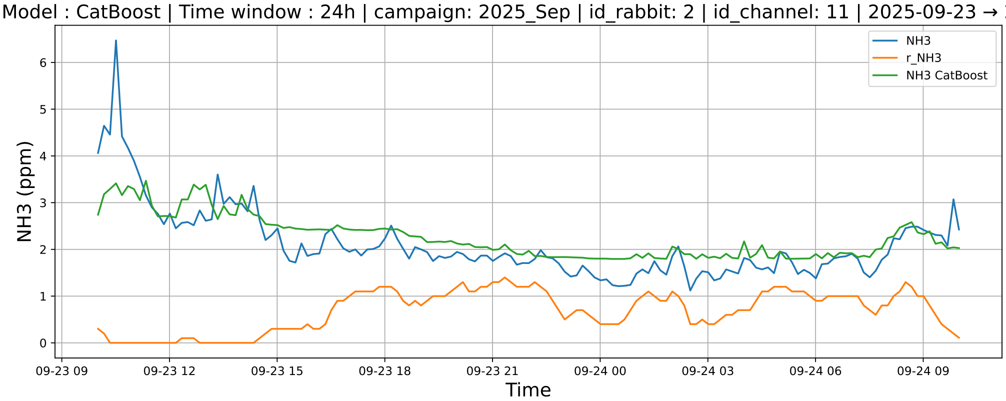
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24

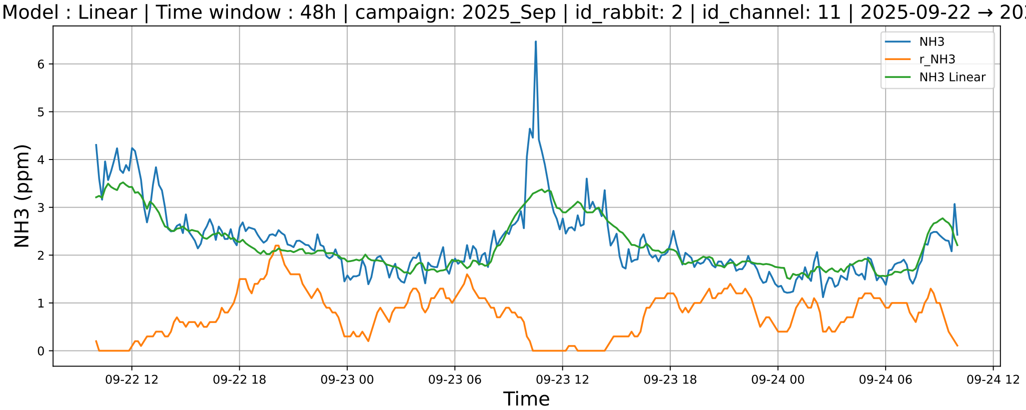


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24

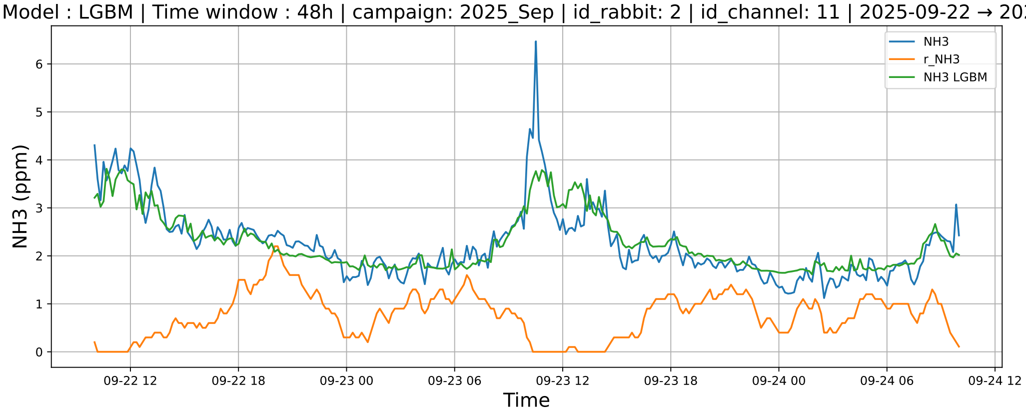


4.1.5.2 Time window : 48

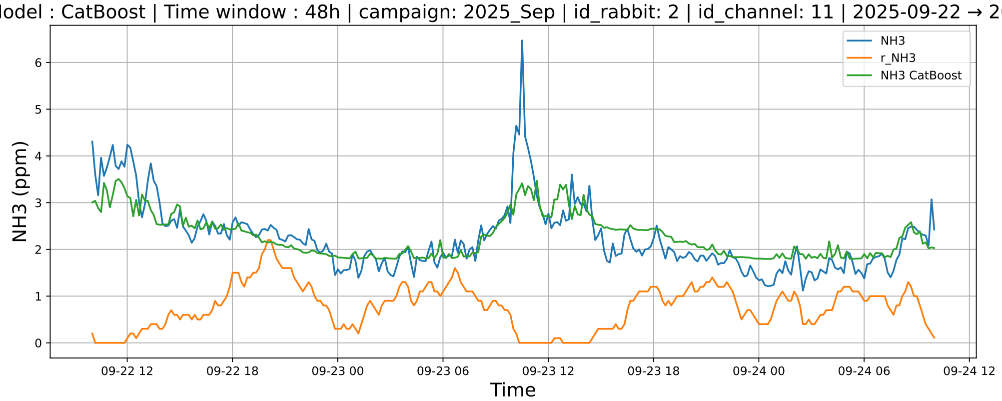
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24



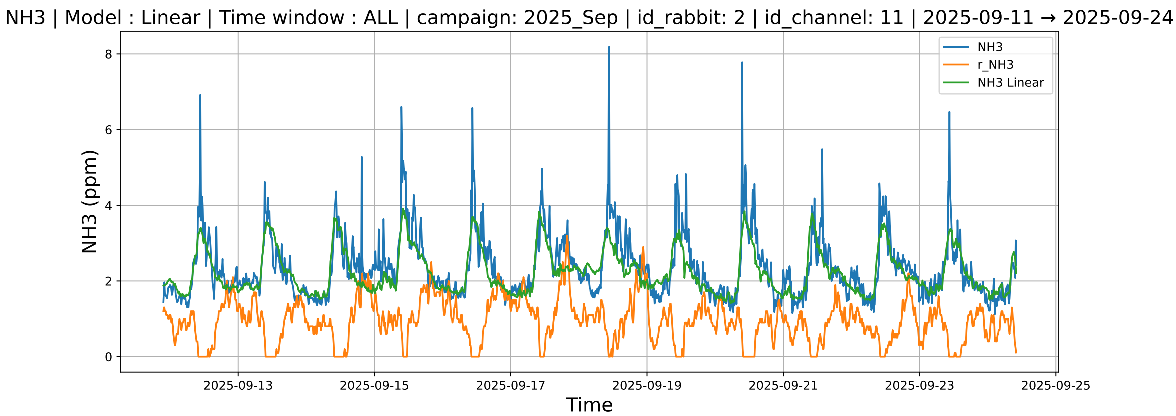
NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24

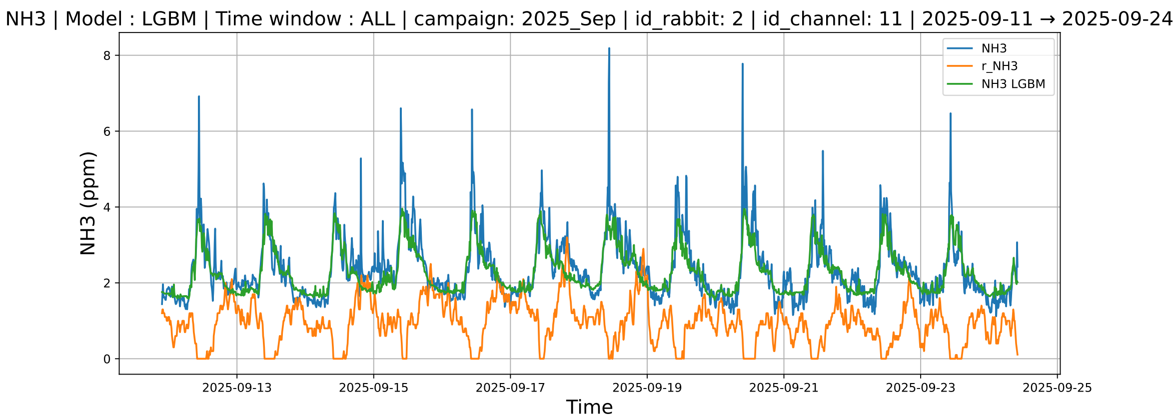


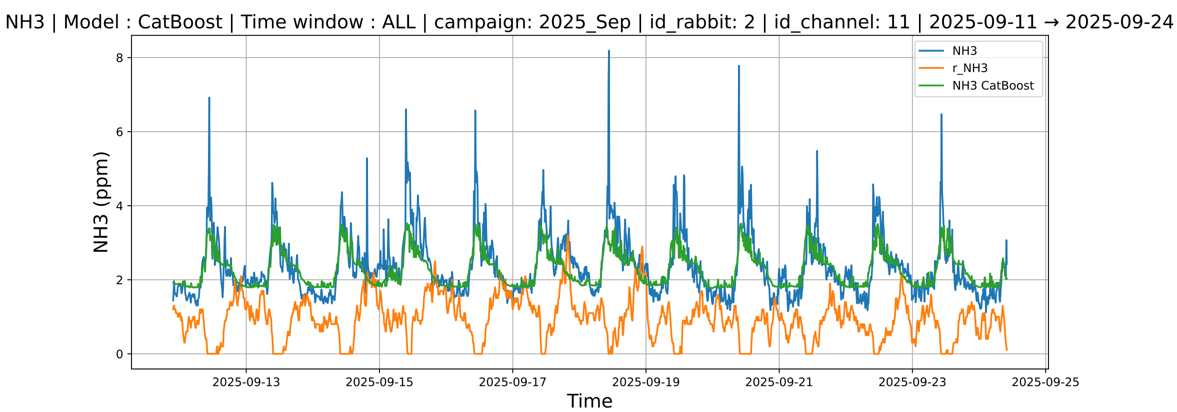
NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24



4.1.5.3 Time window : ALL



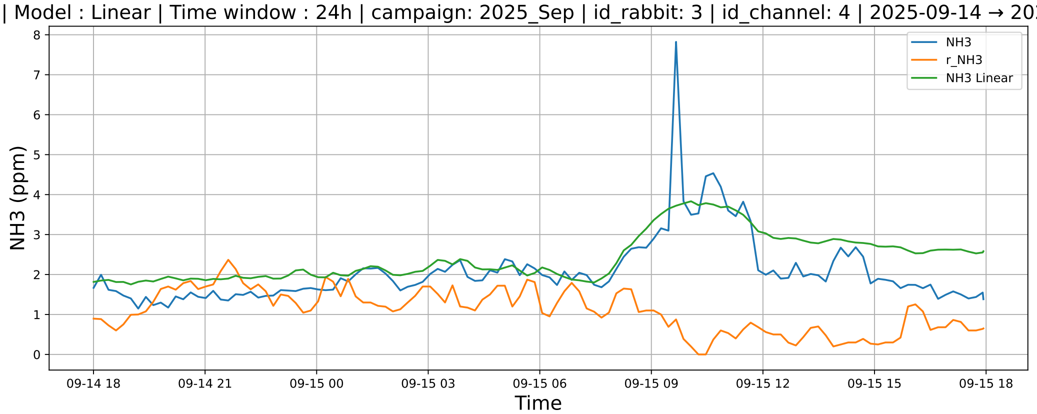




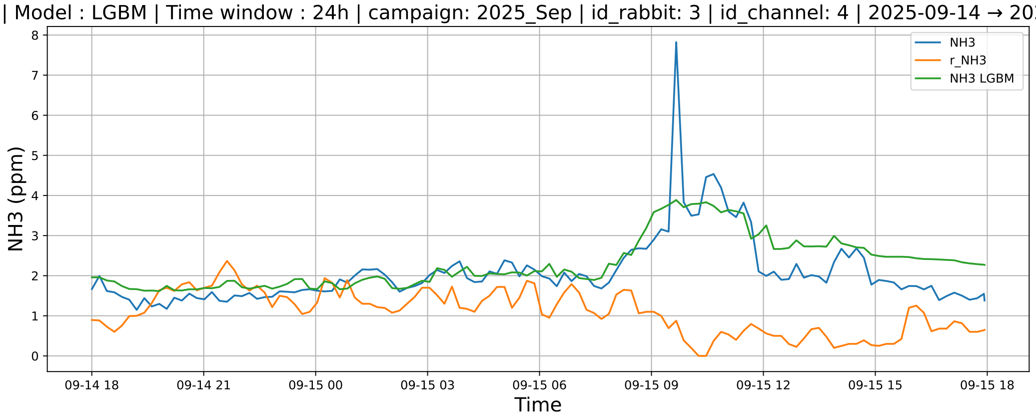
4.1.6 Group: campaign: 2025_Sep,id_rabbit: 3,id_channel: 4

4.1.6.1 Time window : 24

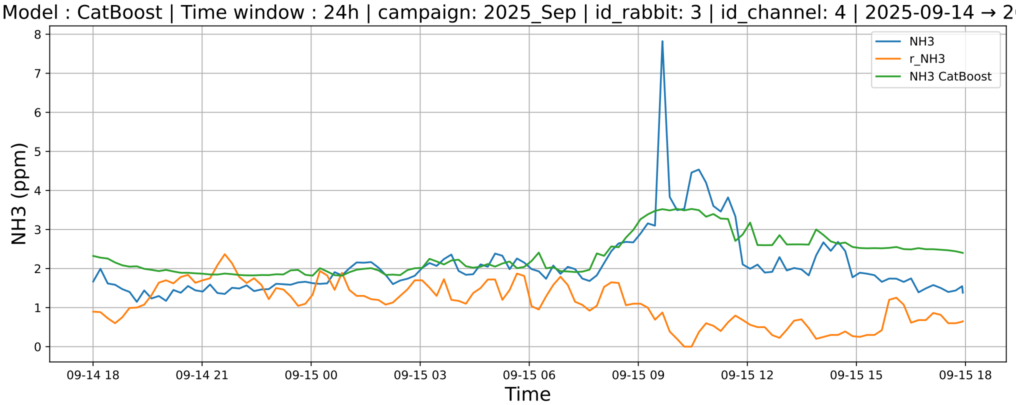
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15



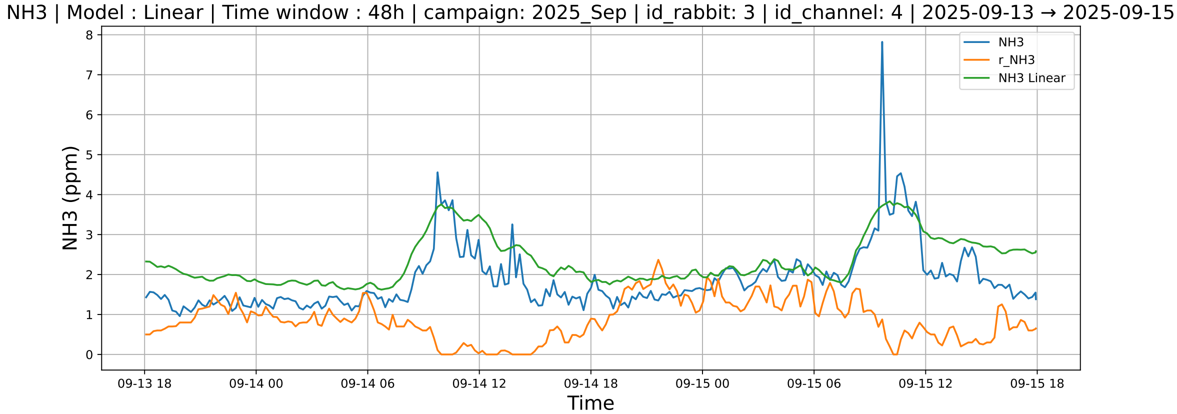
NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15

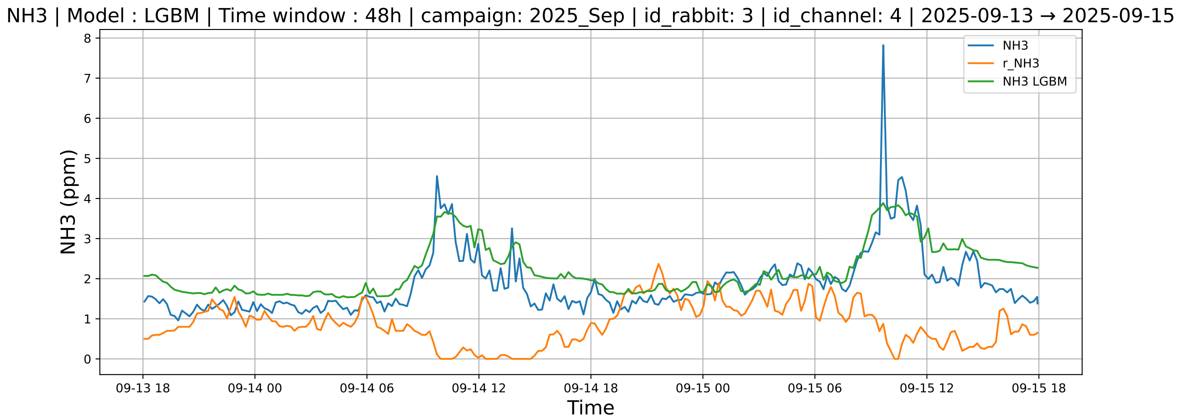


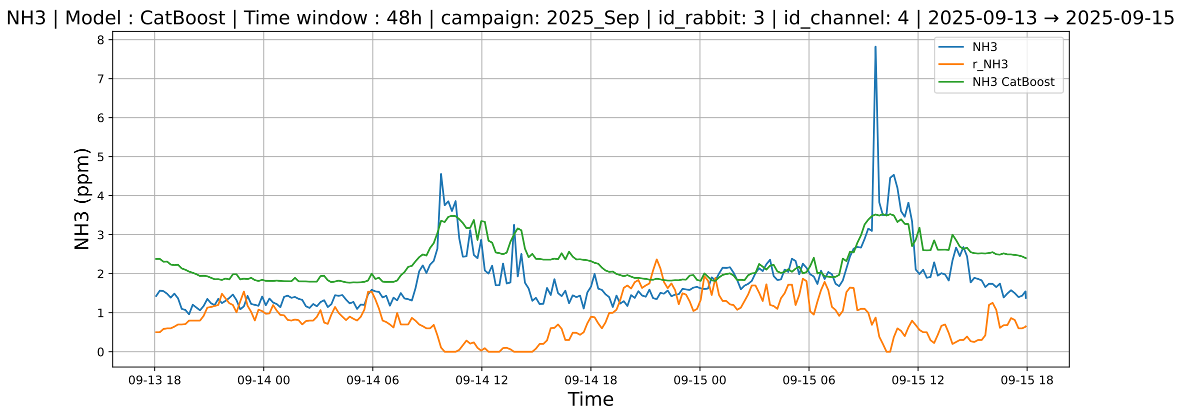
NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15



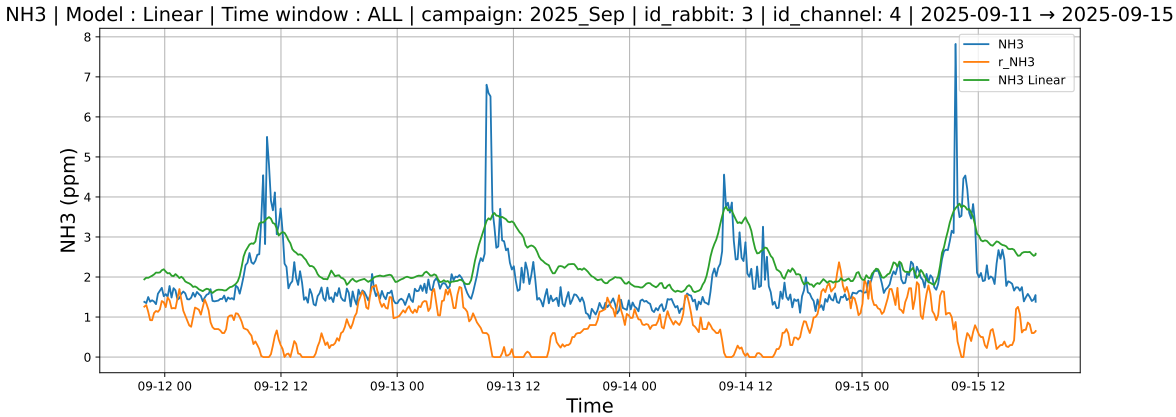
4.1.6.2 Time window : 48

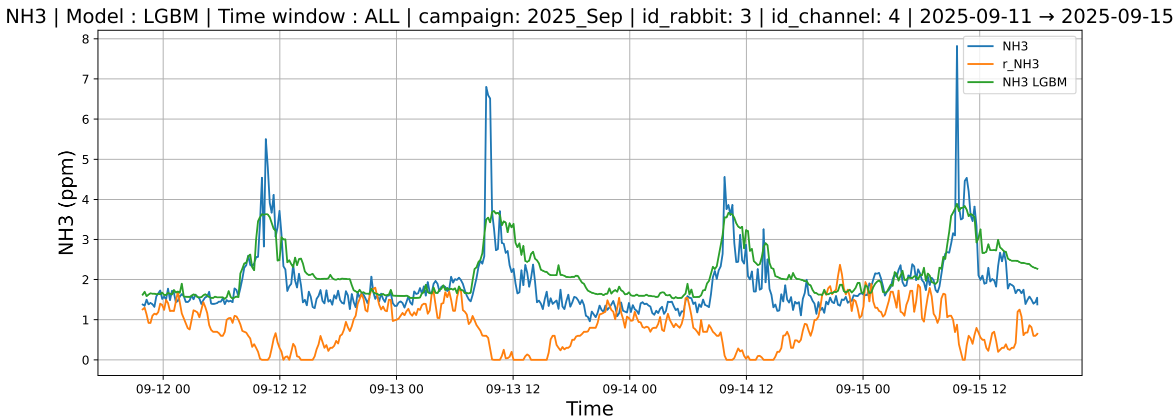


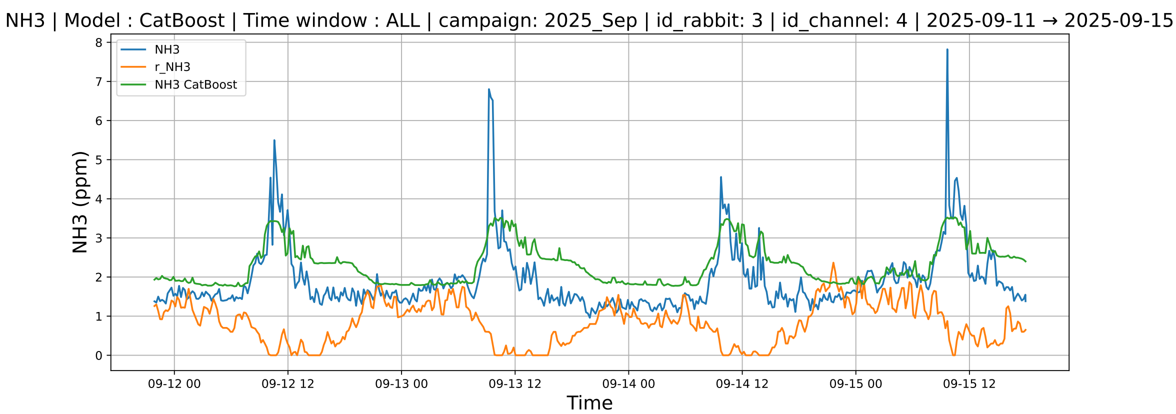




4.1.6.3 Time window : ALL

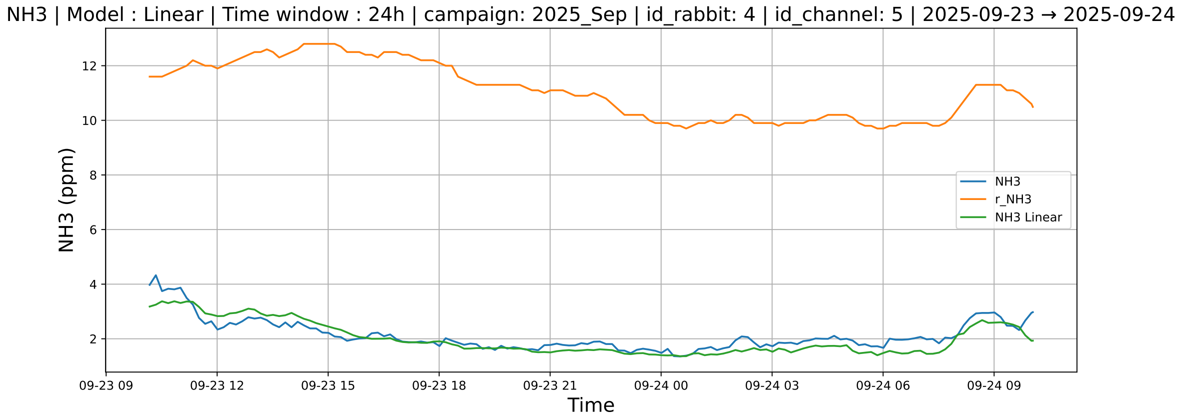


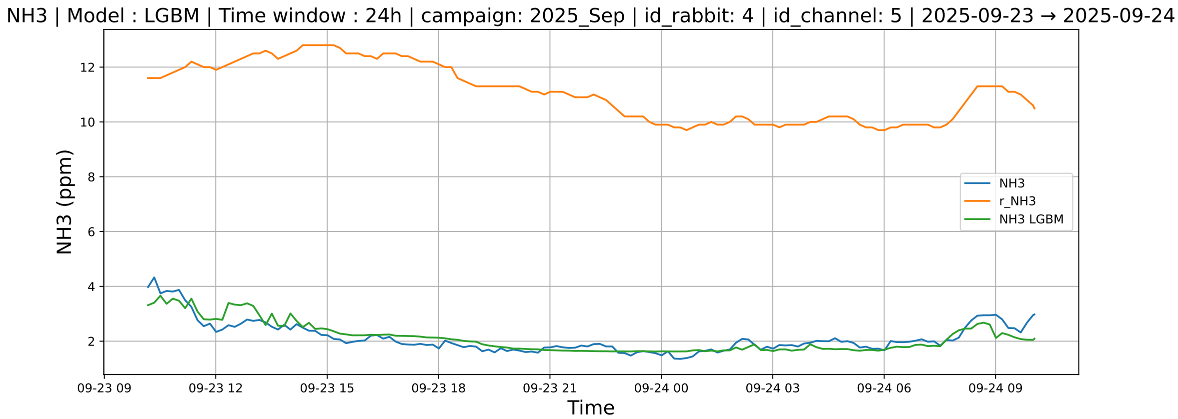


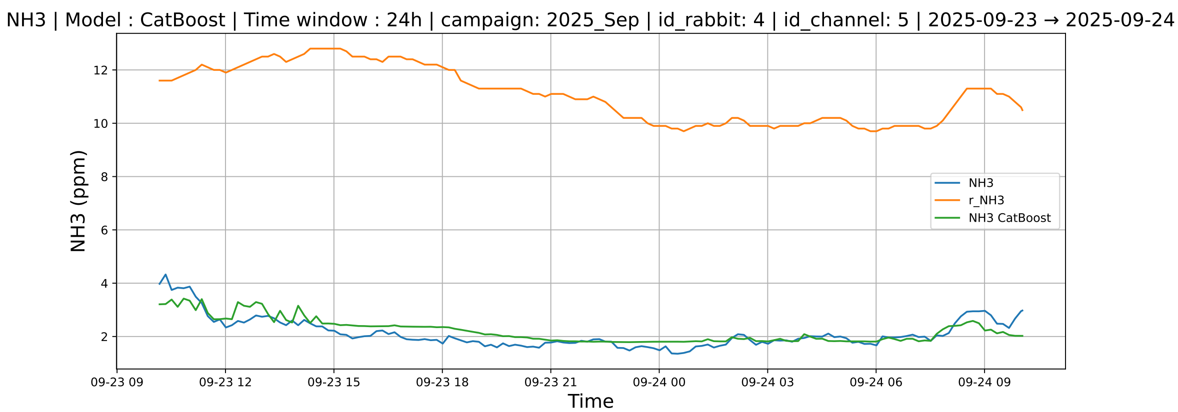


4.1.7 Group: campaign: 2025_Sep,id_rabbit: 4,id_channel: 5

4.1.7.1 Time window : 24

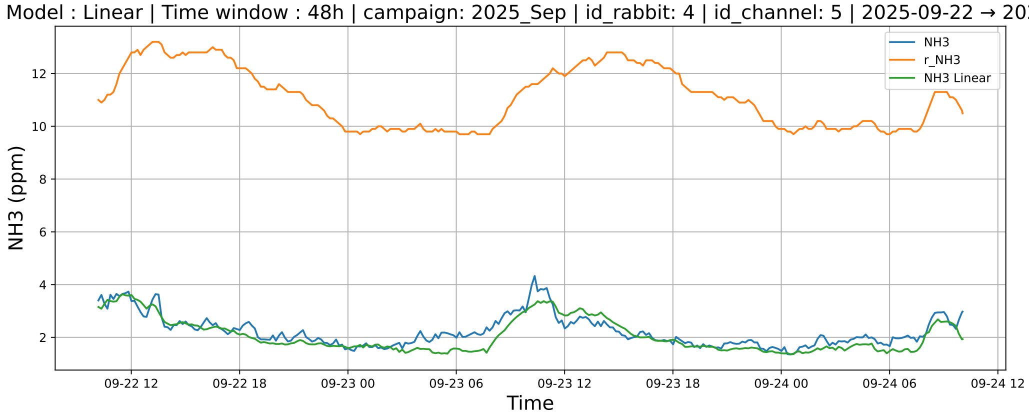




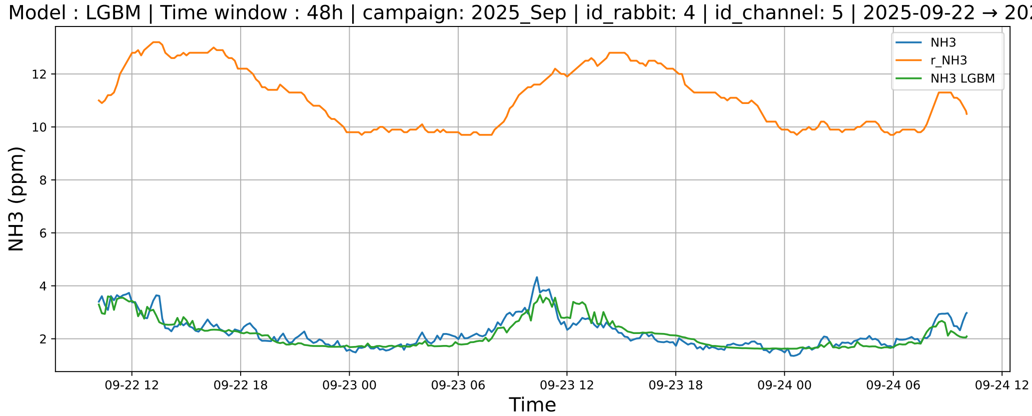


4.1.7.2 Time window : 48

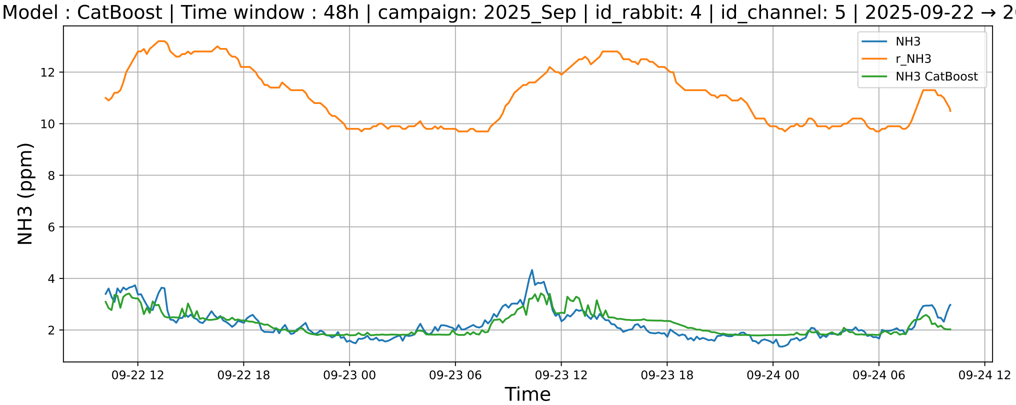
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24

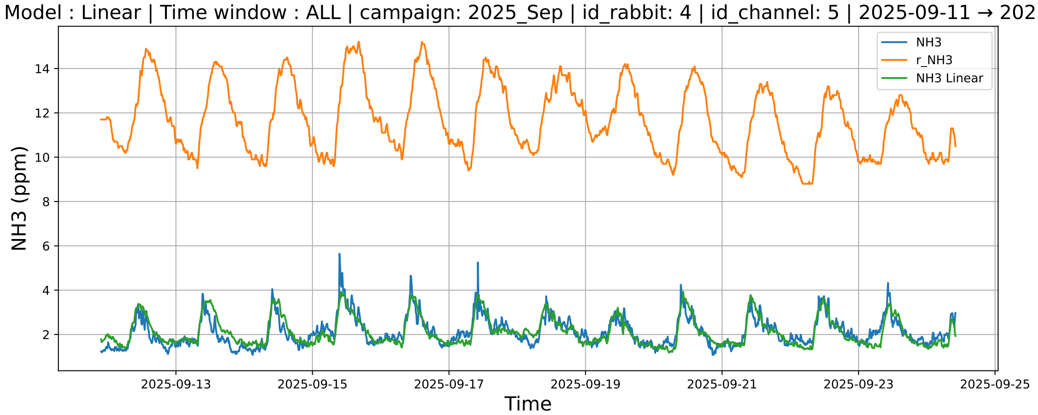


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24

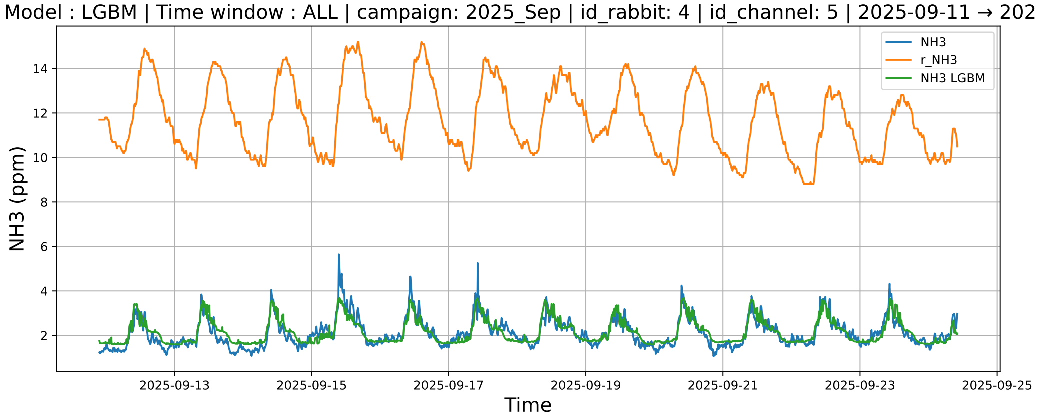


4.1.7.3 Time window : ALL

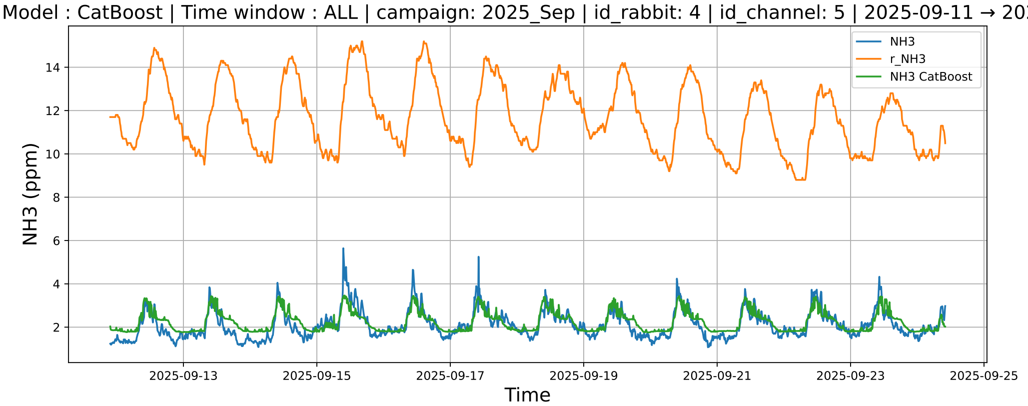
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24

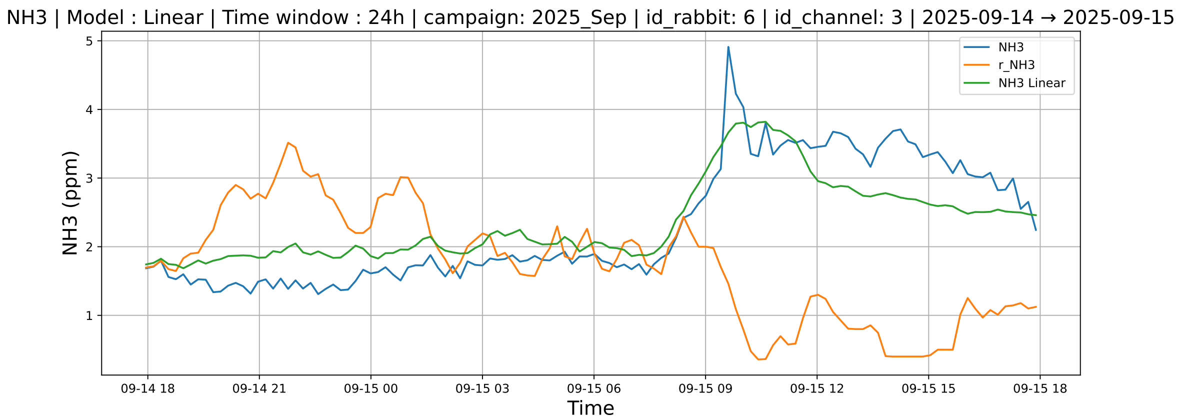


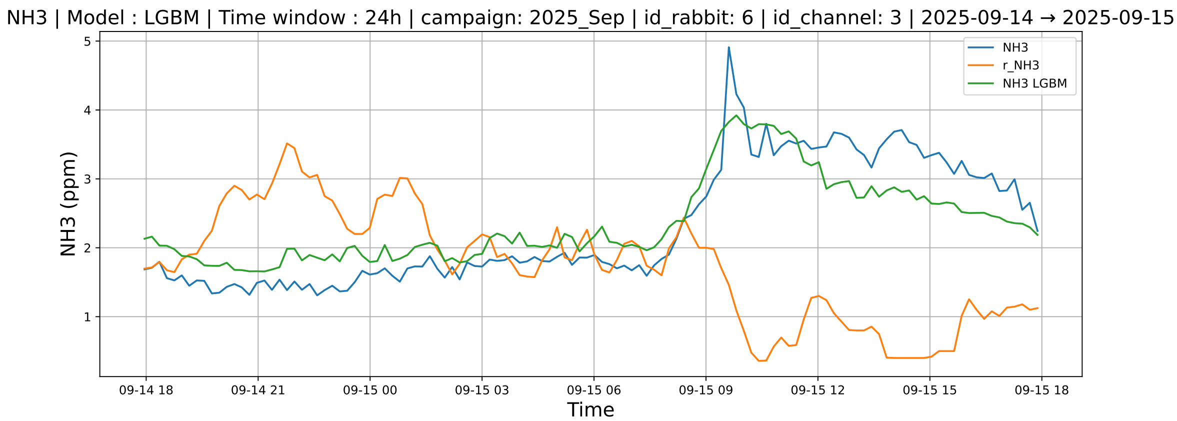
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24

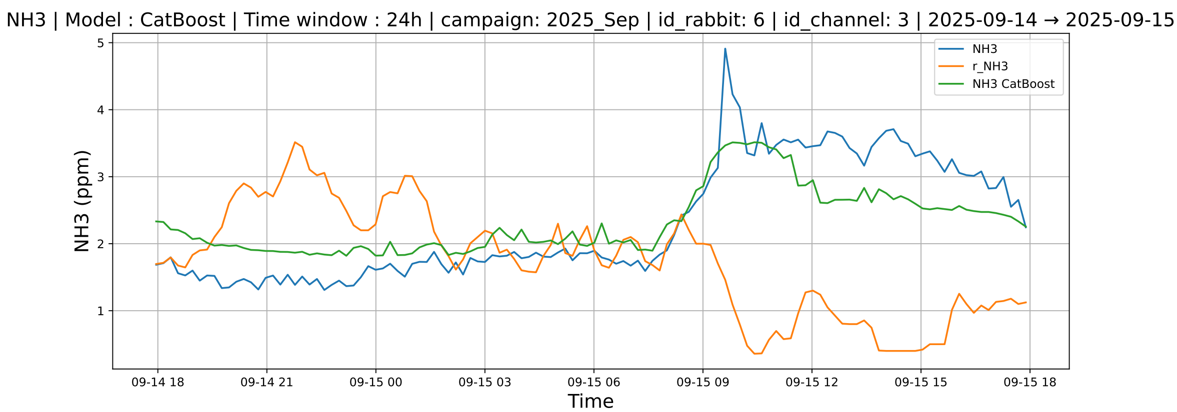


4.1.8 Group: campaign: 2025_Sep,id_rabbit: 6,id_channel: 3

4.1.8.1 Time window : 24

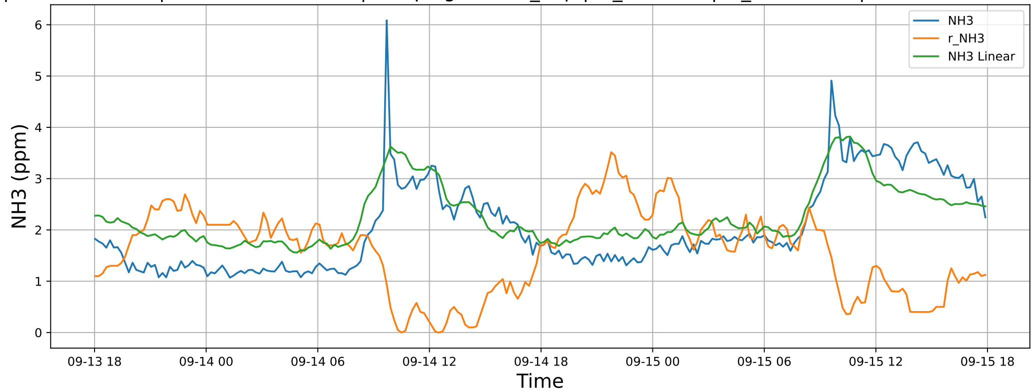




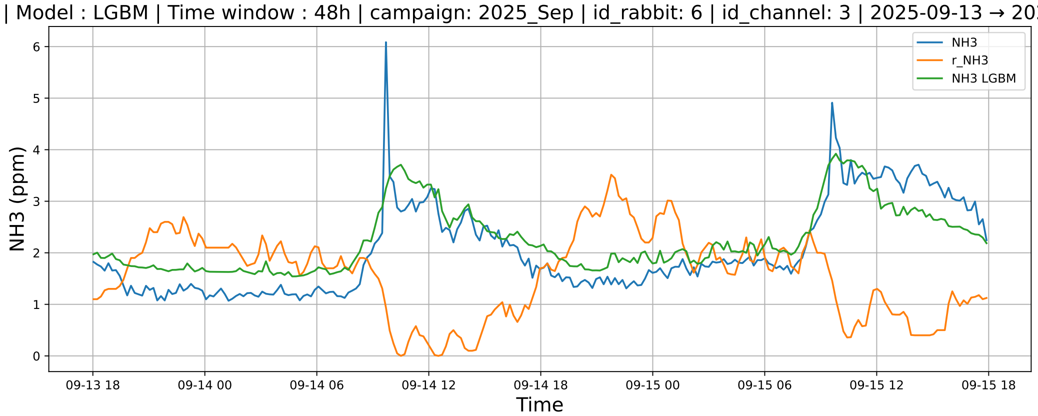


4.1.8.2 Time window : 48

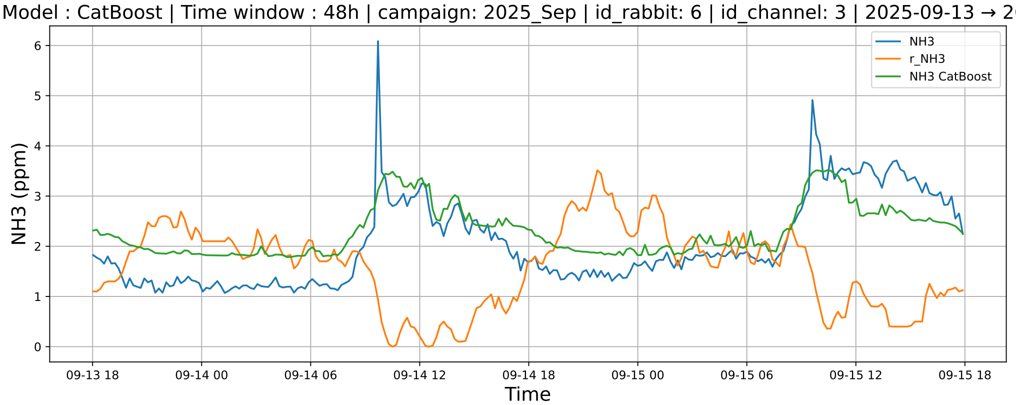
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15

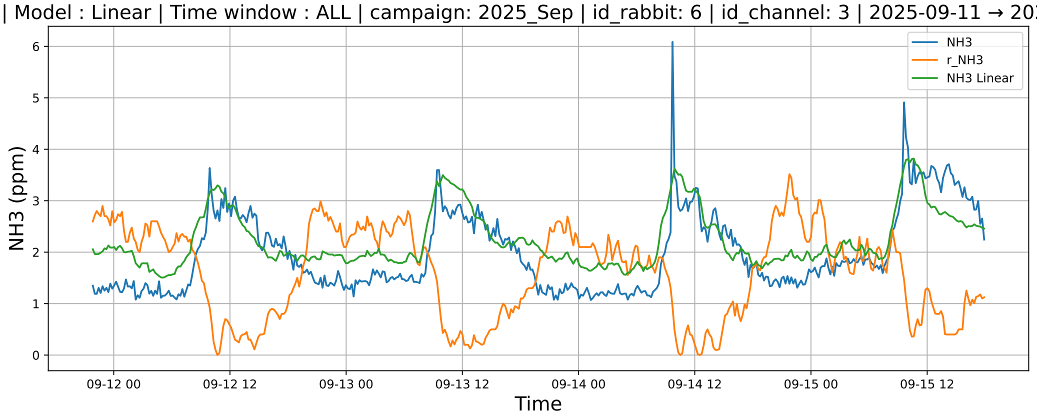


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15

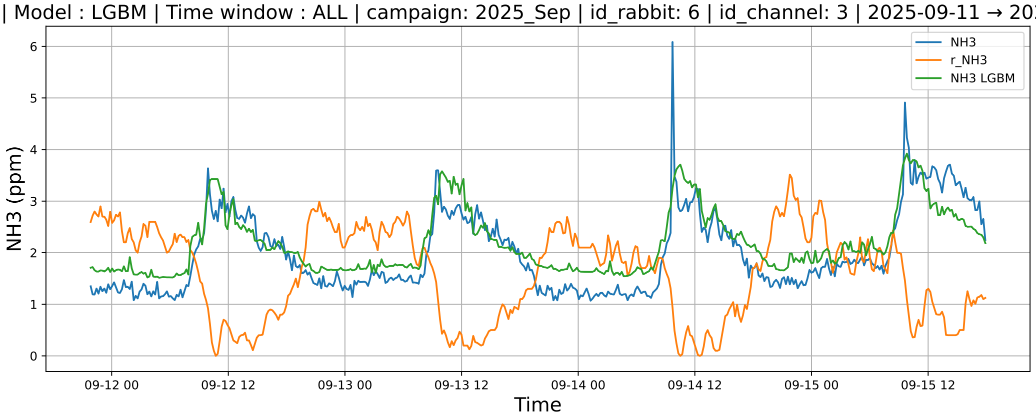


4.1.8.3 Time window : ALL

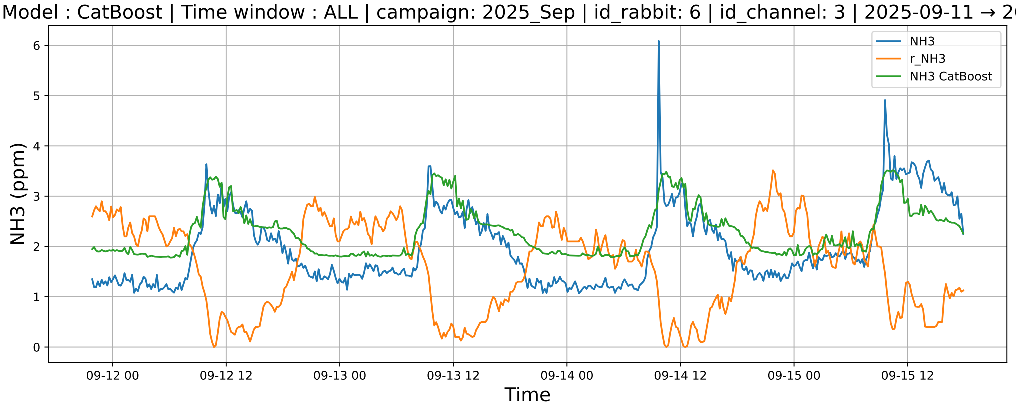
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15



NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15

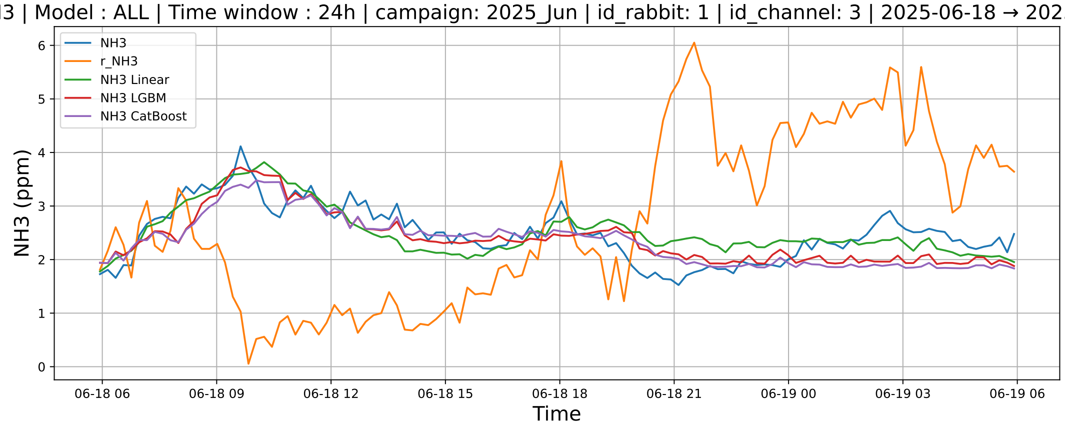


4.2 Compare plots

4.2.1 Group: campaign: 2025_Jun,id_rabbit: 1,id_channel: 3

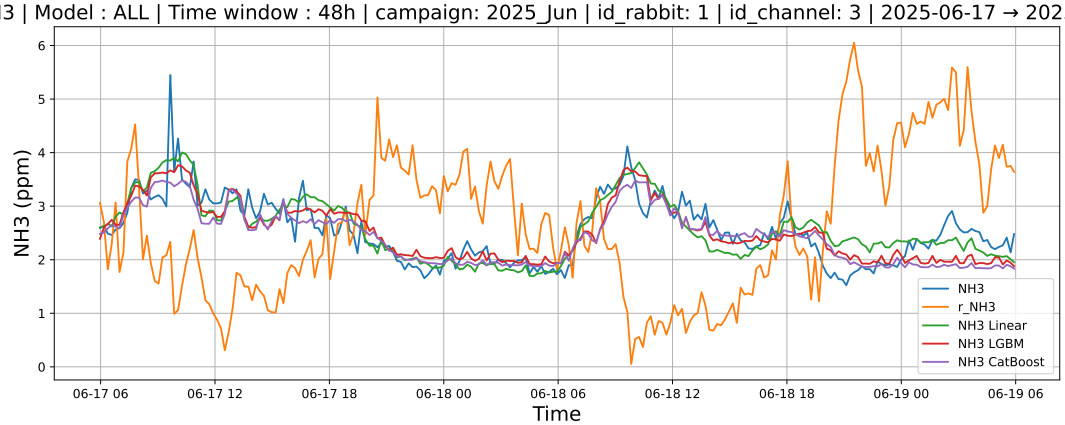
4.2.1.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19



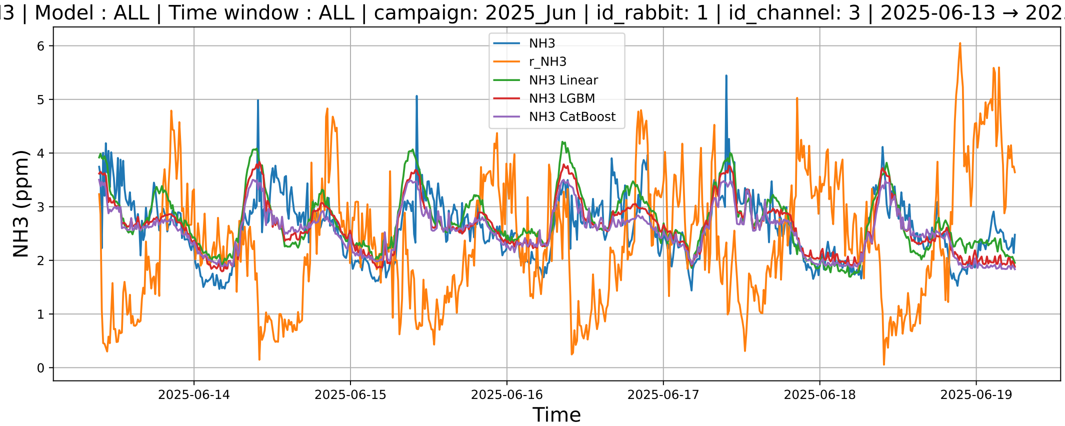
4.2.1.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19



4.2.1.3 Time window : ALL

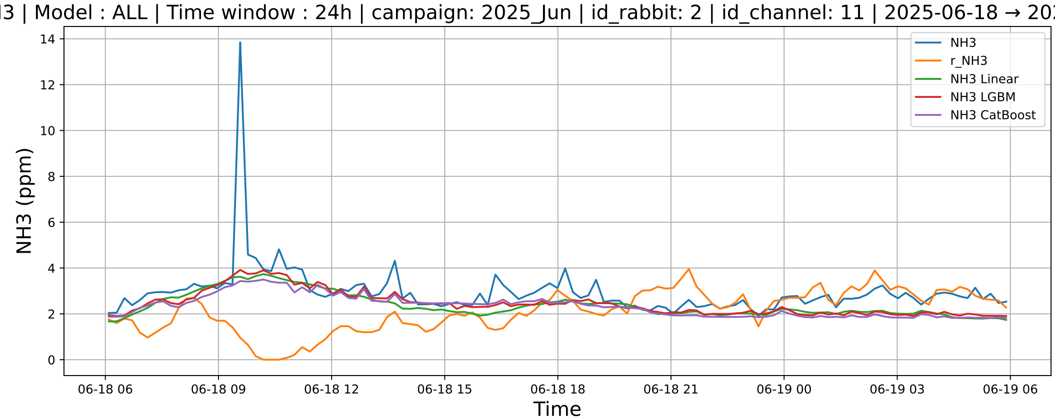
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-13 → 2025-06-19



4.2.2 Group: campaign: 2025_Jun,id_rabbit: 2,id_channel: 11

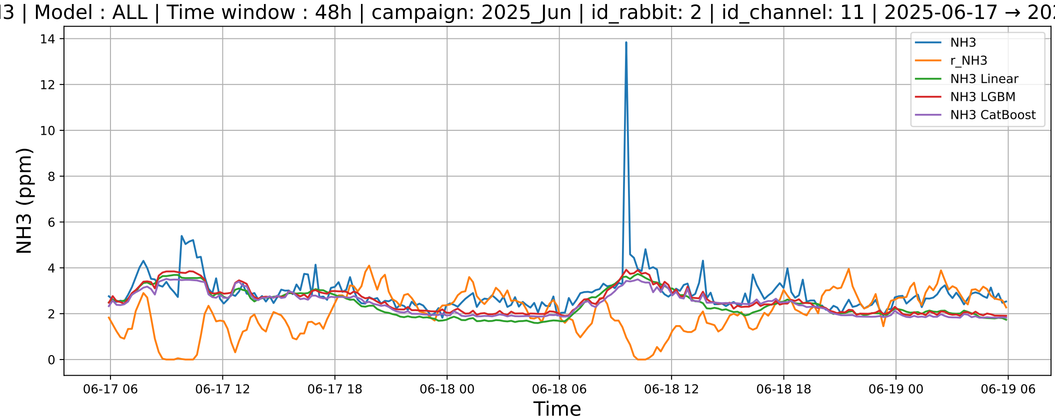
4.2.2.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19

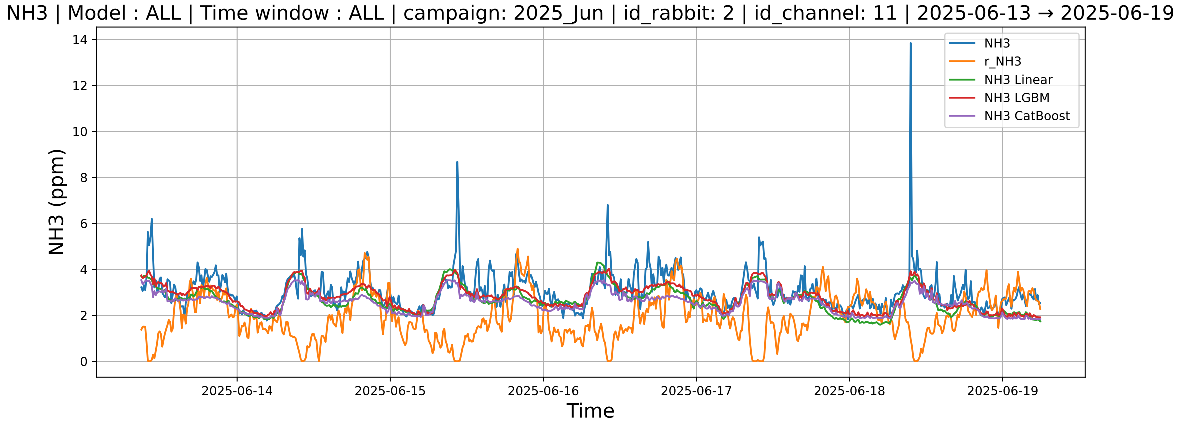


4.2.2.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19

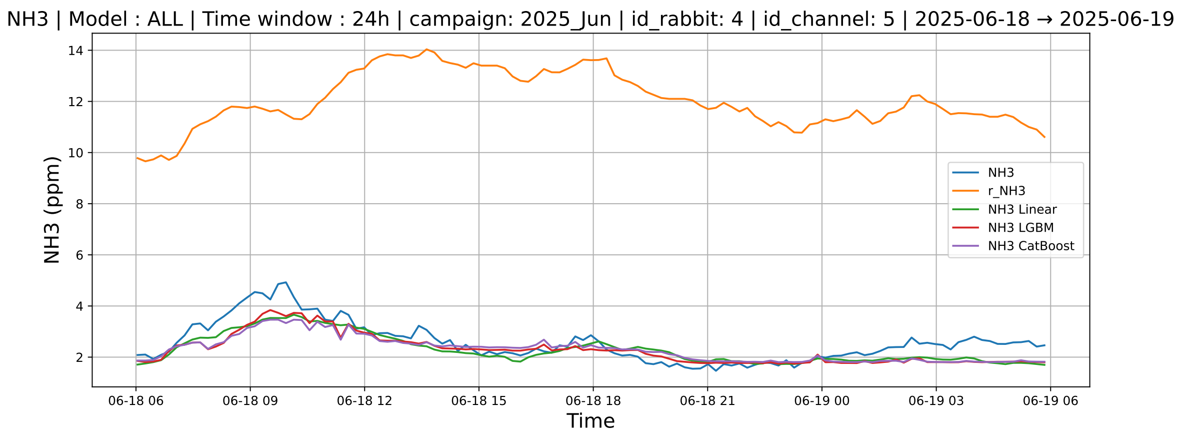


4.2.2.3 Time window : ALL

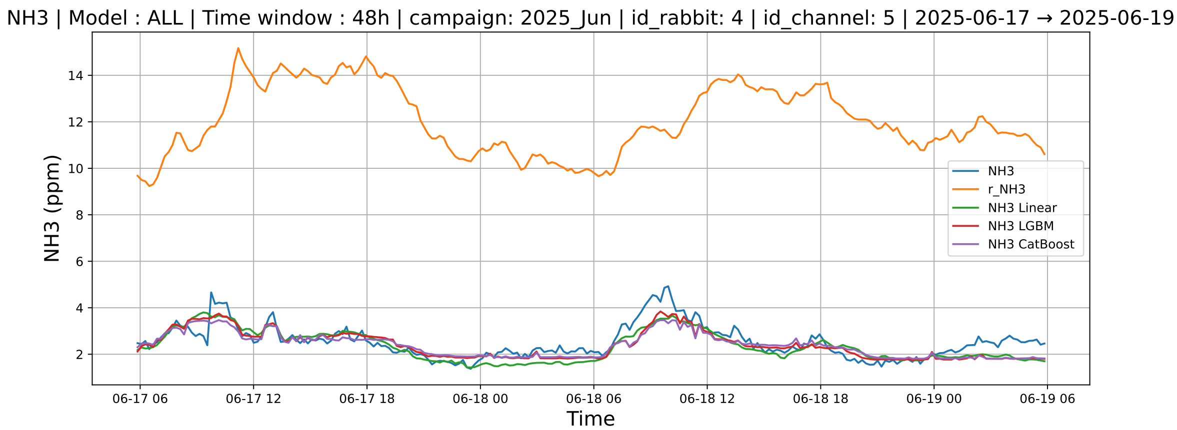


4.2.3 Group: campaign: 2025_Jun,id_rabbit: 4,id_channel: 5

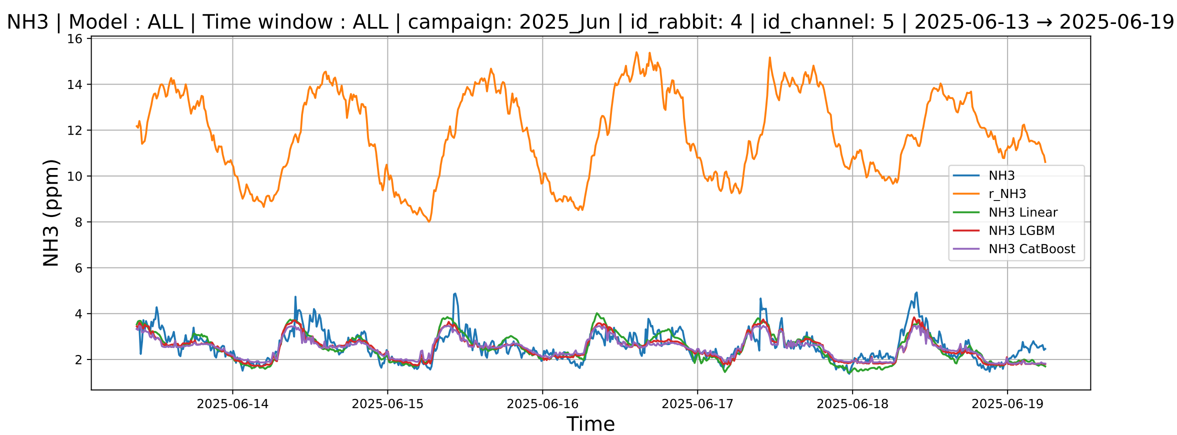
4.2.3.1 Time window : 24



4.2.3.2 Time window : 48

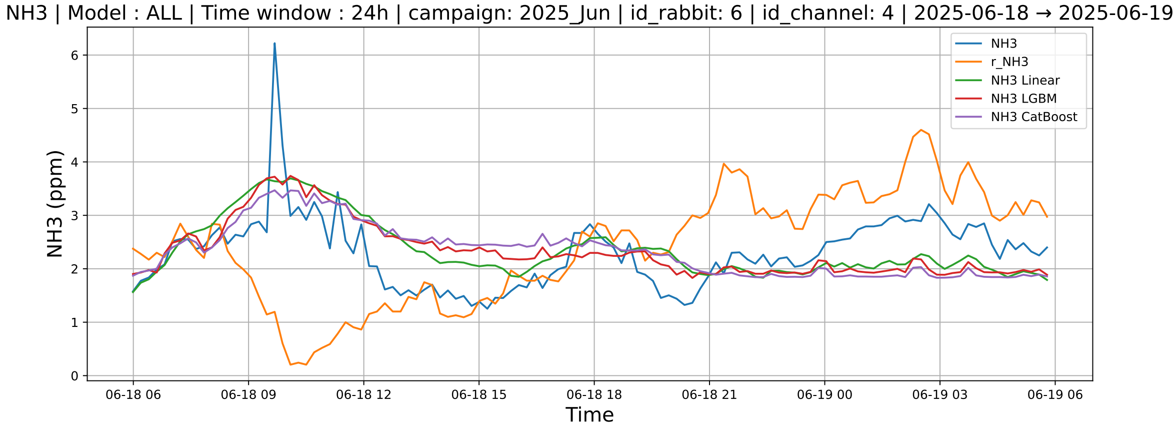


4.2.3.3 Time window : ALL

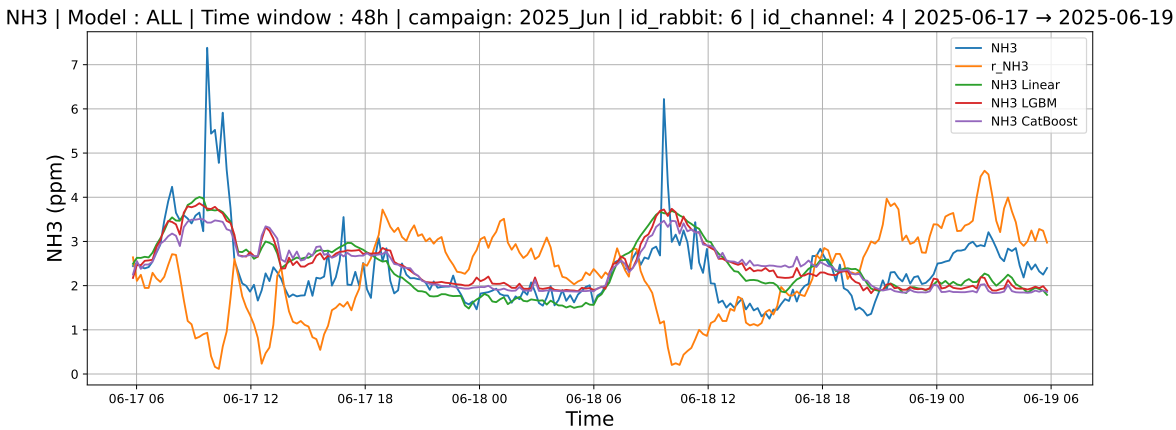


4.2.4 Group: campaign: 2025_Jun,id_rabbit: 6,id_channel: 4

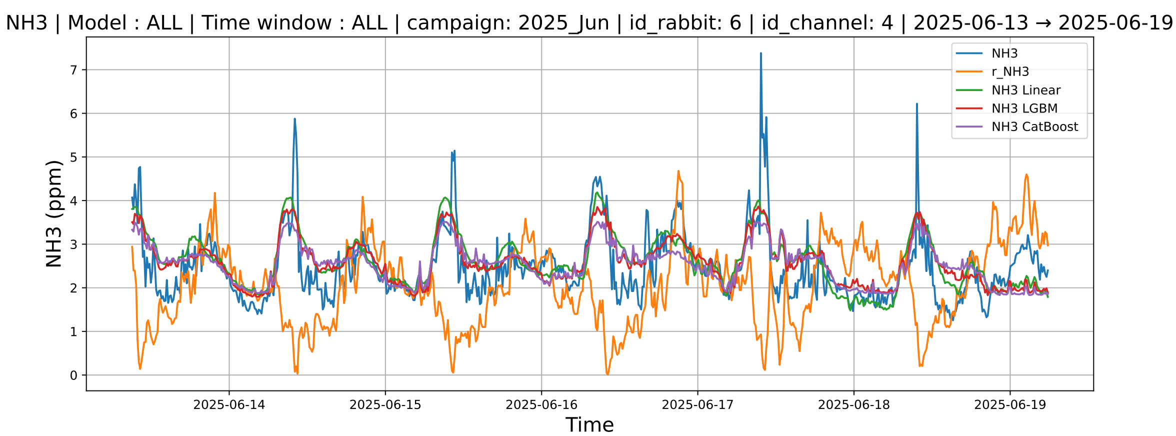
4.2.4.1 Time window : 24



4.2.4.2 Time window : 48



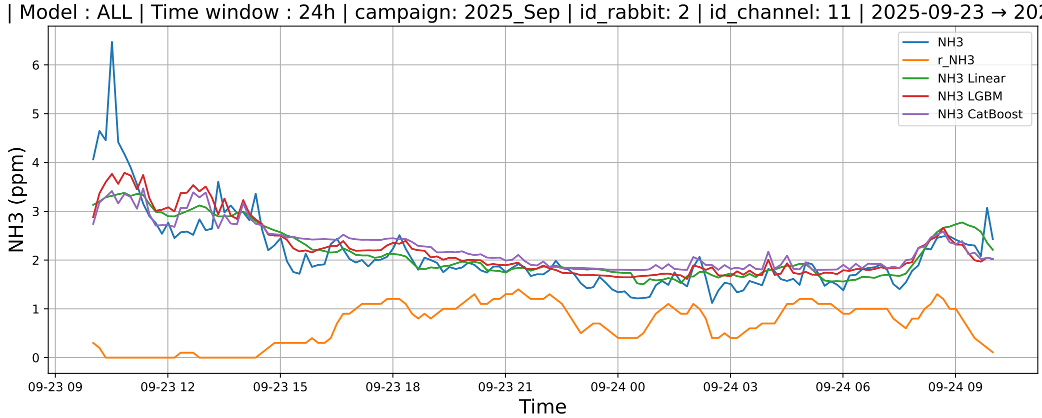
4.2.4.3 Time window : ALL



4.2.5 Group: campaign: 2025_Sep,id_rabbit: 2,id_channel: 11

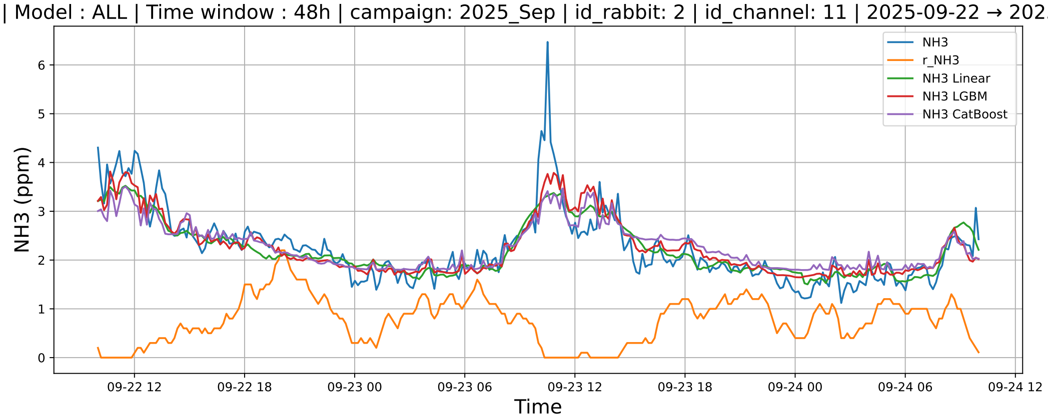
4.2.5.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24



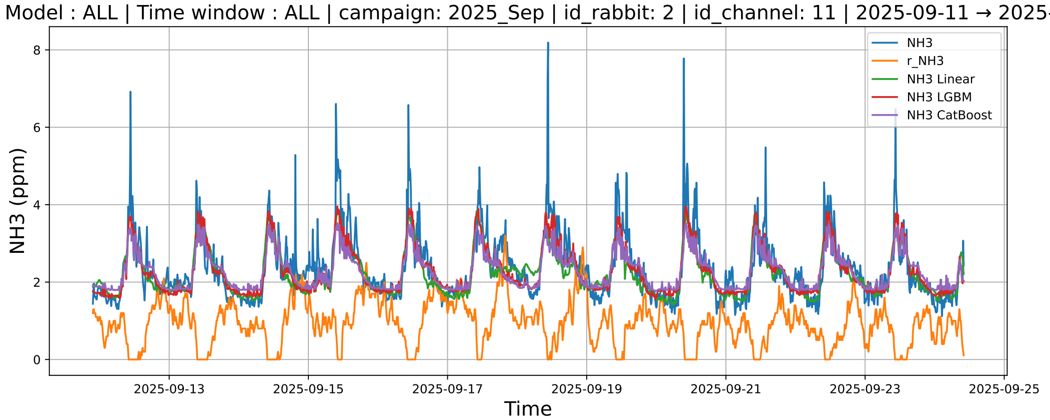
4.2.5.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24



4.2.5.3 Time window : ALL

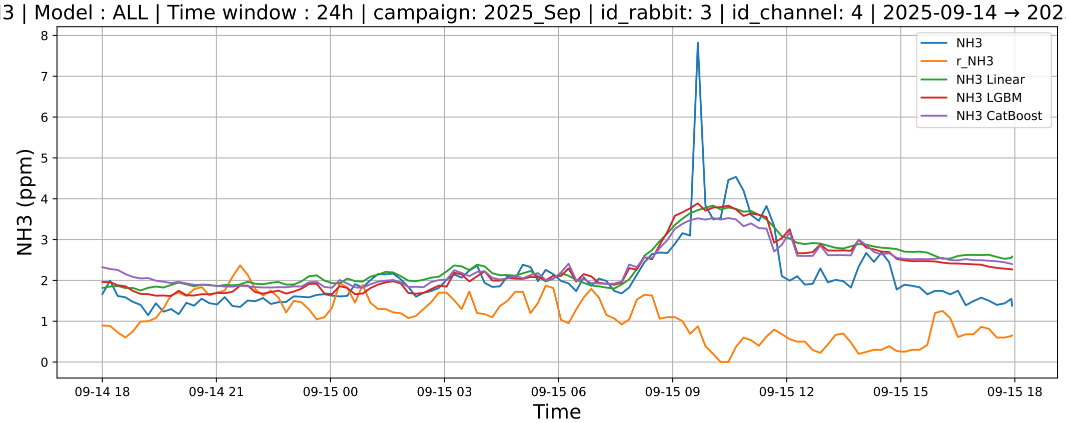
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



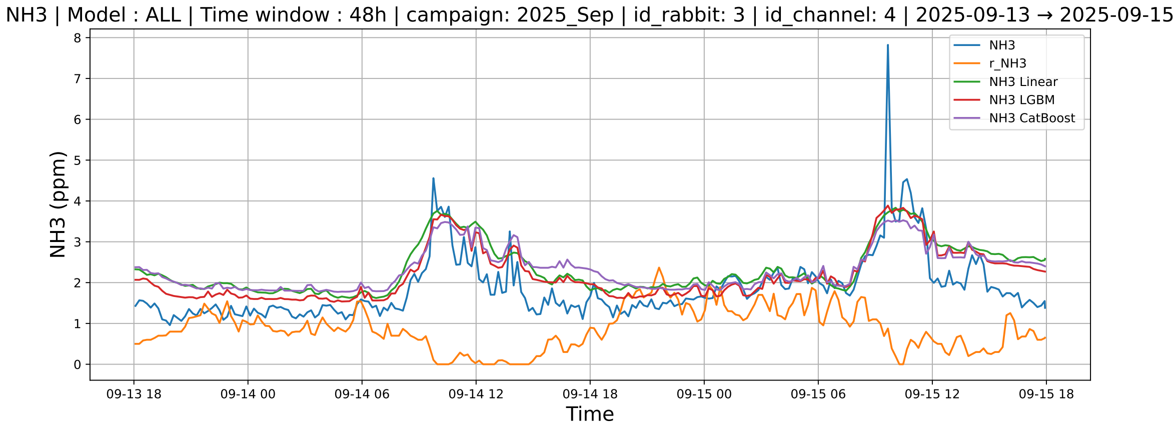
4.2.6 Group: campaign: 2025_Sep,id_rabbit: 3,id_channel: 4

4.2.6.1 Time window : 24

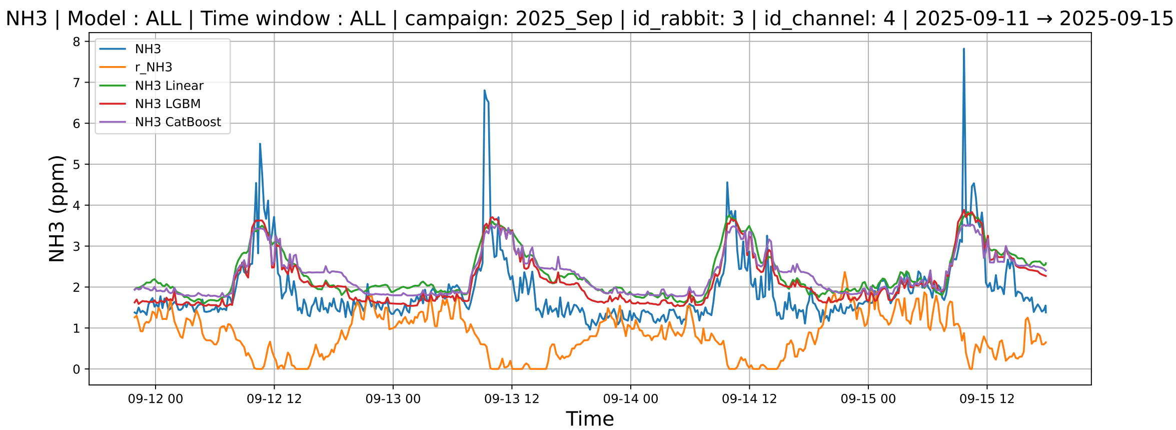
NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15



4.2.6.2 Time window : 48

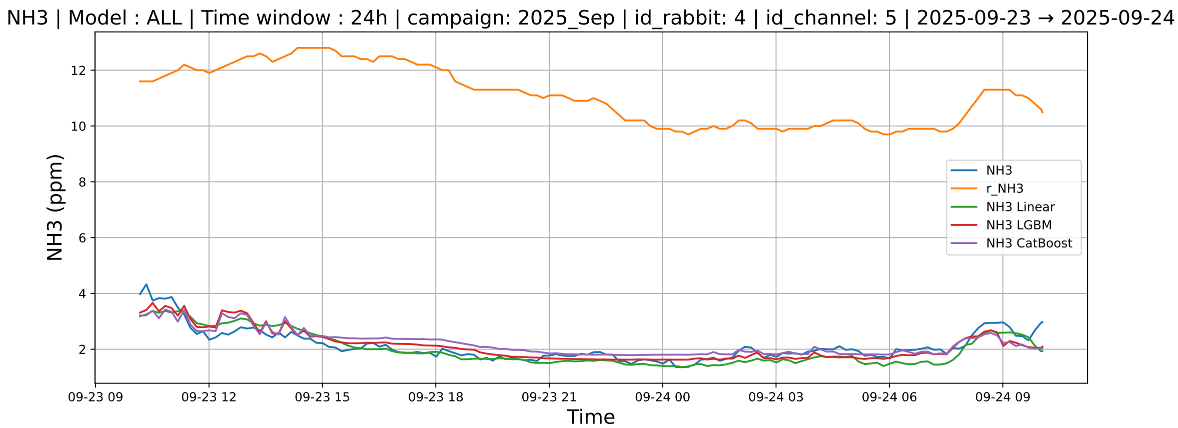


4.2.6.3 Time window : ALL

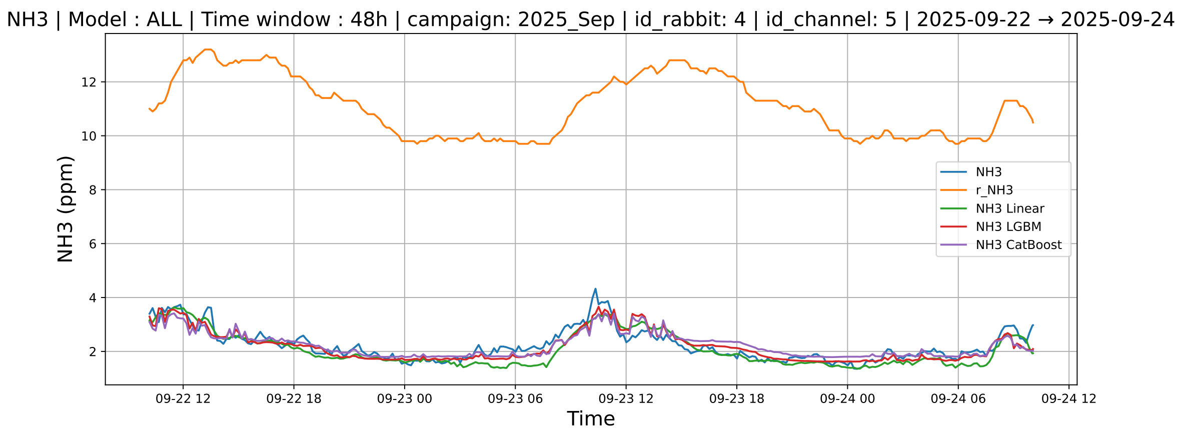


4.2.7 Group: campaign: 2025_Sep,id_rabbit: 4,id_channel: 5

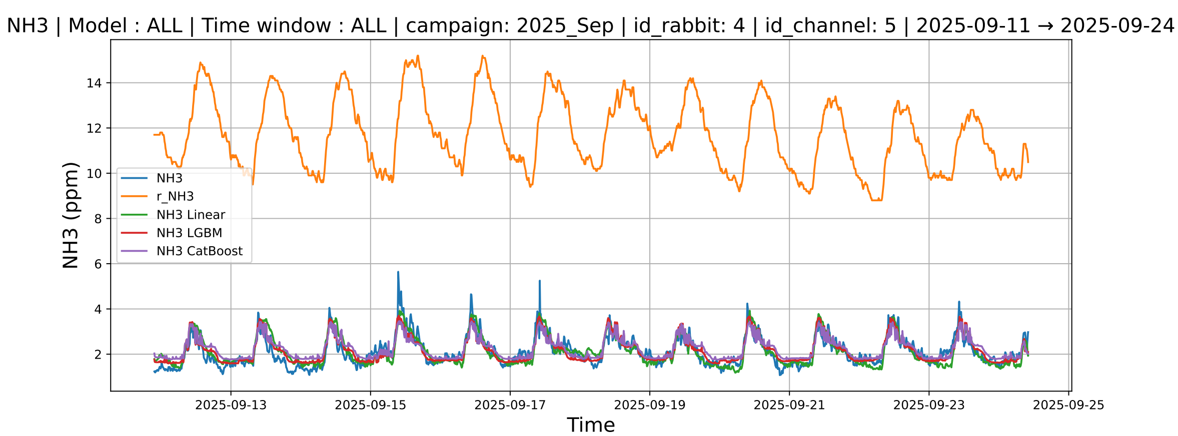
4.2.7.1 Time window : 24



4.2.7.2 Time window : 48



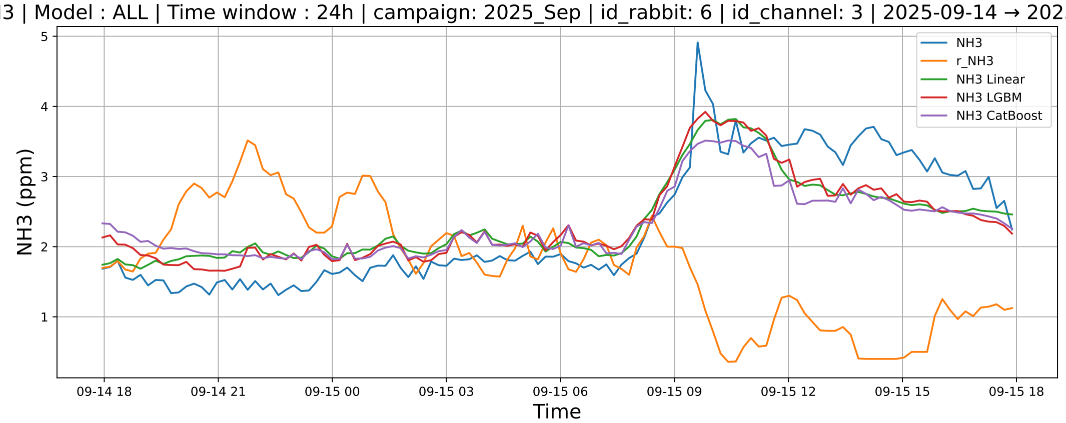
4.2.7.3 Time window : ALL



4.2.8 Group: campaign: 2025_Sep,id_rabbit: 6,id_channel: 3

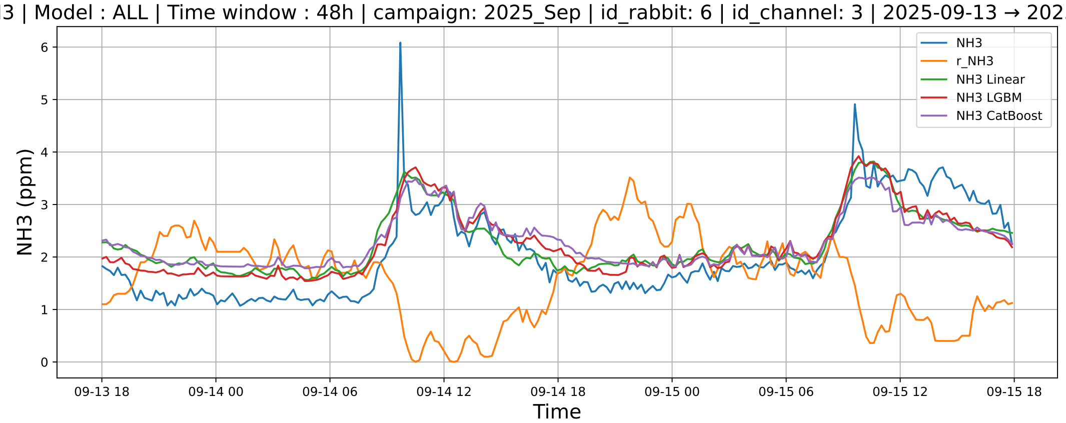
4.2.8.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15

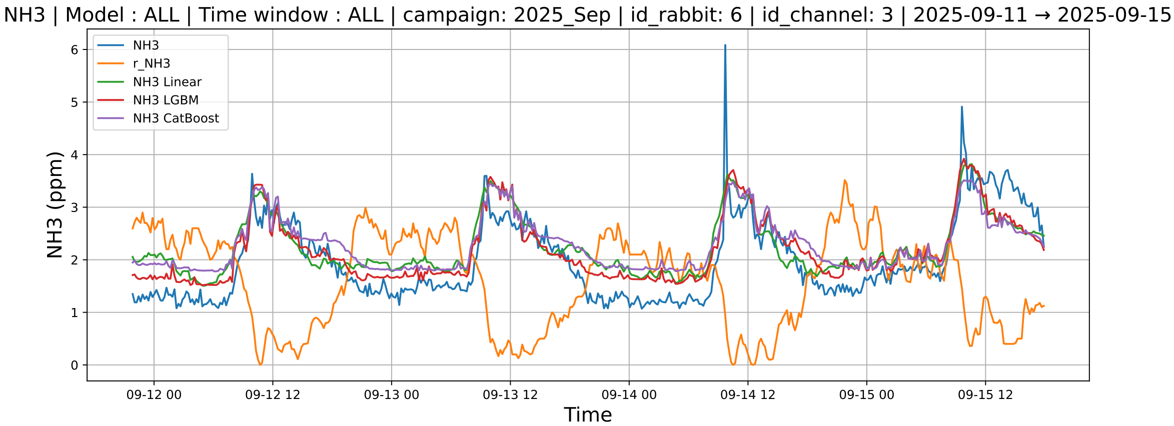


4.2.8.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15



4.2.8.3 Time window : ALL


NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15
NH3
r_NH3
NH3 Linear
NH3 LGBM
NH3 CatBoost
NH3 (ppm)
0
1
2
3
4
5
6
09-12 00
09-12 12
09-13 00
09-13 12
09-14 00
09-14 12
09-15 00
09-15 12
Time